Indian Knowledge System

VII Semester SEE Question Papers

Course codes: HS271TA

- Feb 2026 (HS271TA) - Part A
- Feb 2026 (HS271TA) - Part B

Question Paper ID: 557779

Course Code: HS271TA

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

Common to all Programs

Indian Knowledge System

Time : 1.5 Hours **Maximum Marks : 60**

Instructions to the students

1. Answer all questions from Part A.
2. Encircle the correct answer in OMR sheet only.

Part A

Question No	Question	M	CO	BT
1.1	The motto *“Satyam vada, dharmam chara”* comes from: a) Taittiriya Upanishad b) Bhagavad Gita c) Arthashastra d)Rigveda	01	1	1
1.2	The term *Indian Knowledge System (IKS)* refers to: a) Western interpretations of Indian culture b) Indigenous systems of knowledge evolved in India over millennia c) Only Vedic scriptures d) Modern Indian scientific research	01	1	1
1.3	In Indian Knowledge Systems, *Pratyakṣa* refers to: a) Logical inference b) Scriptural testimony c) Direct perception d) Ethical conduct	01	1	1
1.4	Which combination correctly represents kinds of traditional knowledge? a) Ritualistic and industrial b) Ecological, technical, and social c) Commercial and digital d) Political and military	01	2	1
1.5	Western education in India primarily emphasized: a) Oral transmission b) Integrated learning c) Discipline-centric knowledge d) Guru–Śiṣya system	01	1	1
1.6	*Takshashila* and *Nalanda* were: a) Vedic rituals b) Ancient centers of higher learning c) Holy temples d) Trade centers	01	1	1
1.7	Which text belongs to the Vedic Corpus? a) Purāṇas b) Upaniṣads c) Smṛtis d) Itihāsas	01	1	1
1.8	The concept *“Sa Vidya Ya Vimuktaye”* means: a) Knowledge is wealth b) Knowledge leads to liberation c) Education gives power d) Learning is endless	01	1	1
1.9	Indian philosophical systems differ mainly because they: a) Reject debate b) Share identical metaphysics c) Encourage multiple viewpoints d) Focus only on rituals	01	2	1

1.10	The language predominantly used for transmitting knowledge in ancient India was: a) Pali b) Tamil c) Sanskrit d) Prakrit	01	1	2
1.11	The traditional Indian system of education was centered around: a) Gurukula system b) British schooling system c) Online learning d) Modern universities	01	1	2
1.12	The primary mode of learning in Gurukulas was: a) Debate and memorization (*Shravana*, *Manana*, *Nididhyasana*) b) Written examinations c) Multiple-choice tests d) Internet-based study	01	1	2
1.13	Piṅgala's binary ideas emerged primarily from his study of: a) Astronomy b) Algebra c) Geometry d) Chandas	01	2	1
1.14	How many sūtras are there in Pāṇini's Aṣṭādhyāyī? a) 3000 b) 3500 c) 3983 d) 4200	01	2	1
1.15	In Sanskrit grammar, suffixes used for noun declensions are known as: a) Tiṅ b) Dhātu c) SuP d) Vārtika	01	2	2
1.16	In Sanskrit grammar, the basic element from which words are generated is called: a) Pada b) Dhātu / Prātipadika c) Vākya d) Samāsa	01	2	2
1.17	In Sanskrit phonetics, *Sthāna* refers to: a) Effort of pronunciation b) Tone of voice c) Place of articulation d) Meaning of sound	01	2	2
1.18	The Bhūtasaṅkhyā system represents numbers using: a) Only digits b) Alphabets with place value c) Objects, beings, or concepts d) Binary notation	01	2	2
1.19	In the Kaṭapayādi system, numbers are encoded mainly to: a) Encrypt secret messages b) Perform calculations faster c) Aid memorization through meaningful words d) Replace the decimal system	01	2	2
1.20	Which Pramāṇa involves understanding a new object through similarity with a known one? a) Pratyakṣa b) Anumāna c) Upamāna d) Śabda	01	2	1
1.21	Which of the following is a *means of valid knowledge (Pramāṇa)*? a) Prameya b) Saṃśaya c) Pratyakṣa d) Saṃskāra	01	2	1
1.22	In the Nyāya system, Anumāna (inference) is structured using how many logical steps? a) Three b) Four c) Five d) Seven	01	2	1
1.23	The arrangement of Piṅgala's *Ganas* corresponds in modern computer science to which concept? a) Finite automaton b) Binary tree c) De Bruijn sequence d) Huffman coding	01	2	2
1.24	In Piṅgala's system, *Laghu* and *Guru* syllables correspond respectively to: a) 0 and 1 b) 1 and 0 c) 1 and 2 d) Short and long meters only	01	2	1
1.25	According to Vedāṅga Śāstras, mathematics is compared to which of the following? a) Roots of a tree b) Crown of a king c) Crest on the head of a peacock d) Wings of a bird	01	3	1
1.26	Which system uses consonants to represent numerical values? a) Bhūtasaṅkhyā system b) Binary system c) Kaṭapayādi system D. Decimal system	01	3	1

1.27	Which Indian mathematician formulated rules for zero and negative numbers? a) Āryabhaṭa b) Brahmagupta c) Bhāskara I d) Madhava	01	3	1
1.28	The Bhūtasaṅkhyā system represents numbers using: a) Only digits b) Algebraic symbols c) Names of objects and concepts d) Roman numerals	01	3	2
1.29	.Which text is considered the primary source of Vedic astronomy? a) Sūrya Siddhānta b) Āryabhaṭīya c) Vedāṅga Jyotiṣa d) Brahmasphuṭasiddhānta	01	1	1
1.30	The Naksatra system in Indian astronomy consists of : a) 12 constellations b) 24 constellations c) 27 lunar mansions d) 30 lunar mansions	01	3	1
1.31	The concept that Earth rotates on its axis causing day and night was proposed by: a) Varāhamihira b) Brahmagupta c) Āryabhaṭa d) Madhava	01	3	1
1.32	The Panchāṅga consists of how many elements? a) Three b) Four c) Five d) Seven	01	3	1
1.33	Which of the following is NOT a limb of the Panchāṅga? a) Tithi b) Vara c) Yoga d) Rāśi	01	3	2
1.34	In the lost-wax process, the wax model is primarily made from: a) Resin b) Tree gum c) Beeswax d) Lac	01	3	1
1.35	The yantra used to provide uniform heating using sand is: a) Dola Yantra b) Patana Yantra c) Valuka Yantra d) Dhupa Yantra	01	3	1
1.36	Kapha Doṣa is dominantly composed of which Mahābhūtas? a) Air and Ether b) Fire and Water c) Water and Earth d) Fire and Air	01	3	2
1.37	Improving sleep and nutrition primarily enhances which component first? a) Consciousness b) Intellect c) Body d) Sattva	01	3	2
1.38	Kapha Doṣa, when imbalanced, is most closely associated with which Guṇa? a) Rajas b) Tamas c) Sattva d) None	01	3	2
1.39	Meditation and ethical living primarily help in increasing: a) Rajas b) Tamas c) Kapha d) Sattva	01	3	2
1.40	Which sequence correctly represents information flow in BMI–C? a) Body → Mind → Intellect b) Mind → Body → Consciousness c) Senses → Mind → Intellect → Action d) Consciousness → Mind → Body	01	3	2
1.41	Architecture (Vāstu Vidyā) is classified under which category of Kalas? a) Performing arts b) Nature sciences c) Engineering and craftsmanship d) Games and sports	01	1	2
1.42	The classical list of 64 Kalas represents: a) Religious rituals b) Military sciences c) Holistic life skills d) Only fine arts	01	1	2
1.43	Knowledge of regional languages belongs to which Kala group? a) Performing arts b) Literary and intellectual arts c) Domestic skills d) Martial arts	01	1	2

1.44	Vāstu Śāstra primarily aims at aligning buildings with: a) Political systems b) Modern technology c) Cosmic and natural forces d) Economic efficiency	01	1	2
1.45	Soil testing and land analysis fall under which limb of Vāstu? a) Prāsāda b) Vāstu c) Yāna d) Śayana	01	1	2
1.46	The Vāstu Purusha Maṇḍala is mainly used for: a) Decorative carving b) Structural strength c) Site energy zoning d) Cost estimation	01	1	2
1.47	Which of the following is NOT an limb of Vāstu Śāstra? a) Sthāpatya b) Prāsāda c) Yantra d) Ābharaṇa	01	1	2
1.48	Sustainable consumption-production patterns in IKS relate closely to: a) SDG 6 b) SDG 12 c) SDG 1 d) SDG 14	01	1	1
1.49	Arthaśāstra was written by: a) Vishnugupta / Kautilya b) Patanjali c) Panini d) Kalidas	01	1	1
1.50	Which SDG directly aligns with traditional Indian emphasis on environmental balance? a) SDG 8 b) SDG 13 c) SDG 16 d) SDG 10	01	1	1
1.51	What is the nakshatra in which Sri Ramachandra was born? a) Revathi b) Rohini c) Uttara d) Punarvasu	01	1	1
1.52	Rāma's ability to listen to citizens' grievances reflects: a) Judicial activism b) Participatory governance c) Patrimonial rule d) Neutral administration	01	2	1
1.53	Inspection, espionage, and audit systems in Arthaśāstra relate to: a) Nationalism b) Preventive vigilance c) Ritual governance d) Military conquest	01	2	1
1.54	Which of the following is NOT one of the Saptāṅga of the state? a) Janapada b) Danda c) Kośa d) Yajña	01	2	1
1.55	Which institution in India focuses on protection of traditional healing knowledge? a) NITI Aayog b) Ministry of AYUSH c) RBI d) Election Commission	01	3	1
1.56	Which of the following is incorrect. i)TK is distinctly associated with an indigenous or local community, which preserves and transmits it from generation to generation. ii) TK is old and static. iii) TK is generated, preserved and transmitted in a traditional and intergenerational context. iv) TK is not limited to any specific field.	01	3	1
1.57	Traditional ecological knowledge in India most strongly aligns with which practice? a) Excessive chemical farming b) Sacred groves and community forests c) Mining-based development d) Industrial monoculture	01	3	1
1.58	The term "Daṇḍa" in Arthaśāstra refers to: a) Agricultural surplus b) Law, order, and punitive authority c) Trade networks d) Cultural norms	01	4	1
1.59	Janapada primarily refers to: a) Army b) Territory and population c) Royal palace d) Trade officials	01	4	1
1.60	According to Arthaśāstra, the foremost duty of the king is: a) Expansion of trade b) Promotion of arts c) Protection of subjects (Rakṣaṇa) d) Construction of monuments	01	1	1

Question Paper ID: 257011

Course Code: HS271TA

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VII Semester B.E. Regular Examination February 2026

Common to all Programs

Indian Knowledge System

Time : 1.5 Hours **Maximum Marks : 40**

Instructions to the students

1. Answer FIVE full questions in the answer booklet given.
2. Answer any one full question from 1 and 2, 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Unit-1

Question No	Question	M	CO	BT
1a	Explain any four differences between vedas and vedangas	04	3	3
1b	Discuss the historicity of Indian Knowledge Systems.	04	1	1
	OR			
2a	What are Vedas ? List the four vedas	04	2	2
2b	List four major text types or four portions of Vedas. Explain each briefly	04	2	3

Unit-2

Question No	Question	M	CO	BT
3a	Discuss why Pāṇini is often regarded as the “father of linguistics” from both traditional and modern perspectives.	04	2	2
3b	Explain the development of the *decimal place-value system* in ancient India and its global significance.	04	2	2
	OR			
4a	Explain *Anumāna Pramāṇa* with reference to the Pañca-Avayava (five-step inference) structure.	04	2	3
4b	Describe how Piṅgala’s arrangement of Ganas relates to the modern concept of the *de Bruijn sequence*.	04	2	5

Unit-3

Question No	Question	M	CO	BT
5a	Briefly discuss the contributions of any four ancient Indian mathematicians.	04	3	2
5b	Describe the art of perfume making in ancient India and its cultural significance.	04	3	2
	OR			
6a	Explain the concept of Panchāṅga and its five elements.	04	3	2
6b	Describe the metallurgical knowledge reflected in the Delhi Iron Pillar.	04	3	1

Unit-4

Question No	Question	M	CO	BT
7a	How does Pitta Doṣa influence physiological activities?	04	3	2
7b	Differentiate between Nāgara and Drāviḍa temple styles.	04	3	3
	OR			
8a	Apply the concept of 64 Kalas to modern holistic education	04	3	3
8b	How can Vāstu Śāstra principles be applied in contemporary building design?	04	3	3

Unit-5

Question No	Question	M	CO	BT
9a	Discuss the role of advisors in Rāma's administration.	04	1	3
9b	Name one SDG aligned with Ayurvedic practices.	04	2	1
	OR			
10a	Discuss the role of TKDL in safeguarding traditional knowledge.	04	3	3
10b	What are the sources of income for a state as per Arthasastra?	04	4	2

Product Design and Development

VII Semester SEE Question Papers

Course codes: IM372IA, 21IM72, 18IM72, 16IM73, 12IM7E1

- Feb 2026 (IM372IA)
- Feb/Mar 2025 (21IM72)
- Feb/Mar 2024 (18IM72)
- Feb/Mar 2023 (18IM72)
- Jan/Feb 2021 (16IM73)
- Fast Track Oct 2020 (16IM73)
- Nov/Dec 2019 (16IM73)
- Nov/Dec 2018 (12IM7E1)
- Nov/Dec 2016 (12IM7E1)

IM372IA

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular Examinations February-2026

Industrial Engineering and Management

PRODUCT DESIGN AND DEVELOPMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	**M**	**BT**	**CO**
1	1.1	Define design by innovation.	01	1	1
	1.2	State one objective of design thinking.	01	1	1
	1.3	Mention one characteristic of a successful product development team.	01	1	2
	1.4	What is meant by front-end product development?	01	1	1
	1.5	Define product strategy.	01	1	1
	1.6	What is AI-driven product planning?	01	1	2
	1.7	What is the purpose of Quality Function Deployment (QFD)?	01	2	3
	1.8	Define concept generation.	01	1	3
	1.9	What is concept screening?	01	1	3
	1.10	State one benefit of modular product architecture.	01	1	4
	1.11	What is the role of industrial design in product development?	01	2	4
	1.12	Define human-machine interaction.	01	1	4
	1.13	What is anthropometry?	01	1	4
	1.14	What is concurrent engineering?	01	1	4
	1.15	State one objective of Design for Manufacturing (DFM).	01	1	4
	1.16	Define prototyping.	01	1	3
	1.17	What is virtual prototyping?	01	1	4
	1.18	Mention one AI tool used in product ideation.	01	1	2
	1.19	What is generative AI in design?	01	1	2
	1.20	State one advantage of rapid prototyping.	01	1	3

PART-B

2	a	Differentiate between design by evolution and design by innovation, illustrating each with appropriate examples.	08	4	1
	b	Illustrate the design thinking approach and examine its significance in contemporary product development practices.	08	4	1
3	a	Outline the product planning process and analyze the various methods used to identify and evaluate product opportunities.	08	4	2
	b	Examine the application of design thinking in the AI context and assess the role of generative AI in front-end product development.	08	5	2
		OR			
4	a	Outline the steps involved in identifying customer needs and demonstrate how Quality Function Deployment (QFD) translates these needs into design requirements.	08	3	2

	b	Explain the concept of product specifications, highlighting their importance, and summarize the steps involved in preparing product specifications.	08	2	2
5	a	Explain the concept generation process and discuss the role of generative AI-based concept ideation.	08	4	3
	b	Explain the concept selection process, including concept screening and concept scoring, with suitable examples.	08	3	3
		OR			
6	a	Describe the steps involved in concept testing and explain how AI-assisted benchmarking improves concept evaluation.	08	5	3
	b	Explain product architecture, its implications, and the considerations related to variety and supply chain management.	08	4	3
7	a	Explain the aesthetic, ergonomic, and functional roles of industrial design.	08	2	4
	b	Analyze the application of anthropometric data in the design of consumer products.	08	4	4
		OR			
8	a	Explain the industrial design process and discuss the impact of industrial design on product success.	08	4	4
	b	Assess the roles of AI-enhanced industrial design tools in enhancing creativity and improving design quality.	08	5	4
9	a	Discuss the role of rapid prototyping in design validation.	08	4	4
	b	Explain concurrent engineering and discuss its advantages in reducing product development time and cost.	08	5	4
		OR			
10	a	Explain the concept of concurrent engineering with a block diagram.	08	2	4
	b	Compare virtual prototyping and physical prototyping, highlighting the advantages of virtual prototyping in product development.	08	2	3

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular Examinations Feb/Mar-2025

Industrial Engineering and Management

PRODUCT DESIGN AND DEVELOPMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	State any two objectives of the product design phase in product development.	02	2	1
	1.2	Consider laptop/PC as example and identify modular components in it.	02	2	2
	1.3	Identify two important reasons why user experience (UX) plays an important role in product architecture decisions.	02	3	3
	1.4	Provide two examples of interoperability in any product of your choice.	02	2	2
	1.5	From a business perspective of a product design, designing interfaces that support additional functionality without major redesigns ensures _____ and _____.	02	2	1
	1.6	Comment on the importance of human centered design in product development.	02	2	1
	1.7	Identify the most important criteria for the economic success of a product development company.	02	2	1
	1.8	Identify the three aspects which are the primary reason why a prototype is developed.	02	2	2
	1.9	If you want to call yourself as a product designer with a niche for developing sustainable products, identify two qualities that you should possess or harness.	02	1	1
	1.10	Define 'Lead User'. Identify a segment where you can call yourself as the lead user.	02	1	2

PART-B

2	a	Explain any five methods used to 'gather' and 'prioritize' customer needs and requirements.	10	2	1
	b	Discuss any three roles of benchmarking in establishing target specifications.	06	2	1
3	a	Discuss the critical factors influencing product architecture decisions.	10	4	3
	b	Identify the three advantages and disadvantages of modular versus integral architecture in product design.	06	1	2
		OR			
4	a	Explain the types of modularity from product design context.	10	2	3

	b	How does standardization in product architecture benefit manufacturing and assembly?	06	4	4
5	a	A company is developing a new eco-friendly packaging solution. Evaluate how Design for Sustainability (DfS) principles can be integrated into their product development process and potential impact on their brand and market position.	10	4	3
	b	Discuss the role of human factors in designing safe consumer products. How does ergonomics and usability testing contribute to the safety of products?	06	3	4
		OR			
6	a	A global automotive company is developing an electric vehicle. Discuss how they can incorporate Design for Cost (DfC) and Design for Sustainability (DfS) principles to ensure the vehicle is affordable and environmentally friendly.	08	4	3
	b	Explain the significance of Design for X (DFX) in modern product development.	08	3	4
7	a	Discuss the advantages and disadvantages of using the agile product development approach in a fast-paced technology environment. Provide examples to support your arguments.	06	5	4
	b	Compare and contrast agile product development with the stage-gate model.	10	3	3
		OR			
8	a	A software development company is transitioning from a traditional waterfall model to an agile methodology. Describe the steps they should take to ensure a smooth transition and discuss the potential challenges they might face.	12	3	2
	b	Discuss how concurrent engineering principles contribute to faster product development cycles and cost reduction.	04	2	3
9	a	Explain the role of sustainability in Industrial Design.	08	3	4
	b	Describe how Industrial Design can be integrated with user-centered design principles to create a new consumer electronic product. Provide examples of design features that prioritize user experience.	08	3	4
		OR			
10	a	How does industrial design differ from engineering design?	08	2	3
	b	Examine the role of industrial design in the overall product development process.	08	3	4

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular / Supplementary Examinations Feb/Mar-2024

Industrial Engineering and Management

PRODUCT DESIGN AND DEVELOPMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Comment on the significance of sustainability considerations in contemporary product.	02
	1.2	List any two types of modularity with regard to product architecture.	02
	1.3	What is the goal of concept screening process?	02
	1.4	Engineers working as freelancers are on the rise. If you come across any freelancer working on a new product development, what would be your recommendation to him/her with regard to addressing customer needs? (any 4).	02
	1.5	How important are customer feedbacks in improving product design?	02
	1.6	Comment on the role of market research in shaping successful product design.	02
	1.7	Identify the 3 *F*'s of prototype development.	02
	1.8	Quote two examples of how user centered design is used in any of the gadgets that you use on a daily basis.	02
	1.9	Comment on any two important tasks of manufacturing division during planning stage.	02
	1.10	Identify the attributes marketing division is responsible for concept development stage.	02

PART-B

2	a	Explain the characteristics of Successful Product Development.	10
	b	Discuss the six steps of opportunity identification process.	06
3	a	Discuss the process of establishing the target specifications.	08
	b	Explain the various types of specification used in product design.	08
		OR	
4	a	Illustrate the steps involved in Five step concept generation method. Also provide a brief review of each step.	06
	b	Discuss the five step process for refining the specifications.	10
5	a	Explain any 5 types of survey format which you can use to effectively understand about customer requirement as well as create awareness about a product in the minds of customer.	10

	b	Describe six different ways in which your product ideas can be communicated to various stakeholders.	06
		OR	
6	a	Discuss the four-step method for establishing the product architecture.	08
	b	Explain any four motives for product change.	08
7	a	Discuss in detail the first three phases of Industrial Design Process.	06
	b	Explain five critical goals that industrial designers can help a team to achieve when developing new products.	10
8	a	Explain the role of "Industrial Designer" in enhancing the quality of the product produced.	08
	b	Explain how you would use the understanding of Ergonomics in designing a car seat.	08

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VII Semester B. E. Examinations Feb/Mar 2023

Industrial Engineering and Management

PRODUCT DESIGN AND DEVELOPMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	List any two types of product development projects.	02
	1.2	Mention any two characteristics of successful product development.	02
	1.3	List any two methods used to gather data from the customers.	02
	1.4	Mention any two guidelines used for improving both individual and group internal search during concept generation.	02
	1.5	Differentiate between external decision and product champion methods used for choosing a concept.	02
	1.6	Differentiate between functional elements and physical elements of product.	02
	1.7	List any two aesthetic needs of an industrial design.	02
	1.8	What are the different costs of goods used in design for manufacturing stage?	02
	1.9	Mention different types of anthropometric data used in ergonomics.	02
	1.10	How cognitive ergonomics and product design are related?	02

PART-B

2	a	Identify the various criteria for evaluating fundamentally new product opportunities.	08
	b	Consider any product of your choice and develop a mission statement to be followed by product development team.	08

3	a	"Something that cannot be measured cannot be improved". Further, knowing what metric needs to be measured goes a long way in success of product benchmarking. Choose any digital interactive product (mention clearly) of your choice and identify the metrics for the following aspects.	

Sl. No	Product aspect	Description	Metrics to be identified
1	Happiness	Measures of user attitudes or perceptions	
2	Engagement	Level of user involvement	
3	Adoption	Initial uptake of a product, service or feature	
4	Retention	How existing users return and remain active in the product	
5	Task effectiveness and efficiency	Efficiency, effectiveness and errors	

			08
	b	What are some of the biggest challenges you face as a product designer?	08
		OR	
4	a	Describe a recent product design project you were particularly challenged by and how you approached the problem?	08
	b	What is your approach to making websites and platforms accessible to all user groups, including users with visual, hearing and motor disabilities?	08
5	a	When a client says "I don't like this design", what do you do? Justify your answer with suitable example.	08
	b	List and explain various methods used for choosing a product concept.	08
		OR	
6	a	With suitable example, explain how you test your product designs, what methodologies you use and how you evaluate product usability?	08
	b	List and explain different motives organizations use for product change with some examples.	08
7	a	Choose a set of competing products. Assess each one in terms of the five ID quality categories defined in the guidelines. Which product would you purchase? Would you be willing to pay more for it than for others?	08
	b	With the help of suitable block diagram, explain different elements of the cost of goods.	08
8	a	Take any product of your choice and share your experience dealing with HCD (human-centered design).	08
	b	Differentiate between UX design and product design with a case example.	08

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Jan/Feb-2021

Industrial Engineering and Management

PRODUCT DESIGN AND DEVELOPMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Block diagrams should be drawn wherever necessary to substantiate your answer.

PART-A

1	1.1	Identify the most important criteria for the economic success of a product development company.	02
	1.2	Identify the three aspects which are the primary reason why a prototype is developed.	02
	1.3	If you want to call yourself as a product designer with a niche for developing sustainable products; Identify two qualities that you should possess or harness.	02
	1.4	Examine the relevance of Moore's Law to present day product development firms.	02
	1.5	Comment on any two important tasks of manufacturing division during planning stages.	02
	1.6	Identify the attributes a marketing division is responsible for during concept development stage.	02
	1.7	Distinguish between product and project mindset based on any 2 of the following focus areas: i) Life Cycle. ii) Optimization. iii) Work Structure.	02
	1.8	Define "Lead User". Identify a segment where you can all yourself as the lead user.	02
	1.9	The Legal division investigates patent issues during the _____ phase.	01
	1.10	The Sales division develops sales plans during the _____ phase	01
	1.11	Product development is an interdisciplinary activity. Identify the three central functions or departments.	01
	1.12	Product quality is ultimately reflected in the _____ and the price that customers are willing to pay	01

PART-B

2 a Explain the characteristics of Successful Product Development. 10

b Explain any two competitive strategies used by firms across the globe and also cite a suitable example of a firm that uses such a strategy. 06

3 a Fill in the appropriate details for the development of Mission statement for the product development project considering any handled device of your choice and furnish the information for any 5 aspects.

i)	Product description	iv)	Primary market
ii)	Benefit proposition	v)	Assumptions
iii)	Key business goals	vi)	Stakeholder

10

b "Something that cannot be measured cannot be improved". Further, knowing what metric needs to be measured goes a long way is success of Product Benchmarking. Choose any digital interactive product (Mention clearly) of your choice and identify the metrics for the following aspects.

Sl. No.	Product Aspect	Description	Metric to be identified
1.	Happiness	Measures of user attitudes or perceptions	
2	Engagement	Level of user involvement	
3	Adoption	Initial uptake of a product, service, or feature	
4	Retention	How existing users return, and remain active in the product	
5	Task effectiveness and efficiency		

06

OR

4 a Discuss the 5 step process for refining the specifications. 10

b Illustrate the steps involved in Five step concept generation method. Also provide a brief review of each step. 06

5 a There is a huge debate currently happening with regard to how the corona vaccine such as covaxin, Pfizer, etc. will be distributed. As a product designer it must force you to think about one time use syringes which are going to pile huge amounts of waste. So, you would consider designing a syringe which utilizes minimum amount of material to be built. The development effort certainly will bring about multiple competing concepts. Identify and explain the criterion on which the concept selection will be based on. Also, discuss any three challenges in selecting one syringe concept from many competing concepts. 10

	b	Generate 2 product concepts using preliminary sketches to segregate solid and wet waste. Showcase the concept in concept scoring matrix along with evaluation parameters. (write up not required) Use bullet point or tabular column to showcase advantages of each of the concepts.	06
		OR	
6	a	Scenario: Assume a graduate student worked in an automobile showroom associated with a reputed car manufacturer during the break after 3rd Year of Engineering for two months. During his tenure, he got to hear numerous customers express their opinion regarding their satisfaction and their statements are as mentioned below. Statement 1: Happy with the performance of the car but spare part costs are on the higher side. Statement 2: Initial cost of the car is on the higher side and the drive quality is superior making the pinch of high initial cost lessen. Statement 3: As a customer, if you complain about any malfunction about the car, then the entire sub unit is replaced though it is a minor malfunction adding to higher service cost. Statement 4: Sometimes the spares are not available and need to be imported from other places which increases idle time spent by car at service centre. Now, considering the above scenarios and the statements, along with your knowledge of product architecture. i) Showcase your approach regarding how you want to simplify these scenarios and statements based on positive and negative orientation. ii) Define them appropriately in to problem statement or mission statements. iii) Discuss how some of the complaints from the complaint log can be addressed. iv) Provide insights on implications it will have on the overall design based on modular and integral product architecture.	08
	b	Discuss the four-step method for establishing the product architecture.	08
7	a	List and explain five critical goals that industrial designers can help a team to achieve when developing new products.	10
	b	The shown product in Fig. 7.b was used to operate lift during the initial days of pandemic scare. The advantages being, each person had one and it avoided coming in contact the surface someone might have touched. It was sold at 50Rs/piece near traffic signals. Verify whether the product is worthy of manufacturing even more economically by approximating the material requirement. Product will be made of ABS which has a density of 1.05g/cm3 which costs roughly 130Rs/kg. Thickness of 50mm. All other dimensions shown are in mm.	

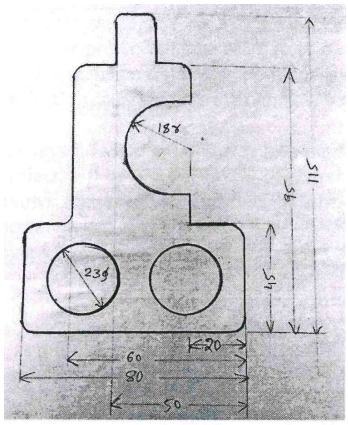
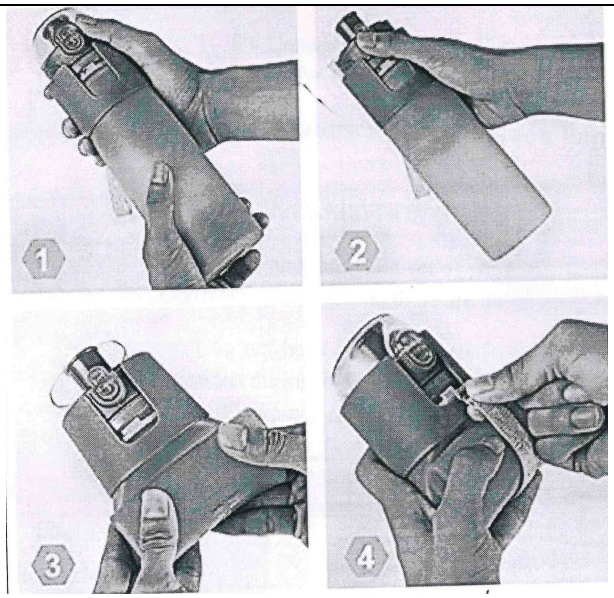


Fig. 7.b 06

8 a



Product name: Collapsible water bottle.

Assume you have conceptualized a product and depicted as shown above using colour rendering.

i) Write any three probable advantages this product will provide to its users?

ii) Sketch the plan to build a prototype "economically" to test the product in the market. 08

b A project manager is working on a single large project for a year. She received a call from her project's sponsor. The two of them have a good business relationship. The project manager is exactly 2 months away from the project's hard deadline and her project's sponsor wants the deadline bumped up by 3 weeks.

He says it is due to an unforeseen market change and it is critical – so much so that the project will be worthless if it is not received by the new deadline. The project manager knows that meeting this new deadline will be particularly challenging due to some personal vacations and training plans that her project team members already have planned during this timeframe. What does the project manager do in this situation? 08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

PRODUCT DESIGN AND DEVELOPMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Mention the three core areas of specialization which are the main contributions to the product design and development.	01
	1.2	Motors, switches, batteries and containers can be grouped under which variant of the generic product development process.	01
	1.3	Expand the acronym *VRIN* as used in the process of opportunity identification.	01
	1.4	Which is the technique used for coordinating technology development with product planning?	01
	1.5	Give any example to emphasize on the importance of timing of product introduction towards the success of the product design and development projects.	01
	1.6	Mention any one of the advantages of the Focus group method of gathering raw data from customers over the interview method.	01
	1.7	What is the basis for fixing the ideal value for target specifications?	01
	1.8	As per Ulrich's product design and development frame work, how many steps are recommended for the concept generation method?	01
	1.9	How many different ways can be there for fixing the value of the metrics for any of the item in the product specifications?	01
	1.10	Mention any one utility of searching patents for the process of concept generation for a new product development or improvement over the existing product.	01
	1.11	In order to make a comparison and rate each concept with the reference concept to arrive at the relative score of concepts, the three point scale used consists of what symbols?	01
	1.12	Mention any one of the advantages of using the face to face interaction as a survey format during concept testing.	01
	1.13	Write down the formula for computing the quantity of product expected to be sold during a time period as part of demand forecasting.	01
	1.14	Illustratively sketch the configuration of the Bus-Modular architecture.	01

	1.15	Mention any one example to demonstrate the impact of choosing between the Modular architecture and the integral architecture keeping product performance as the criteria.	01
	1.16	Define the aspects which are part of appearance during the industrial design of products.	01
	1.17	Justify why a cross functional team is required for analyzing products from the design for manufacturing perspective.	01
	1.18	3D CAD models rendering, finite element analysis and dynamic simulation are all examples of which type of prototypes?	01
	1.19	Mention the two types of analysis used for assessment of the product development economics.	01
	1.20	What is the utility of grant charts for the purpose of managing projects, related to product design and development?	01

PART-B

2	a	Take an example of any product development process and explain the importance of following as the reason for adapting a well-defined development process: i) Quality assurance ii) Planning iii) Management iv) Improvement.	08
	b	List out few of the questions that are part of the Real-Win-Worth(RWW) framework by considering the swing ball car toy example.	04
	c	Give any example for each of the following types of product development projects with respect to the broad area of business category of products (For eg. Machine tool) i) New product development ii) Desirables of existing product platforms iii) Incremental improvements in existing products iv) Fundamentally new products.	04
3	a	Take an example of any product or product variant and explain how you will use the following hints for effective interaction with customers: i) Go with the flow ii) Use visual stimuli and props iii) Have the customer demonstrate the product and /or typical tasks related to the product iv) Be alert for surprises and the expression of latent needs v) Watch for nonverbal information.	08
	b	Outline the processes involved in setting ideal and marginally acceptable target values for defining product specifications during a new product development project with the help of an example.	04
	c	Explain with proper reasoning the usefulness of the following guidelines for improving internal search both at the individual level and at the group level. i) Suspend Judgment ii) Generate a lot of ideas iii) Welcome ideas that may seem infeasible iv) Use graphical and physical media	04

		OR	
4	a	Take the case example of a hand nailer or any other suitable example and outline the process of representing the ideas generated during concept generation phase or a concept classification tree. Bring out clearly the advantages of using a concept classification tree for representing ideas generated systematically.	08
	b	Give the structure of a typical selection matrix used for concept screening by taking any example of your own choice.	04
	c	Mention one advantage for each of the following optional methods of communicating the concept. i) Verbal description ii) Sketch iii) Photos and renderings iv) Story board.	04
5	a	"Concept selection is an integral part of the product development process". Elucidate this statement by taking the example of the concepts related to outpatient syringe or any other example of your choice and detail the entire process of concept screening and concept scanning with the help of the concept selection methodologies.	08
	b	What is product architecture? Give ant two or three reasons to emphasize on the importance of product architecture decisions in the development projects.	04
	c	Discuss the impact of product architecture choices on the product variety and product performance by considering the implications of the architecture on these two aspects.	04
		OR	
6	a	With the help of proper sketches, explain the following types of architecture. Give example for each: i) Integral architecture ii) Modular architecture iii) Slot modular architecture iv) Bus modular architecture v) Section modular architecture.	08
	b	Take the story board approach and explain how you would use it to communicate the concepts to customers to test its acceptance of a new type of toothbrush.	04
	c	Why is that in the concept screening phase a coarse scale is used for rating of the concepts while designing concept screening phase a fine rating scale is preferred to be used? Give an example to clarify.	04
7	a	Using the following five critical goals that industrial designers have to meet, explain the importance of industrial design services for the example of a piece of drawing room furniture or using the example of an office table for work from home scenario: i) Utility ii) Appearance iii) Ease of maintenance iv) Low cost v) Communication.	08

	b	"Design for manufacturing (DFM) is aimed at reducing manufacturing costs while simultaneously improving product quality, development time and development cost" Explain this statement by taking any two or three examples to bring out the advantages of applying DFM principles at the stage of product design and development itself. Sketches wherever necessary should be provided to clarify the ideas.	08
8	a	Explain the following principles of prototyping by giving proper and appropriate examples: i) Analytical prototypes are generally more flexible than physical prototypes ii) Physical prototypes are required to detect unanticipated phenomena iii) A prototype may reduce the risk of costly iterations. iv) A prototype may expedite other development steps.	08
	b	Outline the process of building a Base Case financial model for the economic analysis process of a development project.	04
	c	Give examples for the following types of tasks during the product design and development projects: i) Sequential tasks ii) Parallel tasks iii) Coupled tasks.	04

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

PRODUCT DESIGN AND DEVELOPMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	List any four dimensions commonly used to assess the performance of a product development effort.	02
	1.2	Mention the two variants of matrix organization, which is considered as an appropriate structure for organizations involved in product development initiatives.	02
	1.3	In the context of gathering raw data from the customers, for identification of customer needs, explain the terms 'Lead users' and 'Extreme users'.	02
	1.4	List any four benefits of concept classification tree.	02
	1.5	Concept selection methodology is a two stage process. What are the two stages? Write a line about each.	02
	1.6	A product can be thought of in both functional and physical terms. Distinguish between functional elements and physical elements of a product.	02
	1.7	What is meant by Delayed differention? List any two benefits.	02
	1.8	Distinguish between two types of overhead costs, which are critical components of unit manufacturing cost.	02
	1.9	What is meant by 'Black Box' component procurement strategy in manufacturing industry?	02
	1.10	Distinguish between 'physical Prototype' and 'comprehensive prototype'	02

PART-B

2	a	A leading manufacturer wanted to establish a common product development process structure that would be appropriate for all of the many different operating divisions across the company. Discuss in detail the phases involved in the generic development process.	12
	b	Product plan divides product development projects into four categories. List them and write a brief note on each.	04

3	a	Identifying the customer needs is crucial for the success of any product in the market. This process involves documenting interactions with customers. Four methods are commonly used. What are they? Provide a note about each method.	08
	b	Customer needs statements are the result of interpreting the raw data gathered from the customers. Review the guidelines for writing the need statements.	08
		OR	
4	a	The target specifications are established after the customer needs have been identified. Analyze the four steps involved in the process of establishing the target specifications.	08
	b	Illustrate the steps involved in *FIVE STEP* concept generation method. Also provide a brief review of each step.	08
5	a	Discuss the potential benefits offered by a structured concept selection method. (Any four benefits)	08
	b	When is 'concept scoring' used in Product Development process? Provide a brief overview of the steps involved in concept scoring process.	08
		OR	
6	a	Modular Architectures comprise three types. What are they? Through suitable illustrations, bring out the conceptual differences among these types of architectures.	08
	b	Decisions about dividing the products into chunks and the extent of modularity to be imposed on the architecture are linked to several issues of importance to enterprise. List all the issues. Also review any four issues in detail.	08
7	a	Tabulate the responsibilities of Industrial Design during each phase of the product development process and also show how they relate to the other activities of the development team.	08
	b	Exhibit the categorization of the elements of manufacturing cost. Also provide a brief note on the broad three categories of costs.	08
8	a	Review four basic purposes, for using prototypes in product development projects.	08
	b	Write brief note on the following technologies that have emerged important in the past two decades: i) $3D\ CAD$ modeling and Analysis ii) Free Form Fabrication.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

PRODUCT DESIGN AND DEVELOPMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical table is permitted.

PART-A

		Identify the process, procedure, technique or the step in the method based on the following description pertaining to the system of product design and development.	
1	1.1	The capability of the firm to develop future products as a result of their experience with a product development product. This capability is an asset to the firm.	01
	1.2	The phase in the product development that includes the definition of the product architecture decomposition of the product into subsystems and components and preliminary design of key components.	01
	1.3	The category of opportunities that push out into less known territory in one or both the dimensions of the technology. (Designed as horizons).	01
	1.4	One specific approach used within established companies is the method developed originally by $3M$, to select exceptional opportunities.	01
	1.5	The type of specifications that represent the hopes and aspirations of the team but they are established before the team knows the constraints the product technology will place on what can be achieved.	01
	1.6	The technique of representing the concept of generation of space that is used to divide the entire space of possible solutions into several classes that will facilitate comparison and pruning.	01
	1.7	The method used for concept screening developed in the 1980$'s$. The purposes of this technique are to narrow the number of concepts quickly and to improve the concepts.	01
	1.8	The type of model that are known as “lookslike” models, vividly display the form and appearance of a product.	01
	1.9	The process of assignment of the functional elements of a product to the physical building blocks of the product, to define an approach to product design and development.	01

	1.10	A diagram representing the development teams understanding of the constituent elements of the product. This is the first step in the establishment of the product architecture.	01
	1.11	The professional service of creating and developing concepts and specifications that optimizes the function, value and appearance of product and systems for mutual benefit of both user and manufacturer.	01
	1.12	The type of product appeal from consumer's perspectives, which is achieved in part through appearance, feel, sound and smell. The rating on this appeal is used to assess the quality of industrial design.	01
	1.13	The process of design that expands the traditional manufacturer's focus to consider the full product life cycle and its relationship to the environment.	01
	1.14	The design approach that begins with the concept development phase and system-level design phase; in these phases important decisions must be made with manufacturing cost implications in mind.	01
	1.15	The approximation of the product along one or more dimensions. The definition of any entity exhibiting at least one aspect of the product that is of interest to the development team.	01
	1.16	Examples of these types of prototype include computer simulations, systems of equations encoded within a spreadsheet and computer models of three-dimensional geometry.	01
	1.17	The technology that creates physical objects directly from $3D\ CAD$ models and can be thought of as "three dimensional printing". This ensures quick creation of prototypes.	01
	1.18	The types of intellectual property which is a temporary monopoly granted by a government to an inventor to exclude others from using an invention.	01
	1.19	A measure of the degree to which inflows are greater than outflows in product development project which represent the value in today's monetary units (say dollars) of all the expected future cashflows.	01
	1.20	The traditional tool for representing the terms involved in the design and development projects in the form of a chart. The chart contains a horizontal time line created by drawing a horizontal bar representing the start and end of each task.	01

PART-B

2	a	Discuss some of the characteristics that make product development challenging. Also discuss the several intrinsic attributes of the product development that contribute to its appeal as an serious area of work	08
	b	Give any five examples for the following type of product development process. Give the description of these types of product development. i) Technology-push products. ii) Process-intensive products. iii) High-risk products. iv) Quick build products.	08
		OR	

3	a	"A balanced portfolio product development may include investments in breakthrough products, new platforms, derivatives and current product support" Elaborate on the need for any organization to have a balanced portfolio.	08
	b	Present in a standard format the Mission statement for the development of a multifunctional office document machine. What could be some of the assumptions and constraints in the development of a multifunctional printer? List them.	08
4	a	Translate the following customer statements about a student book bag into proper needs statements: i) "See how the leather on the bottom of the bag is all scratched its ugly". ii) "When i am standing in line at the cashier trying to find my chequebook while balancing my bag on my knee. I feel like a stork" iii) "The bag is my life; If I lose it i'm in big trouble" iv) There's nothing worse than a banana that's squished by the edge of a textbook. v) I have never used both straps on my knapsack; I just sling it over one shoulder."	08
	b	Consider development of a "coffee maker" or a "brewer for ice tea" or "popcorn making machine". How would you go about setting target specifications and seeking the final specifications after concept selection and concept testing?	08
		OR	
5	a	Develop[the functional decomposition diagram for the generation of concepts related to the building of a product such as a "smart phone hand set" or a "vegetable cutter" or a "washing machine". Identify the primary and secondary functions.	08
	b	Outline how the process of explaining concepts systematically can be facilitated through the concept classification tree or concept combination table. Take the example of a portable hand tool for pumping nails into different surfaces as a case study and elaborate.	08
6	a	Prepare the selection matrix for shortlisting few exceptional concepts among the different mechanisms developed for a pencil. Sketch atleast five different concepts that are available for the mechanisms for a pencil and carry out the concept screening through the help of selection matrix.	08
	b	"Concept screening uses weighted selection criteria and a finer rating scales" Justify. Present in a tabular format the concept scoring matrix for shortlisting concepts for the development of a disposable syringe used for administrating small dosages of medicine to patients in the OPD section of a hospital.	08
		OR	
7	a	Explain the procedure for the concept testing focusing on the need to assess the scales potential of the produced concept for a bicycle or any other personal transport vehicle? How would you use the outcome of the consumer survey assessing the sales potential?	08

	b	Outline the procedure for establishing the product architecture for a functional product such as development of a $3D$ printer or a rapid prototyping machine. Explain the process of finalizing the product architecture.	08
8	a	Distinguish between technology driven products and user driven products. How is this discretion useful deciding the importance of industrial design effort in the development of such products?	08
	b	"For products that are characterized by a high degree of user interaction and the need for aesthetic appeal, Industrial design should be involved throughout the product development process". Take the case of designing the exterior of a luxury segment car and elaborate on the role of industrial design in the product development process.	08
		OR	
9	a	Sketch a simple input-output model of a manufacturing system and also indicate the different components of the manufacturing cost of a typical product.	08
	b	"During the 1980s design for manufacturing practices were put in place in thousands of firms. Today DFM is an essential part of almost every product development effort". Elaborate by taking the case of a developing a plastic part processed in the injection molding machine or alternatively developing a casting for engine block assembly of a automotive vehicle.	08
10	a	The four main reasons for the development of a prototype are learning, communication, integration and serving of milestones. Discuss the appropriateness of different types of prototypes for different purposes.	08
	b	Discuss the relevance of the following principles useful in guiding prototypes during product development. i) Analytical prototypes are generally more flexible than physical prototypes. ii) Physical prototypes are required to detect unanticipated phenomena. iii) A prototype may reduce the risks of costly iterations.	08
		OR	
11	a	How do you approach the process of building a Base-Case financial model through estimating the timing and magnitude of future cash inflows and outflows? Take a case example and explain.	08
	b	Take the case of developing a solution for home automation of project planning. Highlight the following i) Preparing a task list. ii) Developing a project schedule. iii) Identifying staffing requirements. iv) Propering a project budget. v) Developing a risk plan.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

PRODUCT DESIGN AND DEVELOPMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical table is permitted.

PART-A

1	1.1	Define product design.	02
	1.2	The first three phases in Morphology of design belong to _____.	01
	1.3	_____ strategy aims at giving only high quality, whatever may be the cost.	01
	1.4	Define Rapid prototyping.	02
	1.5	In the operational aspect of the product characteristic, _____ consideration is most important.	01
	1.6	Define specialization.	02
	1.7	_____ is the catalyst used for speeding up chemical reactions in polymerization.	01
	1.8	Define Vulcanization.	02
	1.9	In _____ strategy, the total production costs remain unchanged but try to improve the ratio of value to price.	01
	1.10	Define concurrent design.	02
	1.11	Stereo-lithography is one of the technologies used in _____.	01
	1.12	Define project planning.	02
	1.13	Establishing the economic existence comes under _____ study of primary design phase.	01
	1.14	In _____ process of Rapid prototyping, a thin layer of powder is applied at the workplace with a roller and a laser is used to sinter the layers of powder.	01

PART-B

2	a	Explain in brief the different factors that are essential for product design.	10
	b	Explain the planning and distribution phase of the Morphology of Design in detail.	06
		OR	

3	a	Briefly explain the different strategies adopted for product design.	08
	b	Explain the different product characteristics to be considered for analysis of the product.	08
4	a	Discuss Quality Function Deployment strategy in detail with an example.	11
	b	Explain in brief the different basic considerations for industrial design.	05
		OR	
5	a	Explain the various types of specifications of a product.	08
	b	Explain the activity of concept generation in detail.	08
6	a	Explain in detail the process of concept screening with an example.	10
	b	Discuss about the different survey formats used in concept testing.	06
		OR	
7	a	Explain the product architecture implications in detail.	08
	b	Explain the supply chain aspect in the product architecture.	08
8	a	Explain in brief the different problems faced by the industrial designer.	06
	b	Discuss about the various types of models designed by industrial designers.	10
		OR	
9	a	Give some examples of guidelines for design for manufacturing of Injection molded products.	08
	b	Explain the properties of Ceramics and Glass in detail.	08
10	a	Explain the different strategies that are adopted by the organization to increase its profit.	06
	b	Explain the process of Stereolithography with a neat sketch.	10
		OR	
11	a	Define project planning. Explain the different phases of project planning in detail.	10
	b	Brief about the project management environment.	06

Total Quality Management

VII Semester SEE Question Papers

Course codes: IM373TA

- Feb 2026 (IM373TA)

Question Paper ID: 379356

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM373TA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Total Quality Management

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Define the term 'Total Quality Management' **(**TQM**).**	02	1	1
1.2	What is the purpose of a Vision Statement in an organization?	02	1	1
1.3	What is the '8D' approach used for in TQM?	02	2	1
1.4	Differentiate between Reactive Improvement and Proactive Improvement.	02	2	2
1.5	Applying Deming's point on "Eliminate management by objective" (MBO), what practical step should a manager take in a production setting?	02	3	3
1.6	A manager wants to use Hoshin Management to implement a new TQM strategy. What is the first *phase* the manager must execute?	02	3	3
1.7	Differentiate between Internal Benchmarking and Competitive Benchmarking.	02	3	2
1.8	A hospital wants to apply TPM to its medical equipment. Illustrate one way this could be done.	02	4	3
1.9	A company's service center has been receiving complaints about the long hold times on the phone. Which dimension of service quality (SERVQUAL) is most affected, and how can they apply a TQM principle to fix it?	02	4	3
1.10	What is the purpose of the 'Control' phase in the Six Sigma DMAIC cycle?	02	2	1

Part B

Question No	Question	M	CO	BT
2a	What are the stages in the evolution of quality methods? Explain with examples.	08	2	3
2b	Compare Quality of Conformance and Quality of Design.	08	2	3
3a	Explain the the importance of the concept of "Problem Identification" in continuous improvement.	08	2	2
3b	Outline the 7 QC Steps for reactive problem-solving.	08	2	2
	OR			
4a	What are the steps in proactive improvement according to Shoji Shiba? List and elucidate.	08	2	2
4b	Discuss the role of semantics in proactive improvement.	08	2	2
5a	What are the main phases of Hoshin Management? List and explain.	08	3	2
5b	Explain the concept of "Quality Circles" and their benefits.	08	3	2
	OR			
6a	Describe a model of Total Quality Management for skill development.	08	3	2
6b	Explain how Total Quality Management acts as a learning system in an organization.	08	3	2
7a	Discuss the relationship between Six Sigma and company culture in the context of Service Organizations.	08	4	3
7b	Explain how Six Sigma supports small business competitiveness in India.	08	4	3
	OR			
8a	Present a case example of successful Six Sigma implementation.	08	4	3
8b	What happens during the "Analyze Phase" of Six Sigma methodology implementation? Provide insights from an implementation perspective	08	4	3
9a	How does Design for Six Sigma complement traditional product development? Summarize.	08	4	4
9b	What is the role of Quality Function Deployment (QFD) in Design for Six Sigma? Develop the skeleton of the QFD for an example of your choice.	08	4	4
	OR			
10a	What are the main differences between DMAIC and DMADV methodologies? Elaborate.	08	2	2
10b	Explain the "Define" phase in the DMADV methodology.	08	2	2

Artificial Intelligence in Industrial and Systems Engineering

VII Semester SEE Question Papers

Course codes: IM374TFB

- Feb 2026 (IM374TFB)

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Question Paper ID: 764719

Course Code: IM374TFB

USN | | | | | | | | | | |

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Artificial Intelligence in Industrial and Systems Engineering

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Define Artificial Intelligence.	02	2	2
1.2	What is the role of a critic in a learning agent?	02	1	1
1.3	Mention two applications of AI in logistics.	02	1	1
1.4	Define production scheduling.	02	3	2
1.5	What is the purpose of root cause analysis?	02	3	3
1.6	Name two types of sensors used in predictive maintenance.	02	3	3
1.7	What is pose estimation in human motion analysis?	02	2	2
1.8	Name two industries where AI-based fatigue monitoring is used.	02	2	2
1.9	What is AI for sustainability in industrial operations?	02	2	2
1.10	What role does AI play in the circular economy?	02	2	2

Part B

Question No	Question	M	CO	BT
2a	Explain the evolution of Artificial Intelligence from early symbolic systems to modern data-driven learning.	08	2	2
2b	Explain inductive learning with a suitable industrial engineering example.	08	2	2
3a	Discuss case studies of AI implementation in warehouse slotting, order picking, and space optimization.	08	3	3
3b	Discuss AI-based production scheduling and resource allocation techniques with examples.	08	3	2
	OR			
4a	Compare traditional optimization techniques with AI-powered optimization in operations management.	08	3	4
4b	Describe AI applications in end-to-end supply chain visibility and control.	08	2	3
5a	Discuss predictive maintenance techniques using machine learning and their benefits.	08	3	3
5b	Explain end-to-end predictive maintenance using ML: data, models, diagnostics, forecasting, and decision-making.	08	4	3
	OR			
6a	Discuss equipment failure prediction using sensor data analytics.	08	3	3
6b	Describe AI-based surface defect detection techniques in manufacturing.	08	3	3
7a	Evaluate case studies where AI enhanced ergonomic performance in manufacturing workplaces.	08	3	4
7b	What are the ergonomic challenges solved by human–robot collaboration systems?	08	4	2
	OR			
8a	Elaborate the workflow of an AI-based human motion analysis system.	08	4	2

8b	Explain how AI-powered human motion analysis improves ergonomic evaluation.	08	3	3
9a	Explain challenges and barriers to AI adoption in industries with examples.	08	3	2
9b	Explain safety challenges associated with AI deployment in industrial systems.	08	4	4
	OR			
10a	Analyse the AI applications for sustainability and circular economy in industrial operations.	08	4	4
10b	Explain future research directions of AI in Industrial Systems Engineering.	08	3	3

Global Supply Chain Management

VII Semester SEE Question Papers

Course codes: IM375TGN

- Feb 2026 (IM375TGN)

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Question Paper ID: 590515

Course Code: IM375TGN

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

Global Supply chain management (Institutional Elective)

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of Normal distribution table is permitted.

Part A

Question No	Question	M	CO	BT
1.1	Why are supply chain decisions important for an organization?	02	2	2
1.2	The most limited scope over which the strategic fit is considered is one operation within a functional area within a company. This is referred to as _____ scope	01	1	2
1.3	Movement of goods across national borders is managed under _____ logistics.	01	1	2
1.4	Two major factors that influence distribution network design are _____and _____.	02	2	2
1.5	The main role of IT in transportation is to improve _____ of goods movement.	01	2	2
1.6	Among the modes of transportation, the _____ mode is most suitable for high-value and urgent shipments.	01	2	2
1.7	Safety inventory is held to protect against _____ in demand and supply.	01	3	2
1.8	Real-time information sharing enabled by IT helps reduce the _____ effect in inventory management	01	3	2
1.9	The portion of demand satisfied from on-hand inventory is known as _____.	01	3	2
1.10	Higher service levels require _____ safety stock.	01	3	2
1.11	State the effect of seasonal demand on pricing decisions in supply chains.	02	3	2

1.12	Explain the role of pricing in coordinating supply chain decisions.	02	3	2
1.13	A firm reduces lead time but sees no inventory reduction. Identify the dominant inventory driver.	02	3	2
1.14	Two supply chains use identical IT systems, yet only one exhibits low Bullwhip. Identify the differentiating factors.	02	4	2

Part B

Question No	Question	M	CO	BT
2a	Sketch the framework for structuring the drivers of supply chain performance. Choose an Indian organization and discuss the structuring of its drivers.	08	1	3
2b	Explain the obstacles to achieving strategic fit in modern supply chains.	08	1	2
3a	Explain the role of distribution in supply chains. Analyze the factors influencing distribution network design	08	2	2
3b	Enumerate and explain any four factors influencing distribution network design.	08	2	2
	OR			
4a	Discuss the impact of centralized and decentralized transportation networks on supply chain performance.	08	3	2
4b	Discuss how transportation planning decisions interact with inventory and facility planning decisions in a supply chain.	08	3	2
5a	Discuss the role of IT and modern technologies in inventory management.	08	3	2
5b	Illustrate the relationship between service level and safety inventory.	08	3	2
	OR			
6a	Weekly demand for HP printers at a sam's club store is normally distributed, with a mean of 250 and a standard deviation of 150. The store manager continuously monitors inventory and currently orders. Assume that whenever Sam places an order there is a fixed ordering cost of Rs. 45000/- The cost of a printer for Sam is Rs. 5000/-, and annual inventory holding cost is about 18% of the product cost. HP currently takes two weeks to fill an order. Evaluate the order quantity. Given that Sam would like to ensure 93% service level, how much safety inventory does the store carry? What shall be the re-order point? Assume the value of Z=1.48.	08	3	3
6b	Weekly demand for a product follows a normal distribution with a mean of 400 units and a standard deviation of 120 units. The lead time is 2 weeks, and the firm desires a 95% service level to avoid stockouts during lead time. The corresponding standard normal value (Z) is 1.65. Assuming demand during lead time is independent and normally distributed: i). Determine the mean demand during lead time. ii). Compute the standard deviation of demand during lead time. iii). Calculate the safety inventory (safety stock) required. iv). Briefly explain the role of safety inventory in inventory management.	08	3	3
7a	Analyze the use of auctions in supplier selection and sourcing decisions.	08	3	4
7b	Illustrate the conditions under which in-house production is preferred over outsourcing using suitable examples.	08	3	2

	OR			
8a	Despite its benefits, revenue management is not easy to implement across supply chains. Explain the key operational, behavioral, and ethical challenges involved in implementing pricing and revenue management systems.	08	3	4
8b	In many industries, pricing decisions are closely linked with production, inventory, and distribution planning. Explain how coordination among supply chain partners supports effective pricing and revenue management.	08	3	3
9a	Describe Collaborative Planning, Forecasting, and Replenishment (CPFR). Explain its advantages over traditional planning.	08	4	2
9b	Explain Continuous Replenishment Program (CRP) and Vendor-Managed Inventory (VMI). Discuss their benefits.	08	4	2
	OR			
10a	Discuss the managerial levers available for achieving coordination in a supply chain and evaluate the role of information sharing.	08	4	4
10b	Discuss the contribution of forecasting errors and order batching to the bullwhip effect in digitally enabled supply chains.	08	4	3

Predictive Analytics

VII Semester SEE Question Papers

Course codes: IM374TFA, 16IM7G5

- Feb 2026 (IM374TFA)
- Jan/Feb 2021 (16IM7G5)
- Nov/Dec 2019 (16IM7G5)

Question Paper ID: 305949

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM374TFA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Predictive Analytics

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	What type of regression is used for categorical outcomes?	01	1	1
1.2	What is over fitting in model training?	01	1	1
1.3	Which step typically initiates the data mining process?	01	1	1
1.4	Which metric is commonly used to evaluate classification models?	01	1	1
1.5	An industrial engineer is building a model to predict the *time* until machine failure. What type of machine learning analysis is most appropriate for this continuous outcome variable?	02	2	2
1.6	Describe the key assumption that the Naïve Bayes classification algorithm makes about the relationship between the features in the dataset. How does this assumption simplify the calculation of probabilities?	02	2	2
1.7	A data scientist needs to visualize the stability and tightness of clusters resulting from k-means. They calculate the Silhouette Score for every data point. A point P has a Silhouette Score of -0.5. What is the *most critical* interpretation of this score for point P?	02	1	2
1.8	An association rule is generated: ${{Coffee}} →{Sugar}} with a Lift of 0.8. Given the minimum support and confidence thresholds were met, how should a retailer interpret this specific Lift value for store placement strategy?	02	1	2
1.9	What does TF-IDF stand for?	01	1	1
1.10	Which ensemble method is more prone to over fitting: random forest or gradient boosting?	01	1	1

1.11	Data mining in supply chain helps optimize what?	01	1	1
1.12	Anomaly detection aims to identify what?	01	1	1
1.13	Discussing the responsible use of data mining in industry often involves mitigating algorithmic bias. What is the most common source of bias in a deployed data mining model?	02	4	2
1.14	List three essential Key Performance Indicators (KPIs) that a data visualization dashboard for a manufacturing production line should communicate to management.	02	4	2

Part B

<table>
<tr><th>Question No</th><th>Question</th><th>M</th><th>CO</th><th>BT</th></tr>
<tr><td>2a</td><td>Briefly explain its primary goal in the context of business or scientific research. Discuss the six common types of data mining techniques that are used for data driven decision making in the industries.</td><td>08</td><td>2</td><td>2</td></tr>
<tr><td>2b</td><td>Justify the use of a classification algorithm over a clustering algorithm when an industrial quality control process already has historical data labeled with known defect categories (e.g., 'Scratch', 'Warp', 'Pass').</td><td>08</td><td>4</td><td>2</td></tr>
<tr><td>3a</td><td>Explain accuracy, precision, recall, and why accuracy alone can be misleading.</td><td>08</td><td>2</td><td>2</td></tr>
<tr><td>3b</td><td>Describe ROC and AUC curves and their significance</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4a</td><td>Describe the confusion matrix and interpret each component.</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td>4b</td><td>Explain cross-validation and its importance in model evaluation.</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td>5a</td><td>Transactions: Apriori Algorithm. Generate item sets with frequencies.

<table><tr><td>TID</td><td>Items</td></tr><tr><td>1</td><td>A, B</td></tr><tr><td>2</td><td>B, C</td></tr><tr><td>3</td><td>A, C</td></tr><tr><td>4</td><td>A, B, C</td></tr></table></td><td>08</td><td>3</td><td>4</td></tr>
<tr><td>5b</td><td>Cluster the following 2D points into 2 clusters using k-means:

<table><tr><td>Point</td><td>X</td><td>Y</td></tr><tr><td>A</td><td>1</td><td>2</td></tr><tr><td>B</td><td>2</td><td>1</td></tr><tr><td>C</td><td>5</td><td>4</td></tr><tr><td>D</td><td>6</td><td>5</td></tr></table></td><td>08</td><td>3</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>6a</td><td>K-means centroid update in 1D

Data points: (x={1,2,3,10,11,12})

Initial centroids: (μ_1=2, μ_2=11)

Task: Assign points and compute updated centroids.</td><td>08</td><td>3</td><td>3</td></tr>
</table>

<table>
<tr><td>6b</td><td>Given Transactions:

<table><tr><td>TID</td><td>Items</td></tr><tr><td>1</td><td>A, B</td></tr><tr><td>2</td><td>A</td></tr><tr><td>3</td><td>B</td></tr><tr><td>4</td><td>A, B</td></tr></table>
Compute Support (A), Support(B), Support (AB), Cofidence (A↔B)</td><td>08</td><td>3</td><td>4</td></tr>
<tr><td>7a</td><td>Review: “packing damaged, delivery late and support unresponsive” Lexicon weights: packing damaged = (-2) (bigram), delivery late = (-2) (bigram), nresponsive = (-2), support = (0). Compute total and average sentiment across these scored tokens.</td><td>08</td><td>4</td><td>4</td></tr>
<tr><td>7b</td><td>Throughputs: ([95, 100, 98, 97, 96, 150, 99, 101]). Compute (Q1, Q3), IQR, and flag outliers using (x < Q1 - 1.5*cdot *mathrm{IQR}) or (x > Q3 + 1.5*cdot *mathrm{IQR}).</td><td>08</td><td>4</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>8a</td><td>True cycle times (y = [20, 22, 19, 21]). Current predictions ({y_{hat}}^ {(0)} = [18, 23, 20, 22]). Residual learner outputs per instance (h = [2, -1, -1, -1]). Learning rate (η= 0.2). Compute updated predictions.</td><td>08</td><td>4</td><td>4</td></tr>
<tr><td>8b</td><td>A node has 30 parts: 9 defective, 21 non-defective. A candidate split on “humidity > 60%” yields:Left: 12 samples, 6 defective, 6 non-defective.Right: 18 samples, 3 defective, 15 non-defective.Compute information gain using entropy.</td><td>08</td><td>4</td><td>4</td></tr>
<tr><td>9a</td><td>Manufacturing defect rate monitoring with EWMA

A line’s hourly defect counts have mean baseline (μ_0 = 2 per hour). Use EWMA with (λ= 0.3). Starting ($Z_0 = \mu_0$). Observations: ([3, 2, 5, 4]). Compute EWMA (Z_t) for (t=1 to 4).</td><td>08</td><td>4</td><td>4</td></tr>
<tr><td>9b</td><td>Two plants’ defect rates (%): Plant A ([3, 4, 2]), Plant B ([6, 5, 7]). Compute z-scores for each measurement relative to its plant mean and standard.</td><td>08</td><td>4</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>10a</td><td>You need to communicate the key findings of a machine learning-based equipment failure prediction model to non-technical factory floor managers. Assess the primary design principle you would follow when constructing the dashboard to ensure effective communication.</td><td>06</td><td>4</td><td>3</td></tr>
<tr><td>10b</td><td>A plant has frequent defects after a tooling changeover. Propose a data mining approach to locate hotspots and quantify impact.</td><td>10</td><td>4</td><td>4</td></tr>
</table>

16IM7G5

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

PREDICTIVE ANALYTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Define a variate.	01
	1.2	List atleast two facial characteristic used in chernoff faces to categorize variables.	01
	1.3	Calculate the statistical distance for the point (4,5) from (1,1).	01
	1.4	Define random vector.	01
	1.5	What is the purpose of using coefficient of determination in regression analysis.	01
	1.6	Which matrix is used as input to obtain the principal components of a multivariate data set.	01
	1.7	Suppose the eigen values of 3 variables are 5.1,3.2 and 0.5 compute the variance explained by the first principal component.	01
	1.8	The eigen value Eigen vector pair for the first Principal component is given λ_1 = 5.8 and eigen vector by [0.8,0.51,0] , obtain an equation for the first principal component.	01
	1.9	The loadings of variable 1 on two factors are 3 and 2 and the total variance is 16. Find the specific variance.	01
	1.10	Define communality.	01
	1.11	List atleast two types of factor rotation methods.	01
	1.12	Define factor scores.	01
	1.13	Represent a confusion matrix.	01
	1.14	What is the error rate used in classification problem?	01
	1.15	What is the goal of classification procedure?	01
	1.16	When conducting fisher approach to classify two population what is the assumption about the population data.	01
	1.17	Explain the term in bi in biplots.	01
	1.18	What is corresponding analysis?	01
	1.19	What does a dendogram explain?	01
	1.20	List at least two methods used for hierarchical clustering.	01

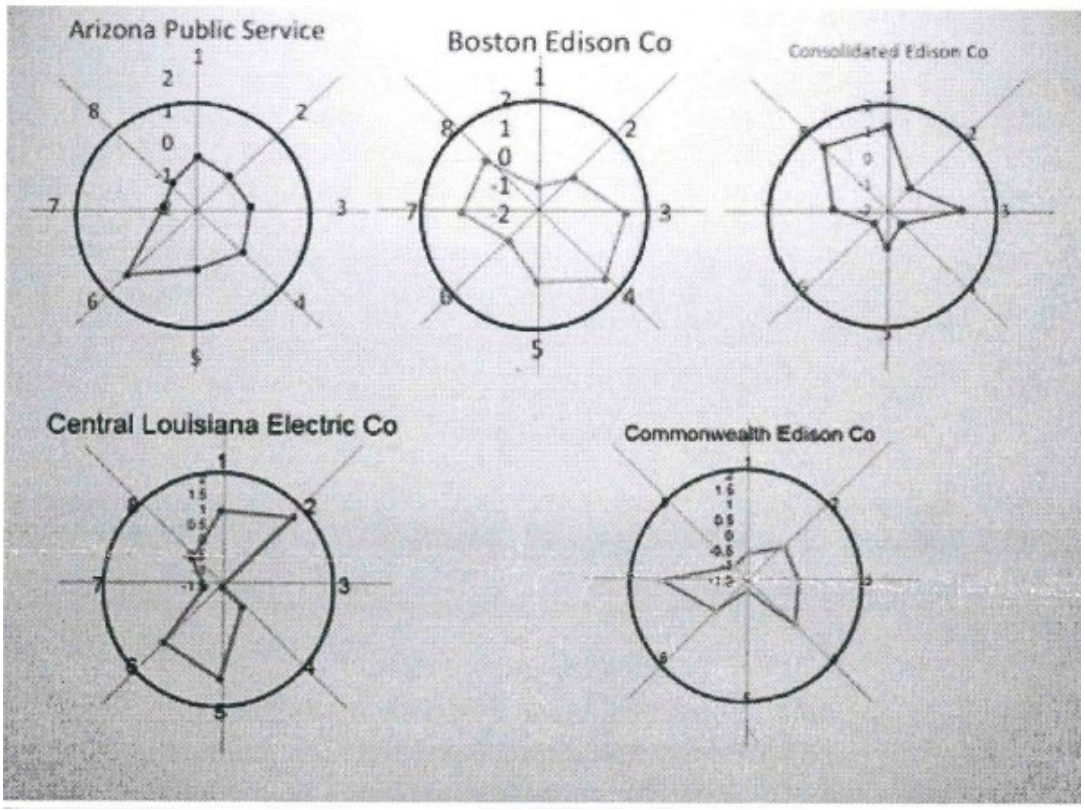
PART-B

2 a Compute the ρ and $V^{\frac{1}{2}}$ from the covariance matrix

$$\Sigma = \begin{matrix} 12 & 17 & 29 \\ 18 & 20 & 38 \\ 14 & 16 & 30 \end{matrix}$$

05

b For a spider (stars) graph given below 8 variables are plotted for the 5 companies using spider graphs, based on the spider (stars) graph which companies can be grouped together for each of the variables **(answer for any three variables, note group in terms of low, medium and high values)**



3

c Consider the following data for 10 largest companies which provide the sales (x1) , profit (x2) and assets (x3) , draw the scatter diagram accordingly (please note different combinations of plots) and comment on the relationship.

	X1	X2	X3
Citigroup	108.28	17.05	1,484.10
GE	152.36	16.59	750.33
American Int	95.04	10.91	766.42
Bank of America	65.45	14.14	1,110.46
HSBC	62.97	9.52	1,031.29
Exxon Mobil	263.99	25.33	195.26
Shell	265.19	18.54	193.83
BP	285.06	15.73	191.11
ING	92.01	8.1	1,175.16
Toyota	165.68	11.13	211.15

08

3	a	A regression model is build using thrust of a jet turbine engine (y) as response variable and six other predictors primary speed of rotation (x1) , secondary speed of rotation (x2) , fuel flow rate (x3), pressure (x4), exhaust temperature (x5) , ambient temperature at time of test (x6). The Minitab output is show below:	

Predictor	Coef	SE Coef	T	P	VIF
Constant	-4738	2445	-1.94	0.061	
x1	1.1185	0.2865	3.90	0.000	289.1
x2	-0.03018	0.03823	-0.79	0.435	71.8
x3	0.2306	0.1180	1.95	0.059	168.1
x4	3.850	2.686	1.43	0.161	220.0
x5	0.8219	0.3507	2.34	0.025	32.4
x6	-16.946	2.620	-6.47	0.000	8.5

S = 26.5088 R-Sq = 99.8% R-Sq(adj) = 99.7%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	6	9908846	1651474	2350.13	0.000
Residual Error	33	23190	703		
Total	39	9932036			

Based on the output

i. Conclude which variables are significant
ii. Provide a multiple linear regression model
iii. What would you comment on the model adequacy
iv. Would you do further modeling and if so which variables would you prefer to do further modeling? 08

b Define the following and explain its uses in factor analysis
i. Factor loading
ii. Communalities 04

c The following mathematical equation is provided for a model with m common factors

$$X = \mu + LF + \in$$

For a p variable system, and each variable represents a vector, describe each of the terms in the equation. 04

OR

4 a For 14 variables principal component analysis were conducted and the following eigen value results were obtained as below:

Eigenvalue	5.6181	3.1111	1.8278	1.6567	1.02	0.8178	0.6257	0.5356
Proportion	0.33	0.183	0.108	0.097	0.06	0.048	0.037	0.032
Cumulative	0.33	0.513	0.621	0.718	0.778	0.827	0.863	0.895

Eigenvalue	0.4518	0.3468	0.2631	0.219	0.1892	0.1282	0.1056	0.0752
Proportion	0.027	0.02	0.015	0.013	0.011	0.008	0.006	0.004
Cumulative	0.921	0.942	0.957	0.97	0.981	0.989	0.995	1

a. Draw a scree plot
b. Based on the scree plot, how much would be the factors which you would use to conduct factor analysis
c. Would you agree based on the scree plot that a cutoff value of 1 would be appropriate to decide upon the number of factors to be extracted? 08

b	Transient points of an electronic inverter are influenced by many factors. The factors which affect transient tpoint(y,in volts) of PMOS-NMOS inverters have been included in regression modeling and include x1:width of the NMOS device, x2:length of the NMOS device , x3:width of the PMOS device , x4:length of the PMOS device, and x5:temperture (°C). The minitab output is shown below: Predictor Coef SE Coef T P VIF Constant 2.855 1.869 1.53 0.143 x1 -0.2905 0.1174 -2.47 0.023 1.2 x2 0.20572 0.07506 2.74 0.013 1.4 x3 0.4544 0.1877 2.42 0.026 1.3 x4 -0.5942 0.2125 -2.80 0.012 1.4 x5 0.00464 0.01817 0.26 0.801 1.3 S = 2.19553 R-Sq = 55.8% R-Sq(adj) = 44.2% Analysis of Variance Source DF SS MS F P Regression 5 115.817 23.163 4.81 0.005 Residual Error 19 91.587 4.820 Total 24 207.404 i. Conclude which variables are significant ii. Provide the regression model iii. Is the model significant for an alpha of 0.1 iv. Do we need to model further if so why?	08
5 a	Suppose there are 20 variables and you would like to know condense the set of 20 variables into few factors which would encompass the data set into few factors, write a step by step procedure on how you would conduct the analysis.	08
b	Twelve different variables were collected for obtaining information on the perception of a particular company and the variables include Product Quality (X1), Technical Support (X2), Compliant resolution (X3), Product Line(X4), Advertising (X5), Competitive pricing (X6), Salesforce image (X7) , Warranty claim (X8), Packaging (X9) , Orders and billing (X10) , Ecommerce (X11), Delivery Speed (X12). Based on the factor loading what variables can be related and what would you provide the names of for the common factor among these grouped variables. Variable Factor1 Factor2 Factor3 Factor4 Factor5 Factor6 Communality X1 0.082 0.932 0.1 0.011 -0.037 -0.104 0.897 X2 0.025 -0.012 -0.113 0.941 -0.073 0.025 0.906 X3 0.932 0.084 -0.06 0.052 0.005 -0.044 0.883 X4 0.138 0.362 0.063 -0.033 0.052 -0.892 0.953 X5 0.572 0.029 -0.744 0.091 0.055 0.119 0.908 X6 0.149 0.876 0.166 0.062 0.024 -0.256 0.888 X7 -0.102 0.269 0.778 -0.193 0.043 0.127 0.743 X8 0.102 0.073 -0.144 0.929 0.061 0.006 0.903 X9 0.054 -0.015 0.023 -0.012 0.995 -0.042 0.995 X10 0.873 0.05 0.032 0.108 0.013 -0.058 0.781 X11 0.476 0.099 0.783 -0.11 0.033 -0.251 0.927 X12 0.945 0.128 -0.031 -0.007 0.06 -0.106 0.926	08
	OR	

6 a The following values of factor loading are given for the following set of variables

Variable	Factor Loading	
	F_1	F_2
Skull Length	0.602	0.200
Skull Breadth	0.467	0.154
Fermur Length	0.926	0.143
Tibia Length	1.000	0.000
Humerus Length	0.874	0.476
Ulna Length	0.894	0.327

Provide a graph of factor loading pairs, based on graph suggest whether you would prefer to conduct factor rotation, If so just provide graph on the same plot on how much you would prefer to rotate the factors. 06

b Alpha car manufactures want to release a small car into the market. For this purpose, 20 respondents were asked about their perception and image attributes of small cars. Ten questions were given to each of them and all answered on a scale of 1 to 7 (1-completely agree and 7-Completely disagree). The following were the questions about V1-Price of the car should be reasonable, V2-Fuel mileage of car should be at least 17km/hr, V3-Small car should be easy to maintain and serviced, V4-seating should be comfortable for all adults, V5-small car should be adequate leg space and headroom, V6-Brakes are the most critical part of small car, v&-Collapsible steering column should be standard equipment in all new cars, V8-Power steering is a must, V9-Interior accessories of a small car should be attractive and V10-adequate space for luggage, Factor analysis was conducted using two methods of extraction (i)Principal components and other maximum likelihood. The table provides the minitab output of loading principal components and second table provides based on maximum likelihood. Based on both output group the various variables and accordingly provide which method is more suitable.

Variable	Factor1	Factor2	Factor3	Factor4	Factor5	Communality
V1	0.97	-0.057	0.009	0.008	-0.013	0.945
V2	0.96	0.122	-0.052	-0.16	-0.086	0.971
V3	0.967	-0.117	-0.031	0.095	0.045	0.961
V4	-0.168	0.923	-0.065	-0.185	-0.094	0.928
V5	0.127	0.951	-0.013	-0.144	0.093	0.951
V6	0.101	-0.014	-0.96	-0.071	-0.147	0.958
V7	0.083	-0.002	-0.179	-0.003	-0.979	0.997
V8	-0.21	-0.233	0.095	0.926	-0.035	0.966
V9	-0.209	0.349	-0.593	-0.565	-0.16	0.861
V10	0.965	-0.004	0.014	-0.166	-0.079	0.966
Variance	3.8783	1.9655	1.3225	1.2977	1.0399	9.5039
%Var	0.388	0.197	0.132	0.13	0.104	0.95

10

Variable	Factor1	Factor2	Factor3	Factor4	Factor5	Communality
V1	0.961	-0.061	0.013	0.011	-0.03	0.929
V2	0.95	0.127	0.071	0.09	0.195	0.97
V3	0.98	-0.119	0.003	-0.041	-0.154	1
V4	-0.172	0.878	0.115	0.082	0.09	0.829
V5	0.125	0.942	0.091	-0.094	0.094	0.928
V6	0.106	-0.035	0.613	0.183	0.023	0.422
V7	0.078	-0.008	0.231	0.969	-0.04	1
V8	-0.211	-0.296	-0.413	0.083	-0.568	0.632
V9	-0.189	0.347	0.891	0.064	0.213	1
V10	0.955	0.002	-0.008	0.093	0.28	1
Variance	3.8422	1.9012	1.4207	1.0176	0.5282	8.7099
%Var	0.384	0.19	0.142	0.102	0.053	0.871

7 a The following matrix for cost of misclassification is provided

	π_1	π_2
π_1	0	12
π_2	8	0

If the p(observation misclassified as π_1) is 0.35 and p(observation misclassified as π_2) is 0.30 , find the expected cost of misclassification. 4

b An anthropologist was interest in studying two sets of tribal groups and had measured the head length, Nasal Height, Nasal breadth and other anthropometric measurements, the following confusion matrix was obtained after fitting discriminant model. Determine the apparent error rate based on the following confusion matrix.

	π_1	π_2
$\pi_1(Tribe\ 1)$	55	12
$\pi_2(Tribe\ 2)$	17	50

04

c Explain with various examples where discriminant analysis can be used and how different is from linear regression analysis and logistic regression. 08

8 a Using Canberra metric and Euclidian distance find the measures of associate for the following data set.

Variable	1	2	3	4	5	6	7
Item 1	1	1	1	0	1	0	1
Item 2	1	0	1	1	1	1	1

06

b Draw the dendogram using simple linkage method based on the following distance matrix given

$$\begin{bmatrix} 0 & \dots & & & \\ 4 & 0 & & & \\ 6 & 9 & 0 & & \\ 1 & 7 & 10 & 0 & \\ 6 & 3 & 5 & 8 & 9 \end{bmatrix}$$

10

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

PREDICTIVE ANALYTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of statistical tables permitted.

PART-A

1	1.1	What measure of association is independent of the units of the two variables considered?	01
	1.2	When a data array consists of 3 variables with 5 observations each, the sample correlation matrix R will have the size _____.	01
	1.3	The inverse of the matrix $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ is _____.	01
	1.4	One key difference between Euclidian distance and statistical distance is the consideration of _____ along each of the coordinate axis.	01
	1.5	A single numerical value for the variation derived from the sample variance –covariance matrix S is _____ which is computed as $\|s\|$.	01
	1.6	In a multiple linear regression model $y = z\beta + \epsilon$, where y, z, β, ϵ are all matrices, the coefficient $\hat{\beta}$ can be computed from z and y as _____.	01
	1.7	Adequacy of a regression model can be checked by examining a plot of _____ against predicted values, one or more predictor values, or versus time.	01
	1.8	Why is the principal component analysis called a variable reduction technique?	01
	1.9	The covariance matrix Σ for two variables x_1 and x_2 is given as $\begin{pmatrix} .69 & -0.08 \\ -0.08 & 0.16 \end{pmatrix}$, comment on the relation between x_1 and x_2.	01
	1.10	With covariance matrix as in question 1.9, the variance of the variable x_1 is _____.	01
	1.11	In factor analysis, the loadings are rotated in order to improve or increase _____.	01
	1.12	Sets of linear equations used to specify phenomena in terms of cause and effect variables which cannot be directly measured are _____.	01
	1.13	The normalized eigen vector e corresponding to the eigen vector $x' = (\sqrt{3}, -2, 3)$ is _____.	01
	1.14	An issue with the Apparent Error Rate ($APER$) is that it usually _____ the Actual Error Rate (AER) in classification of populations into two groups.	01
	1.15	For the confusion matrix: the apparent error rate, AER is _____.	01

		classification π_1	classification π_2
actual	π_1	15	3
	π_1	2	12

1.16	Multivariate observations x are transformed to univariate observations y and used for classification. This is done using linear combinations of x, called _____ function.	01
1.17	In _____ method of clustering, an initial group of all objects is divided into two groups and sub groups further divided until each object is in a group.	01
1.18	Hierarchical clustering is used to cluster both _____ and _____. While non–hierarchical clustering is used to cluster only _____, usually.	01
1.19	Clustering of items 1,2 and 3 is taken up using the distance matrix $D = \begin{array}{c\|ccc} & 1 & 2 & 3 \\ \hline 1 & 0 & & \\ 2 & 10 & 0 & \\ 3 & 14 & 9 & 0 \end{array}$ Items 2 and 3 are grouped 1st. The next level distance matrix entry 'a' in $\begin{array}{c\|cc} & (32) & 1 \\ \hline 1 & 0 & \\ & a & 0 \end{array}$ Using single linkage is _____ and using complete linkage is _____.	02

PART-B

2	a	A multivariate data set is given as : $X = \begin{array}{c\|ccc} & V_1 & V_2 & V_3 \\ \hline X_1 & 1 & 5 & 6 \\ X_2 & 7 & 2 & 4 \\ X_3 & 4 & 8 & 3 \end{array}$ For 3 variables V_1, V_2, V_3. Compute sample covariance matrix S_3 and sample correlation matrix R and mean vector $\overline{x}$.	10
	b	Provide a layout of a matrix of a scatter plots for the above data, along with box plots of individual variables, assuming that there are around 100 rows of data. What can you infer from such a plot?	06
3	a	Using the data and the regression output shown below in fig 3 i) Write the normal equations of the regression relation using only first 5 rows of input. ii) What is the relation relating price 'y' to color 'x_1' and quality 'x_2'? iii) Is the relation significant? Why? What is the strength of the relation? iv) What is the residual when color and quality values are 8 and 7 respectively? v) What is the estimate of variance of the error terms? How can this be obtained from two different output quantities reported?	

Color	*Quality*	*Price*
7	5	65
3	7	38
5	8	51
8	1	38
9	3	55
5	4	43
4	0	25
2	6	33
8	7	71
6	4	51
9	2	49

SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.922330727
R Square	0.850693971
Adjusted R Square	0.813367463
Standard Error	5.888084465
Observations	11

		ANOVA					
			df	*SS*	*MS*	*F*	*Significance F*
		Regression	2	1580.280054	790.1400271	22.79061267	0.000496946
		Residual	8	277.3563093	34.66953867		
		Total	10	1857.636364			
			Coefficients	*Standard error*	*t Stat*	*P − value*	*Lower* 95%
		Intercept	1.751403659	6.960202671	0.251631129	0.807669624	−14.29885248
		Color	4.895288365	0.820229778	5.968191467	0.000335084	3.003835104
		Quality	3.758415483	0.756510987	4.968091073	0.00109572	2.013898018
		Fig 3: Input and output of multiple regression					16
		OR					
4	a	Random variables X_1, X_2, X_3 have the covariance matrix $S = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ with eigen values $(\lambda_1, \lambda_2, \lambda_3)$ as $5.83, 2.0, 0.17$ respectively. The eigen vectors corresponding to λ_2 and λ_3 are $l_2' = (0,0,1)$ and $l_3' = (0.924, 0.383, 0)$ respectively. i) Compute the eigen vector corresponding to λ_1 ii) Derive the principal components. iii) What is the proportion of variance accounted for by the principal components? iv) How might the principal components of the 3 random variables help in further regression analysis, if a total of 100 observations are available (i.e. input data is 100×3 matrix)?						16
5	a	Outline a general strategy to perform a complete factor analysis.					08
	b	Factor loadings for indicators of economic status are given in table 5b. Compute the communalities. Group the variables logically into two groups to aid better interpretation of the factors.					
		Table 5b					
		Variables			*Factor* 1	*Factor* 2	
		1. *Income*			0.65	.11	
		2. *No. of public parks*			0.13	.57	
		3. *Education of resident*			0.59	0.25	
		4. *No. of reported crimes per year*			0.23	0.55	
		5. *occupation*			.48	0.19	
		What do the communalities represent?					08
		OR					
6	a	Starting from a matrix of input data of say 200×8 size representing 200 data on 8 variables, how is a factor analysis performed to derive loadings and communalities?					06
	b	Given eigen values and eigen vectors $\lambda_1 = 1.96, e_1' = (0.625, 0.593, 0.507), \lambda_2 = 0.68, e_2' = (0.219, -.491, 0.843), \lambda_3 = 0.36, e_3' = (0.749, -.638, -.177)$, compute the factors loadings. Assuming a two factor model compute communalities, specific variances and provide your interpretations.					10
7	a	What aspects should a classification procedure consider? How do you evaluate any classification procedure using a confusion matrix?					06

	b	The minimum ECM rule is: "Allocate X_0 to π_1 if $(\overline{X_1} - \overline{X_2})` s_{POOLED}^{-1} X_0 - \frac{1}{2}(\overline{X_1} - \overline{X_2})` s_{POOLED}^{-1}(\overline{X_1} + \overline{X_2}) \geq \ln\left[\frac{c\left(\frac{1}{2}\right).P_2}{c\left(\frac{2}{1}\right).P_1}\right]$". If samples $X_1 = \begin{pmatrix}3 & 2 & 4\\ 7 & 4 & 7\end{pmatrix}$ and $X_2 = \begin{pmatrix}6 & 5 & 4\\ 9 & 7 & 8\end{pmatrix}$ are taken from populations π_1 and π_2, and the pooled covariance matrix $S_{pooled} = \begin{pmatrix}1 & 1\\ 1 & 2\end{pmatrix}$, then classify $X_0 = \begin{pmatrix}2\\ 7\end{pmatrix}$ assuming equal priors and equal misclassification costs. Dose this classification make sense in a scatter plot of the 6 data points?	10
8	a	Given the matrix of distances, cluster the 4 objects using average linkage hierarchical procedure: show the dendrogram. |	1	2	3	4
---	---	---	---	---
1	0			
2	1	0		
3	11	2	0	
4	5	3	4	0		06
	b	Four objects A, B, C, D are measured along two variables (dimensions) (x_1, x_2) as follows. Use k-means clustering to group the objects with initial groups (AB) and (CD). $A = (5,4), B = (1,-2), C = (-1,1), D = (3,1)$.	06
	c	What are some of the issues cropping up with hierarchical and non–hierarchical clustering procedures which an analyst should be aware of?	04

Project Management

VII Semester SEE Question Papers

Course codes: 21IM751M, IM375TGM, 21IM75IM

- Feb 2026 (21IM751M)
- Feb 2026 (IM375TGM)
- Mar/Apr 2025 (21IM75IM)

21IM751M

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Regular Examinations February 2026

PROJECT MANAGEMENT (GLOBAL ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	What are knowledge areas in project management?	02	1	1
	1.2	Mention any two external sources of project ideas.	02	1	1
	1.3	What is Work Breakdown Structure (WBS)?	02	1	2
	1.4	Define organizational process assets (OPA) and Enterprise environmental factors (EEF).	02	1	2
	1.5	What is meant by corrective action and preventive action in Project Integration Management?	02	2	3
	1.6	Define quality in project management.	02	1	3
	1.7	What is Qualitative Risk Analysis and Quantitative Risk Analysis in Project Risk Management?	02	2	4
	1.8	What is meant by resource levelling and resource smoothing in Project Scheduling?	02	2	4
	1.9	Name any two software tools used in project management.	02	1	4
	1.10	What is activity-on-mode (AON) and activity-on-arrow (AOA).	02	2	4

PART-B

2	a	With a neat diagram, explain the relationship among Portfolio Management, Program Management, Project Management, and Organizational Project Management.	08	2	1
	b	Explain the process of generation of project ideas. Discuss various techniques used for idea generation.	08	2	1
3	a	Given a project with multiple stakeholders and changing requirements, explain how the Requirements Traceability Matrix (RTM) can be used to ensure scope control.	08	3	2
	b	With a neat diagram, explain the Project Scope Management process flow.	08	2	2
		OR			
4	a	Explain the concept of project life cycle. Describe the phases of a typical project life cycle with a neat diagram.	08	2	2
	b	Illustrate the formation of a project team for a hospital management system project and explain the roles of project sponsor, project manager and team members.	08	3	2

5	a	With a neat diagram. Explain the flow of Project Integration Management processes.	08	2	3
	b	A construction project for a commercial building has reached completion. As the project manager, describe the steps you would follow to close the project formally. Also, analyze the importance of formal project closure for the organization and future projects.	08	4	3
		OR			
6	a	With a neat diagram, explain the flow of Project Quality Management processes.	08	2	3
	b	A project manager wants to reduce the Cost of Quality (COQ). Explain how quality management processes can be applied to achieve this goal.	08	3	3
7	a	With a neat diagram, explain the Project Risk Management process flow.	08	2	4
	b	During execution of an IT project, several technical risks are identified. Explain how Qualitative and Quantitative Risk Analysis can be applied to prioritize and manage these risks.	08	3	4
		OR			
8	a	Explain the steps involved in preparing a project schedule.	08	2	4
	b	A project is facing cost overruns due to poor scheduling decisions. Analyze the problem and explain how project life cycle costing can help in better planning.	08	4	4
9	a	For a project , the time estimates are: **Activity** | **Optimistic(to)** | **Most likely (tm)** | **Pessimistic(tp)** A | 2 | 4 | 8 B | 3 | 6 | 9 C | 1 | 2 | 3 D | 4 | 5 | 10 **Calculate:** i) Expected time for each activity ii) Variance of each activity iii) Expected project duration	08	4	4
	b	A construction project is delay due to improper scheduling. Explain how Gantt charts and network diagrams can be applied to improve project planning and control.	08	3	4
		OR			
10	a	For an infrastructure project, explain how different project management tools and techniques can be integrated for effective scheduling and control.	08	4	4

b	A project consists of the following activities: **Activity** | **Predecessor** | **Duration(days)** A | - | 3 B | A | 4 C | A | 5 D | B | 6 E | C | 2 F | D, E | 4 Determine: i) Network diagram ii) Critical path iii) Project duration	08	3	4

Question Paper ID: 712283

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM375TGM

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

Project Management (Institutional Elective)

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	How does project management support organizational strategy?	02	2	2
1.2	How is project management different from program management?	02	2	2
1.3	Name any two techniques used for generating project ideas.	02	2	2
1.4	What is meant by scouting for project ideas?	02	2	2
1.5	One tool used for controlling scope.	01	1	2
1.6	Define project scope.	01	1	2
1.7	List two inputs required for Direct and Manage Project Work.	02	2	2
1.8	Give two benefits of preparing a Quality Management Plan.	02	2	2
1.9	Mention two negative risk response strategies.	02	2	2
1.10	What is cloud-based project management?	02	2	2
1.11	What is a product backlog?	02	2	2

Part B

Question No	Question	M	CO	BT
2a	A retail company wants to increase business value through a new billing automation project. Explain how a project manager can ensure delivery of measurable business value.	08	2	3
2b	A company plans to launch a new electric scooter. Explain how portfolio management, program management, and project management would work together to execute this initiative.	08	3	3

<table>
<tr><td>3</td><td>Create a Work Breakdown Structure (WBS) for developing a Mobile Banking Application. Explain how it helps in project management.</td><td>16</td><td>3</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4</td><td>Develop a Work Breakdown Structure (WBS) for organizing a College Cultural Fest (Kannada Habba). Explain the WBS and its importance.</td><td>16</td><td>3</td><td>4</td></tr>
<tr><td>5a</td><td>List the main components of a Project Management Plan.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>5b</td><td>Explain the process of developing a project plan.</td><td>08</td><td>4</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>6a</td><td>Explain the concept of Quality Cost and its components.</td><td>08</td><td>4</td><td>4</td></tr>
<tr><td>6b</td><td>Explain how Control Quality helps in project success.</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td>7</td><td>Using PERT, calculate Expected Time (te), Variance (σ^2), and Project Duration. Also find probability of completing in 25 days.

<table><tr><th>Activity</th><th>Optimistic (O)</th><th>Most Likely (M)</th><th>Pessimistic (P)</th></tr><tr><td>A</td><td>3</td><td>6</td><td>9</td></tr><tr><td>B</td><td>2</td><td>5</td><td>14</td></tr><tr><td>C</td><td>4</td><td>7</td><td>10</td></tr><tr><td>D</td><td>3</td><td>5</td><td>11</td></tr></table></td><td>16</td><td>3</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>8a</td><td>Explain project implementation scheduling and the major steps involved.</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>8b</td><td>Explain the difference between exploit (opportunity) and avoid (threat) with examples.</td><td>06</td><td>3</td><td>3</td></tr>
<tr><td>9a</td><td>A software startup is facing frequent scope changes and customer dissatisfaction.
Apply Agile principles to improve project delivery and stakeholder satisfaction.</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>9b</td><td>Explain the steps involved in the Critical Path Method (CPM).</td><td>06</td><td>3</td><td>2</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>10a</td><td>A startup wants to adopt Agile project management with minimal cost.
Explain how open-source tools can be used to manage Agile projects effectively.</td><td>08</td><td>3</td><td>4</td></tr>
<tr><td>10b</td><td>A software product faces frequent requirement changes from customers.
Explain how Agile Project Management can be applied to handle this situation.</td><td>08</td><td>3</td><td>3</td></tr>
</table>

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular Examinations Mar/April-2025

PROJECT MANAGEMENT (INSTITUTIONAL ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of the statistical table is permitted.

PART-A

			M	BT	CO
1	1.1	What are the key knowledge areas in PMBOK?	02	2	2
	1.2	Why is environmental monitoring important in project selection?	02	2	2
	1.3	Mention purpose of the scope validation help in project management?	02	3	3
	1.4	List the phases of the project life cycle?	02	1	3
	1.5	How does monitoring and controlling project work help in risk management?	02	3	2
	1.6	What is the purpose of quality control in project management?	02	2	3
	1.7	Name two tools used in risk monitoring and control.	02	1	2
	1.8	Differentiate between $PERT$ and CPM.	02	3	3
	1.9	How does a bar chart for combined activities differ from a simple bar chart?	02	3	2
	1.10	What is the purpose of logic diagrams in project management?	02	2	2

PART-B

2	a	Discuss the relationship between Project Management, Operations Management, and Organizational Strategy, and how they contribute to business success.	08	2	1
	b	Evaluate the importance of Preliminary Screening in the project selection process and it impacts on project success.	08	3	3
3	a	A construction company is hired to design and build a new office building for a client. Describe how the project manager should create a Work Breakdown Structure (WBS).	08	2	1
	b	Evaluate the role of the Project Team in the successful execution of a project, considering both functional expertise and collaborative teamwork.	08	3	3
		OR			
4	a	What is project Scope Management? Describe the steps involved in Project Scope Management.	08	2	2
	b	Design a Project Life Cycle model for a construction project, highlighting the key phases and the activities involved in each.	08	4	4
5	a	Create a project charter for developing chat bot which will help first year student queries about the institution.	08	4	2

	b	How does the integration of the documents of project charter help in directing and managing project work effectively? Discuss.	08	4	4
		OR			
6	a	Create a project charter for developing an online learning management system (*LMS*) for your institution.	08	4	3
	b	A food processing company is launching a new line of packaged snacks and wants to ensure consistent quality throughout the production process. Explain how the company can plan quality management.	08	4	2
7	a	Describe the steps involved in identifying potential risks and conducting both qualitative and quantitative risk analysis.	08	2	2
	b	Compare life cycle costing with traditional cost estimation methods with respect to the following aspects: i) Scope ii) Cost component iii) Risk management iv) Decision making	08	3	3
		OR			
8	a	What challenges might arise when integrating PLM into project scheduling, and how can they be addressed?	08	2	2
	b	How would you develop a project implementation schedule for a new product launch? Explain	08	2	2
9	a	A software development company is working on a project with multiple overlapping tasks and deadlines. Explain how a Gantt Chart and a combined activity bar chart can be used to visualize the project schedule and track progress effectively.	08	2	3
	b	A project schedule has the following characteristics: i) Construct the network ii) Compute E_i and L_j values for each activity iii) Find the critical path and total duration	08	2	3
		OR			
10	a	A manufacture firm is planning a complex project involving multiple tasks with uncertain completion times. Discuss how the Project Evaluation and Review Technique(PERT) can be applied to estimate project times lines.	08	2	3
	b	A project schedule has the following characteristics: i) Calculate the expected time for each activity ii) Draw the network diagram iii) Find the critical path and duration	08	2	3

Activity	*Time(Weeks)*	*Activity*	*Time(weeks)*
1 – 2	4	5 – 6	4
1 – 3	1	5 – 7	8
2 – 4	1	6 – 8	1
3 – 4	1	7 – 8	2
3 – 5	6	8 – 10	5
4 – 9	5	9 – 10	7

Activity	10-20	20-30	20-40	20-50	30-60	40-60	50-70	60-70
Optimistic	1	8	3	3	4	7	6	1
Most likely	2	10	4	5	5	8	8	4
Pessimistic	3	12	6	7	6	9	10	7

Sustainability Engineering and Circular Economy

VII Semester SEE Question Papers

Course codes: IM374TFD

- Feb 2026 (IM374TFD)

Question Paper ID: 433196

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM374TFD

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Sustainability Engineering and Circular Economy

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	What is the primary objective of the Paris Agreement?	02	1	2
1.2	Define "Sustainable Engineering."	02	1	2
1.3	Why is data quality important in the LCI phase?	02	2	2
1.4	Distinguish between "Cradle-to-Grave" and "Cradle-to-Gate" boundaries.	02	2	2
1.5	What is a "Conceptual Standard Framework"?	02	2	2
1.6	Define "Standardization" in the context of sustainability.	02	2	2
1.7	Why is "Refurbishment" higher in the circular hierarchy than "Recycling"?	02	4	2
1.8	Explain the "Re-think" stage in Eco-Design.	02	4	2
1.9	What is 'Take-Back' scheme in the Fashion/Textile industry.	02	4	2
1.10	What is 'Predictive Maintenance' in Circular Manufacturing?	02	4	2

Part B

Question No	Question	M	CO	BT
2a	What is life cycle thinking? Illustrate its application in product design.	08	1	3
2b	Describe the concept of systems thinking and its importance in sustainable engineering.	08	1	3

3a	Explain the importance of the Functional Unit and System Boundaries in the Goal and Scope phase of LCA.	08	2	3
3b	Discuss the role of LCI Databases and LCA Software in engineering. Why is it cautioned that software should not be treated as a "black box"?	08	2	3
	OR			
4a	Analyze how Climate Change acts as a primary driver for restructuring modern supply chains. Provide examples of physical and transition risks.	08	3	4
4b	Analyze the impact of resource scarcity (water, rare earth metals) on production strategy and the shift toward "Resource Efficiency."	08	3	4
5a	Explain the "Three Pillars" of Quality Infrastructure and how they support sustainable development.	08	2	3
5b	Metrology is considered as the "foundation" of the three pillars of quality infrastructure. Justify.	08	3	4
	OR			
6a	What are the key elements of a "Standard Framework" for Corporate Sustainability according to Belvedere and Grando?	08	3	2
6b	Urbanization is a "Mega-Trend." Discuss its challenges for sustainable logistics and the concept of the "City Hub."	08	3	4
7a	Explain the three core principles of the Circular Economy as defined by the Ellen MacArthur Foundation.	08	4	2
7b	Discuss how the Circular Economy contributes to SDG 13 .	08	4	4
	OR			
8a	Explain the hierarchy of the 7R model of Circular Eco-Design.	08	4	2
8b	Discuss how Sustainable Engineering differs from traditional engineering in terms of the "Lifecycle Perspective."	08	3	3
9a	Evaluate the role of Remote Sensing and GIS (Geographic Information Systems) in modern resource management.	08	4	5
9b	Businesses measuring their success based on Three P's ensures long term viability. In the context of Sustainability , justify the statement with an example.	08	4	5
	OR			
10a	HungerBox uses AI in corporate cafeterias to predict food demand and reduce overproduction and waste Discuss the role of "Digitalization" in reducing the "Economic Loss" of food waste in India.	08	4	5
10b	Renault converted its Flins plant into Europe's first circular economy factory, focusing on "re-trofitting" used cars and "re-manufacturing" parts like gearboxes and turbochargers. Analyze how the "Refactory" model addresses resource scarcity and creates a new revenue stream for the automotive industry.	08	4	5

Data Mining and Predictive Analytics

VII Semester SEE Question Papers

Course codes: 21IM73GA, 18IM7F2

- Mar/Apr 2025 (21IM73GA)
- Feb/Mar 2024 (18IM7F2)
- Feb/Mar 2023 (18IM7F2)

21IM73GA

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular Examinations Mar/April-2025

Industrial Engineering and Management

DATA MINING AND PREDICTIVE ANALYTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical table is permitted.

PART-A

			M	BT	CO
1	1.1	Define supervised and unsupervised learning in the context of data mining.	02	1	1
	1.2	What is the purpose of using data mining software in the industrial applications?	02	2	2
	1.3	In what type of problems would you prefer to use a decision tree over a logistic regression model?	02	2	2
	1.4	How does linear regression differ from logistic regression in terms of output prediction?	02	2	2
	1.5	Why is it important to standardize data before applying K-means clustering?	02	3	3
	1.6	What does “Support” mean in the context of association rule mining?	02	3	3
	1.7	List two advantages of using random forests over a single decision tree?	02	2	2
	1.8	In what ways can anomaly detection be applied to improve safety in industrial operations?	02	3	3
	1.9	How does a manufacturing company use data mining to improve product quality?	02	3	3
	1.10	What role do association rules play in identifying inefficiencies in industrial processes?	02	2	2

PART-B

2	a	What is Data Mining? What in Predictive Analytics? Bring out the connection between data mining and predictive analytics?	08	2	1
	b	Describe the different stages of the data mining process with examples.	08	2	2
3	a	What is decision tree and state the requirements for using decision trees.	06	3	1
	b	A car insurance company wants to predict whether a customer will purchase an insurance policy based on their age and monthly premium affordability. Here is a small dataset to analyze:			

$Age(Years)$	$Monthly\ Premium(Rs)$	$Will\ Buy(Yes/No)$
23	3,000	No
27	4,500	Yes
45	10,000	Yes
50	7,500	No
42	12,500	Yes
55	5,500	No

Use decision tree splitting based on Age and Monthly Premium to classify customers Determine the best feature to split using Gini Impurity or Entropy. 10 3 2

OR

4 a Outline the procedure for classification and prediction using Logistic regression model. 06 4 3

b A bank wants to classify loan applications as either "Approved" or "Rejected" based on specific factors such as Credit Score, Existing Debt, and Employment Status. Here is a small dataset:

$Application$	$Credit\ Score > 700$	$Existing\ Debt\ (Yes/No)$	$Employed\ (Yes/No)$	$Loan\ Approved\ (Approved/Rejected)$
$App1$	Yes	No	Yes	$Approved$
$App2$	No	Yes	Yes	$Rejected$
$App3$	Yes	Yes	No	$Rejected$
$App4$	Yes	No	No	$Approved$
$App5$	No	No	Yes	$Approved$
$App6$	No	Yes	No	$Rejected$

i) Calculate Prior Probability for Approved or Rejected.
ii) Find the probability of each factor(Credit Score, Debt, Employment) appearing in Approved and Rejected Applications.
iii) Using the Naïve Bayes formula, determine whether a new application is more likely to be Approved or rejected given its features. 10 3 3

5 a Given the following transactional database, construct the Frequent Pattern Tree(FP Tree) with a support threshold of 50% and a confidence threshold of 70%. Show how the tree evolves for each transaction.

$Transaction\ Id$	$Items$
$T1$	$Apple, Banana, Orange$
$T2$	$Apple, Banana$
$T3$	$Apple, Grape, Mango$
$T4$	$Grape, Mango$
$T5$	$Mango, Banana$
$T6$	$Apple, Mango, Grape$

i) Construct the FP-Tree step by step for each transaction.
ii) Identify frequent item sets.
iii) Generate association rules that satisfy the given support and confidence thresholds 08 2 3

	b	For the given data, compute two clusters using K-means algorithm for clustering where initial cluster centers are (1.0 ,1.0) and (5.0,7.0). Execute for two iterations. $Record\ Number$ | $R1$ | $R2$ | $R3$ | $R4$ | $R5$ | $R6$ | $R7$ A | 1.0 | 1.5 | 3.0 | 5.0 | 3.5 | 4.5 | 3.5 B | 1.0 | 2.0 | 4.0 | 7.0 | 5.0 | 5.0 | 4.5	08	2	4
		OR			
6	a	Compare K-Mean and Hierarchical Clustering in terms of Efficiency, Scalability, Cluster Shape and Interpretability	08	2	2
	b	Consider the following data points: $(2,2), (3,3), (6,7), (8,8), (7,6)$. Perform the first iteration of K-mean clustering for $k = 2$ by selecting (2,2) and (8,8) as initial centroids.	08	3	4
7	a	How Random Forest and Gradient Boosting improve predictive performance compared to a single decision tree?	06	2	2
	b	A company applies feature selection but still observes model over fitting. Suggest methods to refine the selection process and justify their effectiveness.	10	3	3
		OR			
8	a	List each four advantages and limitations of principal component analysis (PCA).	08	2	2
	b	Describe how anomaly detection can be integrated into a predictive maintenance system in the manufacturing sector.	08	3	3
9	a	Discuss how data mining techniques can be integrated into industrial engineering processes to improve production efficiency and quality control.	08	2	2
	b	How can predictive analytics, driven by data mining, be utilized in the management of workforce performance in industrial settings?	08	2	2
		OR			
10	a	Explain the challenges of maintain fairness and transparency when applying machine learning algorithms in data mining.	08	2	2
	b	Discuss the ethical considerations involved in using data mining techniques to analyze customer data in the retail or finance sector.	08	2	2

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular / Supplementary Examinations Feb/Mar-2024

Industrial Engineering and Management

DATA MINING AND PREDICTIVE ANALYTICS (ELECIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Give any one significant difference between Data Mining and Statistics.	01
	1.2	Expand the acronym CRISP-DM.	01
	1.3	Which graphical tool is preferred for identifying outliers in a given data set?	01
	1.4	Give an example for missing data for the method of collecting data through survey formats.	01
	1.5	For a face classification system based on $256x256$ pixel images, how many vectors of dimension are required?	01
	1.6	Graphically represent criteria to decide on the number of components to extract.	01
	1.7	What is the term used to represent the proportion of variance of a particular variable that is shared with other variable in PCA context?	01
	1.8	Write down the identity for the factor analysis model.	01
	1.9	What is the meaning of the term overfitting?	01
	1.10	In a two fold cross-validation what are the two types of data set that are created?	01
	1.11	Write down the formula for computing the Euclidian distance.	01
	1.12	What is the significance of K in the K-nearest neighbor algorithm?	01
	1.13	What is the name given to the tree like cluster structure created through recursive portioning in hierarchical clustering?	01
	1.14	The distance that is sought to be minimized among the records in the two clusters is generally which distance?	01
	1.15	Give the formula for computing the support S for a particular association rule.	01
	1.16	Give an example for the use of association focuses in the business or research context.	01
	1.17	Give the structure of a typical decision rule by linking assets to credit risk in banking business.	01
	1.18	Mention any one of the requirements for using Decision free algorithm.	01
	1.19	How many synaptic connections are required to connect neurons in the human brain?	01
	1.20	Why is the sigmoid function preferred as the activation function in the Neural network modeling?	01

PART-B

2 a For each of the following meetings, explain which phase in the CRISP-DM process is represented

i) Managers want to know by next week whether deployment will take place. Therefore analysis meet to discuss how useful and accurate their model is.

ii) The data mining project manager meets with data warehousing manager to discuss how the data will be collected.

iii) The data mining consultant meets with the Vice president for marketing, who says that he would like to move forward with customers relationship management.

iv) The data mining project managers meets with the production line supervisor to discuss implementation of changes and improvement.

v) The analyst meet to disc whether the neural network or decision tree models should be applied

(Give the reasoning for the choice of the process you select in the CRISP-DM process) 10

b The stock price data (in dollars) in as follows:

10 7 20 2 75 15 9 18 4 12 8 14

i) Identify the outlier

ii) Verify that this is an outlier using the Z-score method

iii) Verify that this is an outlier using the IQR method 06

3 a Based on the following information given in table 3a how many components should be extracted using

i) The eigenvalue Criterion

ii) The proportion of the variance

iii) Explained criterion

Components	Initial Eigen Values		
	Total	% of variance	Cumulative (y)
1	1.088	9.890	9.890
2	1.056	9.596	19.486
3	1.040	9.454	28.939
4	1.023	9.296	38.236
5	1.050	9.094	47.329
6	0.989	8.987	56.317
7	0.972	8.834	65.151
8	0.969	8.811	73.961
9	0.963	8.754	82.715
10	0.962	8.747	91.462
11	0.939	8.538	100.00

10

b What is Factor Rotation? Define Quartimax Rotations. Why is factor rotation necessary? 06

OR

4 a What is the theoretical difference between principal component analysis and the factor analysis? When each is preferred over the other? 06

	b	The following factor analysis output of some social psychological data related to preferential and discrimination. Commonalties Initial / Extraction Item 25 – Affirmation action: 0.175 / 0.561 Item 27 – Racial equality: 0.429 / 0.459 Item 28 – Black president of USA: 0.564 / 0.620 Item 30 – Black neighborhood neighbors: 0.522 / 0.671 Item 32 – Foreigners: 0.370 / 0.349 Item 34 – Inter racial: 0.613 / 0.658 Item 36 – Economic group stay in the own place: 0.539 / 0.608 Item 38 – Too many blacks students in the campus: 0.489 / 0.550 Item 39 – Increased equality: 0.380 / 0.425 Item 40 – which superior: 0.531 / 0.641 Item 43 – Bewing: 0.206 / 0.274 Item 46 – A block supervisor: 0.472 / 0.530 Item 48 – Mexican Immigrants: 0.374 / 0.338 Item 50 – Interracial marriage: 0.618 / 0.852 Interpret this factor analysis output. What do you attribute the difference between the initial and extraction columns in the table above?	10
5	a	Explain the difference between supervised methods and unsupervised methods? Which data mining terms are associated with unsupervised methods? Supervised methods? Both?	08
	b	Take a case example and describe the differences between training set, test set and validation set. What are the typical ways of portioning the data set in terms of proportion of the total data set?	08
		OR	
6	a	How is the bias-variance trade-off related to the use of overfitting and underfitting? Is high bias associated with overfitting and underfitting, any why? High variance?	08
	b	Give four point as follows Point / P_1 / P_2 / P_3 / P_4 x / 0 / 2 / 3 / 5 y / 2 / 0 / 1 / 1 Build the matrices of Euclidean distance and write how you would apply K-nearest neighborhood algorithm based on the distance criterion.	08
7	a	Suppose that we have the following data: a / b / c / d / e / f / g / h / i / j (2,0) / (1,2) / (2,2) / (3,2) / (2,3) / (3,3) / (2,4) / (3,4) / (4,4) / (3,5) Identify the cluster by applying k-means algorithm with $K = 2$. By using initial cluster as far apart as possible.	08

	b	Describe the two main methods of representing market basket data. What are the benefits and draw backs of each? Describe support and confidence.	08
8	a	Consider the data in the table below. The target variable is salary. Start by discretizing salary as follows: Less than $35,000 Level 1 $35,000 to less than $45,000 Level 2 $45,000 to less than $55,000 Level 3 Above $55,000 Level 4	

Occupation	Gender	Age	Salary
Service	Female	45	$48,000
	Male	25	$25,000
	Male	33	$35,000
Management	Male	25	$45,000
	Female	35	$65,000
	Male	26	$45,000
	Female	45	$70,000
Sales	Female	40	$50,000
	Male	30	$40,000
Staff	Female	50	$40,000
	Male	25	$25,000

		Construct a classification and regression tree to classify salary based on other variables. Do as much as you can by hand.	10
	b	Give the comparison between a real neuron and artificial neuron model. Given a simple example of a neural network with the input layer, hidden layers and output layers.	08

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Feb/Mar - 2023

Industrial Engineering and Management

DATA MINING AND PREDICTIVE ANALYTICS (Elective)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	"Predictive analysis is the process of extracting information from large data sets in order to make predictions and estimates about future customers" How this definition is applicable to detecting cyber frauds in the digital space.	01
	1.2	Typically how much is the effort that goes into data preprocessing phase of a Data mining project? Express in terms of percentage of duration in relation to the total project duration.	01
	1.3	An alternative to the standard process for data mining CRISP-DM is the SEMMA methodology. What is the expansion of SEMMA?	01
	1.4	Mention any of the technique for identifying outliers under the graphical method of identifying outliers.	01
	1.5	What does a positive covariance between two variable indicate?	01
	1.6	Explain the screen-plot criteria used to determine how many components are extracted during the PCA analysis graphically.	01
	1.7	Mention any of the differences between PCA and factor analysis.	01
	1.8	What does Barlett's test of sphericity support during factor analysis?	01
	1.9	What is the meaning of K-fold cross validation?	01
	1.10	What does this complementary relationship between bias and variance: $MSE = variance + bias^2$, indicate to the analyst?	01
	1.11	Explain in simple terms how k-nearest neighboring algorithm will be useful in the field of law while determining whether a will was written by the person deceased or fraudulently by someone else.	01
	1.12	Mathematically define the Euclidian distance function.	01
	1.13	What does the complete linkage orientation seeks to minimize?	01
	1.14	What is a Dendogram?	01
	1.15	Define confidence C of association rule $A \Longrightarrow B$, as a measure of the accuracy of the rule.	01
	1.16	Write down the formula for determining the lift to quantify the usefulness of the association rule.	01
	1.17	Expand the term CART as applied to decision tree.	01

	1.18	What is the meaning of conditional splits with reference to decision trees?	01
	1.19	Sketch the graph of the sigmoid function $y = f(x) = \frac{1}{(1+e^{-x})}$	01
	1.20	What is Back-propagation?	01

PART-B

2	a	Discuss the six phases of CRISP-DM framework deployed for data mining projects, clearly list the phases and tasks involved in each of the phases.	08
	b	Use the following stock price data 10 7 20 12 75 15 9 18 4 12 8 14 i) Find the min-max normalized stock price for the stock price \$20. ii) Explain why data analyst need to normalize their numeric variables. iii) Find the decimal sealing stock price for the stock price \$20.	08
3	a	The following computer output explores the application of PCA to churn data set. Considering the following information which variable in variables might be well advised to omit from the PCA and why? If we really need all these variables in the analysis then what should we do? Assume that PCA is carried out in this case using seven components.	

Commonalities

	Initial	*Extraction*
ZACCTLEN	1.000	0.606
ZVMAILME	1.000	0.836
ZDAYCALL	1.000	0.528
ZDAYCHAR	1.000	0.954
ZEVECALL	1.000	0.704
ZEVECHAR	1.000	0.621
ZNITECAL	1.000	0.543
ZNITECHA	1.000	0.637
ZINTCALL	1.000	0.439
ZINTCHAR	1.000	0.588
ZCSC	1.000	0.710

Extraction method – Principal component analysis

			08
	b	Write down the formulation of the factor analysis model. What conclusion can be drawn from the application of factor analysis procedure to data collected from the real world?	08

OR

4	a	Let $\Sigma = \begin{matrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{matrix}$ Determine the principal components Y_1, Y_2, Y_3. What can you say about the Eigen Vector (and principal components) associated with eigen values that are not distinct.	08
	b	The eigen values and eigen vectors of the correlation matrix ρ are $\lambda_1 = 1.96, e_1 = (0.625, 0.593, 0.507)$ $\lambda_2 = 0.68, e_2 = (-0.219, -0.491, -0.843)$ $\lambda_3 = 0.36, e_3 = (-0.749, -0.638, -0.177)$ Assuming an $m = 1$ factor model, calculate the loading matrix L and matrix of specific variance Ψ using the principal component solution method. What proportions of the total population variance is explained by the first common factor?	08
5	a	List any four methods under Supervised Learning and two under Unsupervised Learning methods respectively. Clearly bring out the distinction between supervised method and unsupervised methods of data analysis.	08
	b	Consider the following data $\begin{array}{c\|cc} Point\ P & x & y \\ \hline p_1 & 0 & 2 \\ p_2 & 2 & 0 \\ p_3 & 3 & 1 \\ p_4 & 5 & 1 \end{array}$ Plot this on a 2D scatter diagram. Compute the matrix of Euclidean distance based on the above data.	08
		OR	
6	a	Clearly describe what is meant by classification? Explain the K-Nearest Neighbor algorithm with suitable illustrations.	08
	b	Consider the following data: $\begin{array}{lcccc} Record & Age & Na/k & Age_{MMN} & Na/K_{MMN} \\ \hline A\ (dark\ grey) & 16.8 & 12.4 & 0.0467 & 0.2471 \\ B\ (medium\ grey) & 17.2 & 10.5 & 0.0533 & 0.1912 \\ C\ (medium\ grey) & 19.5 & 13.5 & 0.0917 & 0.2794 \end{array}$ Consider a new record with Age 17, Na/K 12.5, Age_{MMN} 0.05, Na/K_{MMN} 0.25. Compute the distance of the new records from the existing records $A, B\ and\ C$. Apply k-nn algorithm on this data taking $k = 3$ (for classification).	08
7	a	Four objects A, B, C, D are measured along two variables (dimensions) x_1, x_2 as follows: $A = (5,4)$ $B = (-1,2)$ $C = (-1,1)$ $D = (3,1)$. Use k means clustering to group to group the objects with initial groups (AB) and (CD)	08

b

Consider the Weather dataset as follows.

Sl no	*Outlook*	*Temperature*	*Humidity*	*Windy*	*Play*
1	*Sunny*	*Hot*	*High*	*False*	*No*
2	*Sunny*	*Hot*	*High*	*True*	*No*
3	*Overcast*	*Hot*	*High*	*False*	*Yes*
4	*Rain*	*Mild*	*High*	*False*	*Yes*
5	*Rain*	*Cool*	*Normal*	*False*	*Yes*
6	*Rain*	*Cool*	*Normal*	*True*	*No*
7	*Overcast*	*Cool*	*Normal*	*True*	*Yes*
8	*Sunny*	*Mild*	*High*	*False*	*No*
9	*Sunny*	*Cool*	*Normal*	*False*	*Yes*
10	*Rain*	*Mild*	*Normal*	*False*	*Yes*
11	*Sunny*	*Mild*	*Normal*	*True*	*Yes*
12	*Overcast*	*Mild*	*High*	*True*	*Yes*
13	*Overcast*	*Hot*	*Normal*	*False*	*Yes*
14	*Rain*	*Mild*	*High*	*True*	*No*

Let $\emptyset = 3$. Generate the frequent 3-elements using 75% minimum confidence and 20% minimum support, generate one-antecedent association rate for predicting play. 08

8 a

The following data is provided

Occupation	*Gender*	*Age*	*Salary*
Service	*Female*	45	$48,000
	Male	25	$25,000
	Male	33	$35,000
Management	*Male*	25	$45,000
	Female	35	$65,000
	Male	26	$45,000
	Female	45	$70,000
Sales	*Female*	40	$50,000
	Male	30	$40,000
Staff	*Female*	50	$40,000
	Male	25	$25,000

The target variable is salary. Start by discretizing salary as follows
Less than $35,000 Level 1
$35,000 to less than $45,000 Level 2
$45,000 to less than $55,000 Level 3
Above $55,000 Level 4
Construct a classification and regression tree to classify salary based on other variables. Do as much as you through hand calculation. 10

b

Clearly describe each of the following characteristics of a neural network

i) Layered
ii) Feed forward
iii) Completely connected 06

Database Management Systems

VII Semester SEE Question Papers

Course codes: 21IM74HD

- Mar/Apr 2025 (21IM74HD)

21IM74HD

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Regular Examinations Mar/April-2025

Industrial Engineering and Management

DATABASE MANAGEMENT SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Define the following terms: data, database, DBMS, database system.	02	1	1
	1.2	Differentiate between Data Definition and Data Manipulation language.	02	4	1
	1.3	What are the different types of database end users?	02	2	1
	1.4	Justify the need for defining Database constraints in DBMS.	02	2	2
	1.5	Mention the features of Foreign Key.	02	2	2
	1.6	Consider the following schema of relations: $Student\ (ssn, name, address, major)$ $Course\ (code, title)$ $Registered\ (ssn, code)$ Write an appropriate query for listing the codes of courses in which at least one student is registered (registered courses).	02	3	4
	1.7	What is logical data independence, and how does it benefit database design?	02	2	2
	1.8	Mention the guidelines for physical database design.	02	2	3
	1.9	Distinguish between the base relation and virtual relation in SQL environment.	02	4	4
	1.10	"Insertion, deletion and modification anomalies are considered bad in database design". Justify	02	2	3

PART-B

2	a	Consider an online retail platform like Flipkart in India. Explain how the database approach ensures data integrity, data security, and supports multiple users in browsing products, placing orders, and updating their information simultaneously. Use this case to illustrate the advantages of the database approach in the Indian context.	08	3	1
	b	Identify the actors on the scene and the workers behind the scene for a healthcare management system in a hospital. Explain their roles and how they contribute to the smooth functioning of the system with examples.	08	2	1
3	a	Discuss the role of high-level conceptual data models in designing a database. Explain how abstraction helps in simplifying complex data structures and its impact on the overall database development process.	10	2	2

	b	Differentiate between data models, schemas, and instances. Provide examples to explain the relationships between these concepts in a database system.	06	4	2
		OR			
4	a	Draw an ER diagram for a small university database, including entities such as students, courses, and instructors, and explain your design.	10	3	2
	b	Elaborate on the three-schema architecture of a $DBMS$ and explain how it helps achieve physical and logical data independence.	06	2	2
5	a	"Whenever update operations are applied on the relations, it is necessary that the integrity constraints are not violated". Evaluate the statement with relevant examples.	08	4	3
	b	From the relations R and S, find the following: i) $R \cup S$ ii) $R \cap S$ iii) $R - S$ iv) $S - R$ R: $Sl.No$, $DEPT$ — $S1$, $PHYS1$; $S2$, PSY; $S3$, $CHEM$; $S4$, $JOUR$ S: $Sl.No$, $DEPT$ — $S10$, $MATH1$; $S3$, $CHEM$; $S15$, ENG; $S16$, $MATH2$	08	2	3
		OR			
6	a	Critically analyze the 12 rules advocate by E.F.Codd for making the database truly relational.	10	2	3
	b	Explain the three types of $UPDATE$ operations involved in relational models.	06	2	3
7	a	Consider the following relation, $BOOK$. $BOOK$: $Acc-No$, $Yr-Pub$, $Title$ — 734216, 1982, *Supply chain management*; 237235, 1995, *Database design*; 631523, 1992, *Machine design*; 543211, 1991, *Quality management*; 376112, 1992, *Personnel management* Write the syntax for applying the following queries on the relation, $BOOK$. i) Select from the relation, $BOOK$ all the books whose year of publication is 1992. ii) Select from the relation, $BOOK$ all the books whose Acc-No is greater than or equal to 56782. iii) List all the Title and Acc-no of the $BOOK$ relation. iv) List all the Title and Yr-Pub of the $BOOK$ relation.	08	3	4
	b	Give an overview of different types of $JOINS$ available in SQL	08	2	4
		OR			

8	a	Consider an $EMPLOYEE$ table and populate it with attributes such as: Name, Department, Salary and Job. Upon populating the table write the SQL queries for the following: i) Display all the employee details. ii) Display employee's name and new salary after the increment of 1000 iii) Select all columns of EMP table iv) Display employee names, department, and annual salary. v) List down unique departments from EMP table.	10	3	4
	b	Consider the following schema: $Suppliers\ (sid : integer, sname : string, address : string)$ $Parts\ (pid : integer, pname : string, color : string)$ $Catalog\ (sid : integer, pid : integer, cost : real)$ i) Find the name of suppliers who supply some red parts. ii) Find the sid's of suppliers who supply every part.	06	3	4
9	a	Why is Normalization necessary while designing the relation data base? Analyze various levels in Normalization process.	10	2	3
	b	Evaluate the guidelines for designing relational systems and propose enhancements for a banking database.	06	4	3
		OR			
10	a	Describe the concept of functional dependencies and demonstrate how they are used in the normalization process.	08	2	3
	b	Illustrate the steps involved in converting a relation into 3NF with and example.	08	2	3

Facilities Planning, Design and Ergonomics

VII Semester SEE Question Papers

Course codes: 21IM74HA, 12IM71

- Mar/Apr 2025 (21IM74HA)
- Nov/Dec 2018 (12IM71)
- Nov/Dec 2017 (12IM71)
- Nov/Dec 2016 (12IM71)

21IM74HA

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Regular Examinations Mar/April-2025

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Facilities planning is the process of determining how an organization's assets are used to achieve its _____.	01	1	3
	1.2	Given a warehouse layout, identify potential bottlenecks in material flow and suggest improvements to operations.	01	3	4
	1.3	Material handling is the movement, storage, protection and control of materials throughout the process of _____.	01	1	1
	1.4	The RAFT in CRAFT algorithm stands for computerized _____ algorithm for facilities planning.	01	1	1
	1.5	In a U-shaped plant layout, material flow follows a _____ path.	01	2	2
	1.6	JIT manufacturing aims to minimize inventory by producing only what is _____.	01	1	1
	1.7	When to use CRAFT over ALDEP in layout design?	01	2	3
	1.8	The main objective of a warehouse is to facilitate storage, retrieval and _____ of goods.	01	1	1
	1.9	Unit load principle in material handling means moving materials in _____units rather than individually.	01	1	1
	1.10	Fixed automation systems are best suited for _____ production.	01	2	2
	1.11	What are main objectives of facilities planning?	01	1	1
	1.12	Name any two factors influencing plant location.	01	2	2
	1.13	Define Just-In-Time (JIT) Manufacturing.	01	2	1
	1.14	What is the difference between flexible and fixed automation?	01	2	2
	1.15	Analyze the impact of poor plant layout design on manufacturing efficiency and suggest modifications to improve productivity.	01	3	4
	1.16	What is the purpose of a dock in warehouse operations?	01	2	2
	1.17	Mention any two principles of plant layout.	01	1	1
	1.18	What does CRAFT algorithm help in optimizing?	01	2	2
	1.19	List any two advantages of a well-planned material handling system.	01	1	1
	1.20	Name any two industries where a straight-line material flow is preferred.	01	1	1

PART-B

2	a	Enumerate the ten steps in facilities planning process. Provide a summary of the process.	08	2	1
	b	Explain the characteristic features of Group layout. List the advantages and disadvantages.	08	2	1

3	a	How do you develop alternative material handling system designs using the material handling system equation? Discuss.	08	2	1
	b	Discuss the significance of "Nine Rights" of material handling in ensuring an efficient and effective material flow system. Provide examples of how these principles can be applied in a manufacturing or logistics environment.	08	2	1
		OR			
4	a	Illustrate with an example the effects of unit load size on job completion time and discuss.	08	3	2
	b	Given the following dimensions of a particular type of plastic reusable containers Inside dimensions 16" 10"10" Outside dimensions 18" 11" 11" Each nested container 18" 11 " 2" A trailer with inside dimensions of 240" 120" 120" is used to transport these containers. The containers are not palletized. Assume that no clearance is needed between containers or between containers and the walls of the trailer. The dimension of the storage opening is 24" 16" 14". Determine the following: i) Container space utilization ii) Storage space efficiency. iii) Container nesting ratio. iv) Trailer space utilization if all containers are stacked in only one orientation. v) Trailer return ratio.	08	4	3
5	a	A manufacturing facility consist of five departments (A, B, C, D, E) arranged in 2 x 3 grid layout as follows: Given Layout A | B | C D | E A B C D E A 0 50 30 20 10 B 50 0 40 10 30 C 30 40 0 60 20 D 20 10 60 0 50 E 10 30 20 50 0 Interdepartmental Material Flow Matric (Units per Day) Area Requirements (in Square Meters) Department A: 60 m^2; Department B: 50 m^2; Department C: 40 m^2 Department D: 70 m^2 Department E: 30 m^2 Assumptions and Cost Factors: • Each cell in the layout is 10m x 10m. • Rectilinear distance is used between department centroids. • Material handling cost = Rs.5 per unit per meter. Answer the following questions: i) Compute the initial total material handling cost based on the given layout. ii) Identify possible department swaps using the CRAFT algorithm to minimize transportation cost. iii) Calculate the total cost for the improved layout and compare it with the initial cost. iv) Minimum one iteration is required.	16	4	4

		OR			
6	a	Examine the importance of each function of a warehouse in ensuring an efficient warehousing operation. Provide real world examples of how these functions contribute to overall supply chain effectiveness.	08	4	4
	b	Compare and contrast the impact of cross-docking, direct put away and load decomposition on warehouse efficiency. Provide real-world examples to support your analysis.	08	3	3
7	a	Examine the key factors that are considered while designing a material flow system aimed at optimizing the movement of materials through the production process.	08	3	3
	b	Evaluate the key features, design considerations and applications of straight line flow layout. Provide examples of such layout.	08	2	2
		OR			
8	a	ABC Tech Solutions is a mid-sized IT and IT-enabled Services (ITES) company providing software development, customer support and data processing services. As the company expanded inefficiencies in its existing office layout led to : • Communication delays between teams. • Underutilization of workspace. • Employee discomfort due to poor ergonomic design Challenges in the Existing Layout • Fragmented Team Structure: Software developers, support staff and data analyst were scattered across different floors, leading to poor coordination. • Limited Collaboration Spaces: The lack of dedicated meeting rooms and breakout zones hindered brainstorming and innovation. • Disruptive Noise Levels: Open workstations created distractions, especially for teams requiring focus. • Inefficient Workflows: Employees had to walk long distances between departments, slowing down processes. • Employee Fatigue and Ergonomic Issues: Poorly designed desks and chairs led to discomfort and reduced efficiency. Propose a layout to enhance productivity and collaboration.	16	4	4
9	a	Analyze the advantages and limitations of dial indexing machines and other production systems that exhibit transfer line characteristics. Compare their effectiveness in different industries such as automotive assembly, chemical processing and steel fabrication. Discuss how automation and material handling impact efficiency in these systems.	08	3	3
	b	Evaluate the suitability of flexible manufacturing systems (FMS) for different manufacturing environments. Provide examples of industries where FMS implementation can significantly enhance efficiency and productivity.	08	4	3
		OR			
10	a	Discuss how principles such as “handling less is best” and “eliminate, combine and simplify” contribute to lean manufacturing.	08	2	4
	b	Analyze the seven types of waste identified by Toyota in the context of lean manufacturing applied to material handling.	08	3	4

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

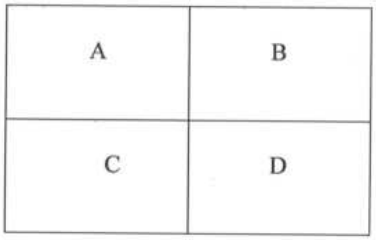
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical table is permitted.

PART-A

1	1.1	In _____ method offline balancing, weightages are assigned to the element's location.	01
	1.2	In $ARC,'E'$ relationship refers to _____.	01
	1.3	Sweep width is typically considered in _____algorithm.	01
	1.4	In _____ layout, machines are arranged in the sequence of operations to be performed.	01
	1.5	_____ distance is measured along paths that are orthogonal.	01
	1.6	What is a serpentine path?	01
	1.7	What are the conclusions that can be drawn from a break-even chart?	02
	1.8	In a layout, both horizontal floor space and vertical space available should be utilized. Justify.	02
	1.9	With suitable examples, classify the different types of plant layouts.	02
	1.10	State the ergonomic principle of material handling with a suitable example.	02
	1.11	What are the attributes of an AGV?	02
	1.12	Mention the difference between travel chart and mileage chart.	02
	1.13	How is plot plan helpful in the process of facilities planning and design?	02

PART-B

2	a	Identify and explain the four step process for carrying out strategic facility planning process.	08
	b	Evaluate the primary objectives of facilities planning.	08
		OR	
3	a	If you are assigned with the task of designing a new engineering college, describe the procedure you would follow to develop the facilities plan.	08
	b	With reference to facilities planning and design, identify any four situations where inadequate planning can be observed in your campus. Use your knowledge of facilities planning and design to technically record the observations.	08

4	a	Explain in detail the procedure adopted for carrying out the analysis related to flow pattern design.	16
		OR	
5	a	Use the activity relationship chart for analyzing activities in your canteen. List significant assumptions that you have made.	10
	b	Discuss with an example the equipment issues and setup time involved in the reduction of manufacturing lead time through facilities design.	06
6	a	Broadly outline the classifications and types of material handling equipments.	08
	b	For each of the following material; handling principles (x) explain what it means and (xi) given a specific example of how it can be applied. i) Standardization principle. ii) Ergonomic principle. iii) Life cycle cost principle. iv) Work principle.	08
		OR	
7	a	With a suitable block diagram and suitable examples construct a material handling system equation. How this is used in the development of alternative material handling system designs.	08
	b	With reference to the facilities planning, discuss the safety considerations which are to be incorporated in a typical crackers manufacturing plant.	08
8		Given the existing layout and flow matrix, assume the cost to move the load/unit distance to be $Rs.10/-$. Carryout 2 iterations using $CRAFT$ algorithm and suggest the minimum cost layout.  Side of each cell : 20 X 20 Flow matrix	16
		OR	
9	a	With a typical block diagram, explain Muther's Systematic layout planning process.	08
	b	With a suitable example, explain the $CORELAP$ algorithm.	08

	A	B	C	D
A	-	8	5	7
B	9	-	8	4
C	6	4	-	8
D	4	4	1	-

10	a	Bring out the difference between conventional and automated storage models.	08
	b	Discuss the advantages of simulation models.	08
		OR	
11	a	Discuss in detail the recent trends in facilities planning that are adopted in factories in order to provide smooth flow of materials, improve quality, increase productivity and optimal space utilization.	08
	b	Is the concept of "automatic factory" still valid in today's manufacturing environment? What are the advantages and disadvantages of operating an automated warehouse?	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical table is permitted.

PART-A

1	1.1	List any four issues that may have a long range impact on the strategic facilities plan.	02
	1.2	Differentiate between the facilities design and facilities location.	02
	1.3	A machined part has a standard machining time of 2.8 min per part on a milling machine. During an $8-h$ shift 200 units are to be produced. Of the 480 min available for production, the milling machine will be operating 80 % of the time. During the time machine is operational, part produced at a rate equal to 95 % of standard time. How many machines are required?	02
	1.4	Write the principle of minimizing the cost of flow.	02
	1.5	State the material handling system equation.	02
	1.6	Define stackability.	02
	1.7	Expand "$LOGIC$" and "$MULTIPLE$" with respect to layout design algorithms.	02
	1.8	Write any two disadvantages of group layout.	02
	1.9	List the most important design consideration in Automatic Storage and Retrieval Systems ($AS\&RS$).	02
	1.10	The distance measured along a straight line path between two points in a facility location problem is _____ distance.	01
	1.11	State the minimax location problem.	01

PART-B

2	a	Discuss a novel approach to contemporary facility planning in the winning facility planning process.	08
	b	List the objectives of facilities planning and explain any two of them.	08
		OR	
3	a	Discuss the facilities planning process in the context of facility life cycle.	08
	b	Analyze the significance of facilities planning typically on new facilities between 1955 and today.	08

4	a	Elaborate on the seven management and planning tools for facility design.	08
	b	Classify the facility layout designs on volume-variety of the product.	08
		OR	
5	a	Give the application of cellular manufacturing facility for a typical component involving primary and secondary machining processes.	08
	b	Explain the concepts of material flow system for fixed material location planning and process planning department.	08
6	a	List the major material handling principles and elaborate on any three of these principles with suitable examples.	08
	b	List the various types of pallets and explain with neat sketches any two types of pallets with proper proportionate dimensions.	08
		OR	
7	a	Explain the classification of material handling equipment for storage and retrieval equipment.	08
	b	Illustrate the container/pallet flow in a conventional system and pooling systems.	08
8	a	Elaborate the steps of the $CRAFT$ algorithm with suitable examples.	08
	b	Discuss the basic layout types of planning department.	08
		OR	
9	a	With the help of an example explain the Muther's systematic layout planning (SLP) procedure.	08
	b	Write a note on algorithmic approaches to layout planning.	08
10	a	Discuss any two facility location model with suitable examples.	08
	b	Discuss the steps of an aisle order picking system for the example of grocery shopping.	08
		OR	
11	a	With suitable example discuss the Rectilinear facility location problems.	08
	b	Discuss the three principles to be used while designing closed loop irreversible conveyors system.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

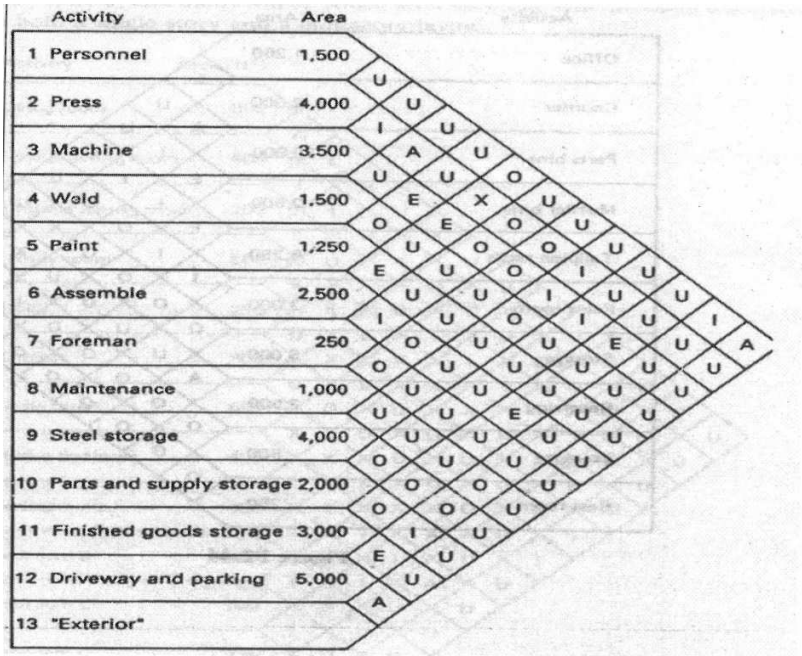
6. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
7. Answer FIVE full questions from Part B.
8. Use of statistical table is permitted.

PART-A

1	1.1	The art and science of employing the _____ of a firm to achieve _____ objectives is Strategy Planning.	02
	1.2	Facilities _____ and facilities _____ problems are two parts of facilities planning.	02
	1.3	In determining the requirements of a facility the three important considerations are _____, _____ and _____.	02
	1.4	The Material Handling System equation is given by _____ + _____ + _____ = Recommended System.	02
	1.5	Layout algorithms are classified on the basis of Objective functions such as _____ objective and _____ objective.	02
	1.6	The fundamental concepts behind Simulated Annealing are based on the analogy between _____ Mechanics and _____ optimization.	02
	1.7	_____ and _____ are the layout algorithms developed for Multi-floor facility layout problems.	02
	1.8	Flexible Manufacturing Systems accommodate changes in _____ and _____.	02
	1.9	The distance measures involved in a facility location problem are _____, _____ and _____.	02
	1.10	QAP stands for _____, _____, _____.	02

PART-B

2	a	Enlist and explain facilities planning process.	08
	b	Mention the objectives of Facilities Planning.	08
		OR	
3	a	How the facilities planning strategies are developed?	08
	b	Mention any two situations of inadequate planning.	08

4	a	How are layouts classified on the basis of Production Volume and Production Variety?	08
	b	The activity relationship chart for Larry's Machine Shop is shown below. Construct *REL* diagram and a Space Relationship diagram. 	08
		OR	
5	a	Describe with neat sketches different pattern of flow: i) Within the workstations ii) Within the departments iii) Between the departments.	10
	b	Explain: i) Cellular Manufacturing and ii) Parts Flow Analysis.	06
6	a	How are Material Handling Systems classified?	06
	b	Explain the concept of Unit Load Design with three examples.	04
	c	List and explain any six principles of Material Handling.	06
		OR	
7	a	Describe the step by step procedure of designing *MH* systems.	10
	b	Briefly explain the problems faced while estimating *MH* costs.	06
8	a	Enlist the objectives of Facilities design function.	06
	b	Explain the salient features of any two of the following layout procedures. i) Nadler's ideal system's approach ii) Immer's basic steps iii) Apple's plant layout procedure.	10
		OR	

9	a	Explain with a line diagram Muther's Systematic layout planning.	10
	b	Elucidate the different algorithmic approaches and its classification to layout problems.	06
10	a	What are the factors to be considered while planning storage requirements?	08
	b	Explain: i) Block stacking ii) $ASRS$.	08
		OR	
11	a	Explain: i) Quadratic Assignment Problem ii) Warehouse Layout Model.	08
	b	Separate electrical cables are to be run from the generator to each machine in a facilities layout. The location of the machines are at $P_1 = (0,0)$, $P_2 = (30,90)$, $P_3 = (60,20)$, $P_4 = (20,80)$, $P_5 = (70,70)$, $P_6 = (90,40)$. Determine the location for the generator that will minimize total required length of the cable assuming rectilinear distance.	08

Innovation and Entrepreneurship

VII Semester SEE Question Papers

Course codes: 21IM78

- Mar/Apr 2025 (21IM78)

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular Examinations Mar/April-2025

Industrial Engineering and Management

INNOVATION AND ENTREPRENEURSHIP

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer THREE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6.

		PART-A	**M**	**BT**	**CO**
1	1.1	Define Nine Windows.	02	1	1
	1.2	When should you use affinity mapping?	02	2	2
	1.3	What's the difference between market segmentation and target markets?	02	1	3
	1.4	List any four traits and characteristics that makes a business model lean.	02	2	4
	1.5	From the perspective of Minimum Viable Product (MPV), what is the primary goal of prototyping a solution?	02	1	4

PART-B

2	a	Explain the concept of Service –Dominant Logic (SDL) and how it differs from the traditional Goods-Dominant Logic (GDL)	08	2	1
	b	Define generic parts technique. Discuss how this technique work & can be used in Entrepreneurship.	08	5	1
3	a	How to use value proposition canvas for a perfect customer profile.	06	3	2
	b	Explain the four types of market segmentation. How does Coco-Cola use the market segmentation?	06	3	2
		OR			
4	a	List and explain seven steps to follow to run a great competitor analysis.	06	2	3
	b	Define Competitive advantage. Explain in detail the key point in identifying your competitive advantage with examples.	06	3	3
5	a	Which strategy is going to give your company an uncontrolled market space separate from other similar business, justify.	06	2	4
	b	Define sales plan. How to make it one step at a time to create a sales plan.	06	3	4
		OR			
6	a	A marketing channel is a medium used to advertise your company. Digital marketing channels include various modes of advertisement where each channel has its own unique mode and purposes. Discuss.	06	3	4

b	Robert has recently launched online jewelry portal. The customer response has been overwhelming, and Robert needs more hands to share his load. He has shortlisted two candidates by name Jimmy and John. While Jimmy is an expert at managing portals, he has little regard for anything other than web-designing. John has a fair knowledge of portals but isn't an expert. He shows great enthusiasm for Robert's innovative designs and even shares some ideas on how to take his venture to the next level. i) Who should Robert choose? ii) Also, Emphasize with Robert to suggest suitable roles that he should give to either Jimmy or John depending on the type of cost.(Hint: Roles as in intern, Freelance, Permanent employee, Contractual workers)	06	3	4

Supply Chain Analytics

VII Semester SEE Question Papers

Course codes: 21IM75IN

- Mar/Apr 2025 (21IM75IN) - Set 1 [DSpace: Mar-Apr 2025]
- Mar/Apr 2025 (21IM75IN) - Set 2 [DSpace: Mar-Apr 2025]

21IM75IN

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular Examinations Mar/April-2025

SUPPLY CHAIN ANALYTICS (INSTITUTIONAL ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	What is the primary goal of Supply Chain Analytics (SCA) in optimizing supply chain processes?	02	2	2
	1.2	Enumerate any four key data sources commonly used in supply chains.	02	1	3
	1.3	How does the use of internet of Things (IoT) technologies improve supply chain visibility and decision-making?	02	2	3
	1.4	What is the purpose of data cleaning in Data Manipulation?	02	2	3
	1.5	A company operates with an average lead time of 4 days and a daily average demand of 100 units. The calculated safety stock is 476 units. Calculate the Reorder Point (ROP).	02	3	4
	1.6	In python, what is the purpose of working with Text and Datetime Data?	02	3	1
	1.7	Identify any four important factors to be considered for warehouse layout design.	02	1	4
	1.8	Differentiate between inbound and outbound logistic activities.	02	4	4
	1.9	What is product value density?	02	2	4
	1.10	State the factors that impact location decisions in a supply chain(any four).	02	2	4

PART-B

2	a	Discuss the potential applications of Supply Chain Analytics (SCA) across various stages of a supply chain and explain the importance of $SMART$ objectives in driving its effectiveness.	08	3	1
	b	Demonstrate the potential application of blockchain technology along a specific supply chain in the sector of your choice. What can be the challenges for its applications?	08	2	1
3	a	Illustrate different techniques for handling missing values in a dataset. How would you decide which method to use in a given scenario?	10	2	2
	b	Discuss the Four-step process for data manipulation.	06	2	3
		OR			
4	a	Compare and contrast the various methods for aggregating GroupBy objects. Evaluate the advantages and potential limitations of each method in different data analysis contexts.	10	4	4

	b	How does effective Data Cleaning and Preparation improve the quality of supply chain decision-making?	06	2	4
5	a	Outline the steps involved in conducting a cohort analysis. Choose a specific business and design custom cohort types, determine the cohort size, select appropriate metrics, and define a data range. Provide reasoning for your choices.	08	3	2
	b	What types of questions does RFM analysis help answer, and how are the Recency (R), Frequency (F), and Monetary (M) values calculated?	08	2	2
		OR			
6	a	Choose a manufacturing firm and a service provider, and compare and contrast which one has a higher degree of vertical integration, and which one has more outsourcing? Discuss the pros and cons of vertical integration and outsourcing with your chosen companies.	10	3	2
	b	Managing relationship in a multi-tier supply network can be very complex and challenging. Discuss the various relationship types in a network and approaches to managing them.	06	2	2
7	a	Discuss the role of a Warehouse Management System (WMS) in optimizing warehouse operations. Highlight its key benefits with examples. Additionally, identify and explain five Key Performance Indicators ($KPIs$) commonly used to measure warehouse performance and their significance in improving operational efficiency.	10	2	3
	b	A company operates with an average daily demand $D_{avg}$of 100 units and an average lead time $LT_{avg}$of 4 days. The standard deviation of daily demand σ_{LT} is 2 days. The company aims to achieve a 95% service level. Using the given data, calculate the safety stock required to meet the target service level. Z value is 1.96.	06	3	3
		OR			
8	a	Identify and describe the various ways of evaluating forecast accuracy.	06	2	3
	b	Analyze the various demand forecasting methods by categorizing them into qualitative and quantitative approaches. Examine the key differences between these two types of methods and assess how they impact decision –making in different business scenarios.	10	3	3
9	a	Compare and contrast the different modes of transport with suitable examples.	10	4	2
	b	Explain the concepts of 3 PL and 4PL in supply chain management with suitable examples.	06	2	2
		OR			
10	a	Explain the meaning and significance of DPU, CIP, FOB and DDP in the context of the 11 Incoterms.	08	2	2
	b	What are the benefit and challenges of implementing Route Optimization techniques in logistics management?	08	2	3

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular Examinations Mar/April-2025

Industrial Engineering and Management

SUPPLY CHAIN ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	**M**	**BT**	**CO**
1	1.1				
	1.2				
	1.3				
	1.4				
	1.5				
	1.6				
	1.7				
	1.8				
	1.9				
	1.10				
		PART-B			
		Unit-I			
2	a				
	b				
		Unit-II			
3	a				
	b				
		OR			
4	a				
	b				
		Unit-III			
5	a				
	b				
		OR			
6	a				
	b				
		Unit-IV			

7	a				
	b				
		OR			
8	a				
2	b				
		Unit-V			
9	a				
	b				
		OR			
10	a				
	b				

Virtual Commerce

VII Semester SEE Question Papers

Course codes: 21IM73GC

- Mar/Apr 2025 (21IM73GC)

21IM73GC

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular Examinations Mar/April-2025

Industrial Engineering and Management

VIRTUAL COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	How does Virtual Reality (VR) enhance the V-Commerce experience?	02	1	1
	1.2	Virtual commerce (v-commerce) uses technologies like _____ and _____ to provide immersive shopping experiences.	02	2	1
	1.3	What is meant by a protocol in an internet architecture context?	02	1	1
	1.4	Mention any two features of B2G e-commerce model.	02	2	2
	1.5	Virtual commerce improves customer engagement by providing a more _____ and _____ experience than traditional online shopping?	02	2	2
	1.6	How does V-commerce differ from Traditional E-commerce?	02	1	2
	1.7	What is the significance of Metaverse in Virtual Commerce?	02	2	2
	1.8	What is the impact of fiber optics in the First mile on the overall performance of internet services, considering bandwidth and scalability.	02	2	2
	1.9	How would you apply guaranteed service levels to ensure low error rates in an IoT network with critical data requirements?	02	1	3
	1.10	What is internet economy?	02	2	3

PART-B

2	a	Examine the role of Virtual Reality (VR) and Augmented Reality (AR) in enhancing the shopping experience in Virtual Commerce. Provide examples of companies successfully implementing these technologies.	10	3	2
	b	Distinguish between B2B (Business-to-Business) and B2C (Business-to-Consumer) models with a focus on serviceability, control, and order completion cycle.	06	3	2
3	a	Analyze the impact of Artificial Intelligence (AI) on Virtual Commerce, focusing on personalized marketing, customer service, and inventory management. How has AI changed the way companies interact with consumers online?	10	3	2
	b	What is HTML? Discuss its role and importance in e-commerce applications, focusing on how it helps in creating user –friendly websites, structuring content, and enabling seamless customer interactions.	06	4	3
		OR			

4	a	Explain the role of Virtual Payment Systems in Virtual Commerce. Discuss the types of virtual payment methods available and their advantages and disadvantages for both business and consumers.	10	2	1
	b	Discuss aspect such as cost, geographical and customer experience in virtual commerce providing relevant examples to illustrate your points.	06	3	2
5	a	Discuss the advantages of integrating V-commerce into a multichannel marketing strategy. How do the key driving factors of B2C online selling impact the Indian market, particularly when targeting rural customers?	10	3	3
	b	Assess the advantages and disadvantages of using Virtual Commerce over M-Commerce for a retail business.	06	3	3
		OR			
6	a	Analyze the impact of influencer trust and credibility on consumer purchasing decision in virtual commerce.	10	3	4
	b	Assess the future potential of V-Commerce in comparison to M-commerce in the Indian market.	06	3	3
7	a	Discuss the role of live-stream shopping in enhancing customer engagement and sales conversion in social commerce.	10	4	3
	b	Develop a Caselet on a company successfully utilizing all three: V-Commerce, E-commerce, and M-Commerce	06	3	2
		OR			
8	a	Discuss the benefits of Internet-based marketing.	08	4	3
	b	Mention the design criteria for selecting Web-enabled supply channel?	08	3	4
9	a	For social commerce to be successful, the distribution channel needs to be redesigned. Once customers get used to the benefits of buying through the Internet, they become a new kind of customer. This new customer expects, if not demands, highly personalized products, personalized services, and immediate delivery. Why do firms need to understand consumer behavior?	08	3	3
	b	Given the increasing deployment of IoT devices in smart homes and industrial setting, apply your knowledge of guaranteed service levels and lower error rates to design an IoT –based solution that ensures optimal performance and reliability. How would you implement a system that balance low error rates with maintaining the cost-effectiveness of the solution, considering the declining costs of IoT devices?	08	3	3
		OR			
10	a	Given increase demand for high speed internet access in remote (i) underdeveloped areas Apply your knowledge of written campus network to design a solution that leverage advanced Internet access for providing connectivity	10	3	3
	b	How Social Commerce evolved overtime? What are the advantages & pitfalls?	06	4	4

Foundations of Robotics and Automation

VII Semester SEE Question Papers

Course codes: 18IM73

- Feb/Mar 2024 (18IM73)
- Feb/Mar 2023 (18IM73)

18IM73

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Regular / Supplementary Examinations Feb/Mar-2024

Industrial Engineering and Management

FOUNDATIONS OF ROBOTICS AND AUTOMATION

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	_____ is a technology concerned with the application of mechanical, electronic and computer-based systems to operate and control production.	01
	1.2	The information-processing cycle, represented by the outer ring, can be described as consisting of four functions, they are Business functions, Product design, Manufacturing planning and _____.	01
	1.3	_____ is the detailed planning process that tracks manufacturing output and matches this against customer orders that have been placed.	01
	1.4	Intermittent transfer is also called as _____.	01
	1.5	Physical structure of robot which moves around is called _____.	01
	1.6	*SCARA* robot is very suitable in which kind of operations _____.	01
	1.7	Radial movement in and out to the manipulator arm is powered by _____.	01
	1.8	Proximity sensors are also called as _____ sensors.	01
	1.9	_____ is a collection of mechanical linkage connected by joints.	01
	1.10	_____ is the name for space inside which a robot unit operates.	01
	1.11	*ARCL* stands for _____.	01
	1.12	The _____ is equipped with a toggle switch or a pair of contact buttons for each joint.	01
	1.13	Define first law of Robotics.	02
	1.14	What is Mechanization?	02
	1.15	List the uses of grippers.	02
	1.16	List the two types of Suction grippers.	02

PART-B

2	a	What are the types of automation that can be used in a production system? Compare them for their features and drawbacks.	10
	b	Identify major socio-economic considerations favoring automation.	06
3	a	Discuss the various types of conveyors used in automated material handling. Explain any two in detail.	10

	b	What are automated assembly lines? What are the reasons for using automated assembly lines?	06
		OR	
4	a	Identify the various types of transfer devices. Explain the working of any one type of transfer device with a sketch.	08
	b	Differentiate between volumetric and gravimetric type feeders.	08
5	a	Categorize the motivating factors of Robotics.	06
	b	What is work envelope of a robot? Sketch and explain two views to indicate the wok envelope of a cylindrical robot.	10
		OR	
6	a	Classify robots on the basis of path movement.	06
	b	Explain *SCARA* robot indicating its joints and degree of freedom with the applications.	10
7	a	Explain the working principle of an end effector with the help of a sketch and give its important applications.	08
	b	Differentiate between contact and noncontact sensors in robots with the help of examples.	08
8	a	Explain the use of teach pendant for robot programming.	08
	b	Discuss the two types of lead through programming technique used in robot programming.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Feb/Mar 2023

Industrial Engineering and Management

FOUNDATIONS OF ROBOTICS AND AUTOMATION

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Define a robot.	02
	1.2	What is meant by automation?	02
	1.3	Identify any 2 rules of robot.	02
	1.4	Identify any four configuration systems.	02
	1.5	Identify any 2 types of automation.	02
	1.6	List any four benefits of industrial automation.	02
	1.7	Distinguish between accuracy and precision of a robot.	02
	1.8	Identify any 4 sub-systems of an Industrial Robot.	02
	1.9	List any two issues in implementing Low Cost Automation in an Industry.	02
	1.10	List any two applications of Vibratory Feeders.	02

PART-B

2	a	List and explain automation strategies that are employed to improve productivity in manufacturing operations.	08
	b	Discuss the advantages of automation in a typical manufacturing industry.	08
3	a	Explain the types of assembly lines. Also, state two reasons for choosing assembly line.	08
	b	Discuss the four types of transfer systems in assembly lines with an example for each.	08
		OR	
4	a	Discuss any three types of feeders with emphasis on their advantages.	06
	b	Explain the advantages and disadvantages of transfer machines.	06
	c	Write short note on classification of feeders.	04

5	a	Consider a manager as a decision maker in a manufacturing industry. Explain the motivating factors from the manager point of view to consider for choosing a robot for his industrial setup.	06
	b	From an application point of view, considering farmer as a "User" belonging to agriculture sector, provide your understanding of what kind of robots can be recommended. Also state the purpose of using such robots.	10
		OR	
6	a	Explain any three advantages and disadvantages of robot.	08
	b	List and explain any ten robot terminologies common to vast majority of industrial robots.	08
7	a	Explain the functionality and application of the following sensors. i. Touch sensor ii. Proximity sensor iii. Range sensor iv. Velocity sensor.	08
	b	Explain the advantages and disadvantages of vacuum/suction grippers.	08
8	a	Explain any two robot programming techniques.	08
	b	List and explain any two types of robot languages.	08

E-Commerce

VII Semester SEE Question Papers

Course codes: 18IM7F5

- Fast Track Nov 2023 (18IM7F5)
- Feb/Mar 2023 (18IM7F5)

18IM7F5

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Fast Track Examinations Nov-2023

Industrial Engineering and Management

E-COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	*B2B* is a type of Ecommerce that refers to transactions that occur as _____.	01
	1.2	eBay, Amazon.com belong _____ segment.	01
	1.3	The process in which a buyer posts its interest in buying a certain quantity of items, and sellers compete for the business by submitting successively lower bids until there is only one seller left _____.	01
	1.4	The study of encryption is called _____.	01
	1.5	The underlying computing equipment that the system uses to achieve its e-commerce functionality is called a _____.	01
	1.6	The set of standards for communication between a browser and a program running on a server that allows for interaction between the user and the server is known as _____.	01
	1.7	e-government arena is being used when governmental agencies send out and accept bids for work is _____.	01
	1.8	Web server functionality allows it to verify usernames and passwords and process certificates and encryption information is _____.	01
	1.9	The ability that determines when your IT system will be available foe knowledge workers to access is _____.	01
	1.10	Unique value auction is mainly applied to _____ products.	01
	1.11	Redundant navigation refers to _____.	01
	1.12	The set of applications and technologies that allow users to create, edit and distribute content online is known as _____.	01
	1.13	Creating multiple versions of information goods and selling essentially the same product to different market segments at different prices is called _____.	01
	1.14	Companies that choose to operate solely in the traditional physical markets are described as _____.	01
	1.15	In the term m-commerce, the “m” refers to _____.	01
	1.16	Large private organizational buyers and government agencies make large-volume or large-value purchases through _____, also known as reverse auctions.	01

	1.17	The combination of the Internet and Web technologies has given rise to a global platform where firms across the world can compete for customers and gain access to new.	01
	1.18	Which products are people most likely to be more uncomfortable buying on the internet _____.	01
	1.19	The component of TCP/IP that provides the internet's addressing scheme and is responsible for the actual delivery of packets is _____.	01
	1.20	Which protocol permits users to transfer files from the server to their client computer, and vice versa?	01

PART-B

2	a	Explain differences between $B2B$ and $B2C$ in terms of serviceability, control and order completion cycle.	10
	b	What are different business models available for e-commerce?	06
3	a	Explain the need to have frames in e-commerce web pages.	06
	b	Discuss the Internet is based on a layered model called transmission control protocol/internet protocol (TCP/IP).	10
		OR	
4	a	How do you transfer data from client to the server and fetch it back to the end user?	10
	b	Differentiate between IPv4 and IPv6.	06
5	a	What are the differences between e-commerce and e-business?	10
	b	What is electronic payment system and how does it vary from traditional payment system?	06
		OR	
6	a	What is m-commerce and how is it related to e-commerce?	10
	b	What is e-tailing? And what are its prospects?	06
7	a	Discuss a product that has differential pricing for E-commerce and for traditional channel.	10
	b	Discuss, if in future, e-commerce can make "build-to-order" cars in India.	06
8	a	What is the concept behind socio-commerce? How is it different from vanilla e-commerce business model?	10
	b	What are the precautions that a marketer should take while using socio-commerce channel?	06

18IM7F5

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Feb/Mar-2023

Industrial Engineering and Management

E-COMMERCE (Elective)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	List the factors that impacted E-commerce.	02
	1.2	Write the characteristics of Web 2.0 with an example.	02
	1.3	Write the four layers of the link.	02
	1.4	What are the web components in E-commerce?	02
	1.5	What are the three primary processes that are enhanced in e-business?	02
	1.6	Elicit the issues with adoption of E-commerce by entrepreneurs.	02
	1.7	Illustrate the E-commerce evolution for entrepreneurs.	02
	1.8	What is the importance of E-commerce in multichannel marketing?	02
	1.9	List the latest trends and the challenges of social commerce.	02
	1.10	Highlight the major driving factors of B2C online-selling.	02

PART-B

2	a	Explain differences between B2B and B2C in terms of serviceability, control and order completion cycle.	10
	b	Illustrate with examples, how the customers are benefitted from E-commerce.	06
3	a	Explain the features of WEB 2.0 for E-commerce.	08
	b	Explain the need to have frames in E-commerce web-pages.	08
		OR	
4	a	The Internet is based on a layered model “transmission control protocol/internet protocol (TCP/IP)”. Discuss with an example.	08
	b	E-commerce cannot exist without internet. Discuss the evolution of internet related to E-commerce.	08

5	a	Elucidate how information and communications technology (ICT) is used in inter-business or inter-organizational transactions in E-commerce.	08
	b	List the factors that influence the growth of e-banking in developing countries.	08
		OR	
6	a	Demonstrate how the buying and selling of goods and services through wireless technology has impacted the digital commerce transactions.	08
	b	List and explain the major factors affecting e-commerce.	08
7	a	Give examples of products that have been designed for hybrid channels.	08
	b	Dell – is it a case of mass customization or using hybrid channel? Discuss your perspective.	08
8	a	Enumerate the social commerce challenges in India.	10
	b	How has social commerce evolved over time? What are its trends and the challenges of social commerce?	06

Project Planning and Control

VII Semester SEE Question Papers

Course codes: 18IM7G2

- Fast Track Nov 2023 (18IM7G2)
- Feb/Mar 2023 (18IM7G2)

18IM7G2

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Fast-track Examinations November-2023

Industrial Engineering and Management

PROJECT PLANNING AND CONTROL (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Block diagrams should be drawn wherever necessary to substantiate your answers.

PART-A

1	1.1	List different types of project life cycle.	02
	1.2	Define business value with example.	02
	1.3	What is the purpose of project charter?	02
	1.4	Differentiate between project and operations.	02
	1.5	What is critical path?	02
	1.6	Differentiate between CPM and $PERT$.	02
	1.7	Define internal failure cost.	02
	1.8	List the various costs of quality.	02
	1.9	Mention various negative strategies used to manage risk in the project.	02
	1.10	List the factors which affect decision making in any organization.	02

PART-B

2	a	Differentiate between activities carried out in initiating and planning process groups.	08
	b	What is project management? Briefly discuss any six project management knowledge areas.	08
3	a	Create a project charter for developing metro facility in your city as a case study.	12
	b	Identify the various drivers of project with suitable examples.	04
		OR	
4	a	Identify the factors used to measure the success of any project.	04
	b	Create a project charter for developing four wheeler electric vehicles as a case study.	12

5		Draw a network diagram of the following schedule of activities and find its critical path. Also calculate slack time for each event.	16

Activity	1-2	1-3	1-4	2-6	3-7	3-5	4-5	5-9	6-8	7-8	8-9
Duration (in days)	2	2	1	4	5	8	3	5	1	4	3

OR

6		Draw the network diagram for the following project and find the critical path and maximum time for completion of the project.	16

Activity	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*	*J*	*K*	*L*
Preceded by	–	*A*	*A*	*B*	*B*	*C*	*C*	*F*	*D*	*G, H*	*E*	*I*
Duration (in days)	10	9	7	6	12	6	8	8	4	11	5	7

7	a	Take any software development project of your choice and identify various quality costs involved in that project.	08
	b	Identify the cost estimation technique you are going to use for the following projects. Justify your answer suitably. i) Student management information for your institution. ii) Construction of metro.	08
8	a	Differentiate with case examples, different types of contracts used in procurement management.	08
	b	Assume that you have been appointed as project manager to develop software to manage student information system. List some of the risks you are going to encounter. What are the tools and techniques you are going to use to identify risks in the project and why?	08

18IM7G2

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Feb/Mar-2023

Industrial Engineering and Management

PROJECT PLANNING AND CONTROL (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Block diagrams should be drawn wherever necessary to substantiate your answers.

PART-A

1	1.1	Define the term Project management.	01
	1.2	What is a Tender?	01
	1.3	Mention the formula for the ROI of a project.	01
	1.4	What is 'Zero date' of the project?	01
	1.5	_____ represents the net benefit over and above the compensation for time and risk.	01
	1.6	The time estimates in $PERT$ is assumed to follow _____ distribution.	01
	1.7	Define slack time.	01
	1.8	What is tailoring?	01
	1.9	What is meant by portfolio management?	01
	1.10	What is meant by control resources?	01
	1.11	Mention any four creativity techniques.	02
	1.12	What are the contents of project family tree?	02
	1.13	List the basic steps of $PERT/CPM$ techniques.	02
	1.14	What is a Gantt chart?	02
	1.15	What is the difference between quality and grade?	02

PART-B

2	a	Explain the typical roles and responsibilities of a project manager.	08
	b	Illustrate the concept of organizational project management and its strategies.	08
3	a	Explain the five phases of project life cycle with reference to a software project.	10
	b	Explain any six factors which are considered as the organizational process assets by the enterprises.	06
		OR	

4	a	Illustrate project management tools/techniques for a project involving construction of a flyover.	08
	b	Develop a project charter for redesigning a website. Show all the elements of a project charter as per the standard format.	08
5	a	Explain any four aspects of a well designed work-breakdown structure.	04
	b	The project composed of seven activities whose time estimates are listed in the task. Activities are identified by beginning node (i) and ending (j) nodes. i) Draw the network diagram of the activity of a project. ii) Calculate the variance and standard deviation of the project length. iii) Find the expected duration and variance of each activity.	12

Activity $(i\text{-}j)$	*Estimated Duration*$(weeks)$		
	Optimistic (to)	*Most likely*(Tm)	*Pessimistic*(Tp)
$1-2$	5	8	10
$1-3$	18	20	22
$1-4$	26	33	40
$2-5$	16	18	20
$2-6$	15	20	25
$3-6$	6	9	12
$4-7$	7	10	12
$5-7$	7	8	9
$6-7$	3	4	5

OR

6	a	Explain four major reasons for ineffective control of projects.	08
	b	Mention and explain any eight techniques that project managers can practice in order to make better use of their time.	08
7	a	Define cost of quality. List and explain any three types of costs.	08
	b	Explain in brief any seven tools for project planning and control. Support your answer with suitable illustrations.	08
8	a	Explain in brief the process of defining how to conduct risk management activities for a project.	10
	b	Elaborate the process of procurement management with suitable examples.	06

Digital Supply Chain Management

VII Semester SEE Question Papers

Course codes: 18IM7G5

- Feb/Mar 2023 (18IM7G5)

18IM7G5

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Feb/Mar-2023

Industrial Engineering and Management

DIGITAL SUPPLY CHAIN MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Block diagrams should be drawn wherever necessary to substantiate your answers.

PART-A

1	1.1	Define e-logistics.	02
	1.2	List the three E's of the roadmap for building e-logistics capability.	02
	1.3	What is the use of telematics cloud?	02
	1.4	Write the full form of the acronym $UNCITRA$.	02
	1.5	Enumerate the basic categories of information covered by post centric ICT.	02
	1.6	Identify the challenges faced by companies in terms of growth of e-commerce.	02
	1.7	What are the three ways to consolidate goods in warehouses?	02
	1.8	List the process blocks of core warehouse management system.	02
	1.9	Compare active and passive $RFID$.	02
	1.10	List the benefits of using $GS1$ standards.	02

PART-B

2	a	Explain the method of building e-logistics capability at micro and macro levels within and between organizations.	08
	b	Discuss the applications of following technologies for e-logistics. i) Social media ii) Big data and Internet of Things.	08
3	a	Enumerate key features of CAN as a technology for enabling efficient road freight transport.	08
	b	Summarize the application of technograph in managing fleet of commercial vehicles.	08
		OR	
4	a	Analyze the functional aspects of bill of lading document.	08
	b	Evaluate the background legal framework with regard to Electronic Data Interchange (EDI) deployed for Bill of Lading.	08

5	a	Summarize the key features of four major port community systems.	10
	b	Enumerate the challenges to the port-centric ICT.	06
		OR	
6	a	Evaluate the performance features of different types of warehouse/fulfillment center.	10
	b	"Connect Distribution; a leading spare part provider for domestic appliances has deployed voice technology to speed up fulfillment process." Provide comments.	06
7	a	Explain inbound management workflows of core warehouse management system.	10
	b	Write a note on complexity of connectivity of WMS solutions into ERP and information management system.	06
8	a	Elucidate the costs and benefits of active $RFID$.	08
	b	Summarize key features of Global Trade Item Number ($GTIN$).	08

Financial Management

VII Semester SEE Question Papers

Course codes: 18IM7F1

- Feb/Mar 2023 (18IM7F1)

18IM7F1

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VII Semester B. E. Examinations Feb/Mar - 2023

Industrial Engineering and Management

FINANCIAL MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of Interest factor tables permitted.

PART-A

1	1.1	List the functions of financial markets.	02
	1.2	Differentiate between corporate income tax and individual income tax.	02
	1.3	If you deposit Rs.1000 today in a bank which pays 10% interest compounded annually, how much will the deposit grow to after 8years?	02
	1.4	List the steps involved in earnings capitalization approach.	02
	1.5	List the 2 ways of defining the relationship between benefit and costs.	02
	1.6	List the 2 conditions that should be satisfied for using the company's weighted average cost of capital (WACC) for evaluating new investments.	02
	1.7	List any 2 key differences between Debt and Equity.	02
	1.8	Differentiate between primary equity market and secondary equity market.	02
	1.9	List any 2 plausible reasons for mergers.	02
	1.10	Distinguish between call option and put option.	02

PART-B

2	a	Discuss the key elements of the regulatory framework applicable to corporate investment and financing decisions.	06
	b	Describe briefly the different ways of classifying financial markets.	06
	c	Explain briefly the profit vs. cash flows.	04
3	a	Explain the key inputs for basic valuation models for securities.	06
	b	Rohan borrows Rs.80000 for a Laptop at a monthly interest of 1.25%. The loan is to be repaid in 12 equal monthly installments, payable at the end of each month. Prepare loan amortization schedule.	10
		OR	

4	a	A Rs.100 par value bond bearing a coupon rate of 12% will mature after 5years. What is the value of the bond if the discount rate is 15%?	05
	b	The stock of XYZ limited performs well relative to other stocks during recessionary periods. The stock of ABC limited, on the other hand does well during growth periods. Both the stocks are currently selling for Rs.100 per share. You assess the rupee return (dividend plus price) of these stocks for the next year as follows.	

	Economic condition			
	High growth	Low growth	Stagnation	Recession
Probability	0.3	0.4	0.2	0.1
Return on XYZ's stock	100	110	120	140
Return on ABC's stock	150	130	90	60

		Calculate the expected return and standard deviation of investing for Rs.1000 in the equity stock of XYZ limited.	05
	c	Differentiate between unique risk and market risk with appropriate examples.	06
5	a	Explain briefly the different phases of capital budgeting process.	08
	b	The capital structure of Info Limited in book value terms are as follows.	

Equity capital (20million shares, Rs.10par)	Rs.200 million
Preference capital, 12% (5,00,000 shares, Rs.100 par)	Rs.50 million
Retained earnings	Rs.350 million
Debentures 14% (1,20,000 debentures, Rs.100par)	Rs.120 million
Terms loans, 13%	Rs.80 million
	Rs.800 million

		The next expected dividend per share is Rs.2.00. The dividend per share is expected to grow at the rate of 12%. The market price per share is Rs.50.00. Preference stock, redeemable after 10years is currently selling for Rs.85.00 per share. Debentures, redeemable after 5years are selling for Rs.90.00 per debenture. The tax rate for the company is 30%. Calculate the average cost of capital.	08
		OR	
6	a	Explain the procedure for determining the weighted marginal cost of capital.	06
	b	ABC motors have a net operating income of Rs.50 million. ABC employs Rs.120million of debt carrying 15% interest charge. The equity capitalization rate applicable to ABC is 16%. What is the market value of ABC under the net income method? Assume there is no tax.	05
	c	The following information is available for Rahul Metals.	

Net operating income	Rs.40 million
Interest on debt	Rs.10 million
Cost of equity	18percent
Cost of debt	12percent

		What is the average cost of capital of Rahul Metals?	05

7	a	Discuss the important features of venture capital.	06
	b	Describe any four important participants in the security market.	05
	c	Explain briefly the factors influencing the working capital requirements of a firm.	05
8	a	List and explain the various steps involved in a merger.	06
	b	Describe briefly the different sources available to an Indian firm for raising foreign currency finance.	10

Lean Manufacturing Systems

VII Semester SEE Question Papers

Course codes: 18IM7G4, 16IM7G2

- Feb/Mar 2023 (18IM7G4)
- Jan/Feb 2021 (16IM7G2)
- Nov/Dec 2019 (16IM7G2)

18IM7G4

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Feb/Mar-2023

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEM (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What is meant by lean manufacturing system?	02
	1.2	Mention the three statements made by Taichi Ohno which when taken together defines his TPS.	02
	1.3	What is meant by Kanban?	02
	1.4	Standard work as defined by Ohno has three elements. What are they?	02
	1.5	Distinguish between $DMAIC$ and $DMADV$.	02
	1.6	What are the three fundamental issues to culture change that needs to be assessed for implementing lean project?	02
	1.7	What does Spaghetti diagram used for?	02
	1.8	What is meant by 'catch ball' in the context of Hoshin Kanri planning?	02
	1.9	What does $PDCA$ stand for?	02
	1.10	Leader of lean initiative must have three requisite skills. List them.	02

PART-B

2	a	Taichi Ohno's waste elimination system has two pillars. What are they? Elaborate.	10
	b	Review seven types of waste identified by Taichi Ohno as part of behavioral definition of TPS.	06
3	a	The total inventory is the sum of three types of inventory. What are they? Explain.	08
	b	What are 'Buffers' in the context of inventory management? Discuss the three forms of Buffers usually considered.	08
		OR	
4	a	What is the primary measure of production effectiveness? Explain the five parameters to be considered for calculating the same.	08
	b	What is total productive maintenance? Describe the five pillars of the same.	08

5	a	Elaborate the three categories of problems typical in a lean plant.	08
	b	The overall approach to make a value stream lean consists of four key strategies. Review the strategies briefly.	08
		OR	
6	a	Lean six sigma uses a five step method to improve processes and solve problems. Discuss the same with appropriate examples.	10
	b	Mention any six benefits of process improvement.	06
7	a	Analyze the five precursors to the successful implementation of lean initiatives.	10
	b	While redesigning the process flow to eliminate waste, future state value stream map is prepared. What are the objectives/outcomes with which this map is drawn?	06
8	a	At the heart of policy deployment is the development of goals. Review the characteristics/attributes of the goals.	10
	b	Critically examine the three enemies of sustaining the gains of lean implementation in the rapidly changing world.	06

16IM7G2

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	What is a Lean Manufacturing system?	02
	1.2	"Lean is a war on waste". Justify.	02
	1.3	Calculate the Takt time based on the following Information: Customer demand is 34,400 parts per month. The company works 5 *days* per week and on average 20 *days* per month. There are two Eight hour shifts. Lunch is 30 *minutes* per shift. Two, 10 *minutes* breaks are taken per shift.	02
	1.4	What does single piece flow mean in a Lean Manufacturing system?	01
	1.5	You are required to determine how many Kanban card sets you need for an operation. You find that average number of units demanded is 1000 *per hour*, the lead time to replenish the order for this item is 10 *hours*, the container size is 10 *units* and the safety stock is estimated to be 5% of the expected demand.	02
	1.6	_____ is that type of inventory kept on hand to cover the variations internal to the plant including line stoppages, raw material stock outs etc.	01
	1.7	A major barrier to inventory reduction and batch size reduction is _____.	01
	1.8	Identify the five precursors to Lean Implementation.	02
	1.9	The Lean Implementation plan will normally be expressed in the form of a _____ chart.	01
	1.10	The primary purpose of setting goals in lean implementation is to guide the _____.	01
	1.11	Mention the 3 real enemies in sustaining the gains in Lean Implementation.	02
	1.12	What are the desirable properties of goals in Lean Implementation?	02
	1.13	In the Toyota production system, what do you mean by "Genchi Genbutsu"?	01

PART-B

2	a	What are the salient features of Toyota production system? Describe them in detail.	10
	b	"All processes and philosophies have their own limitations". State the limitations of lean manufacturing and cite two examples where lean cannot be implemented.	06
3	a	List and explain the six rules of Kanban system. Illustrate with suitable examples.	10
	b	How do you measure stability while controlling quality in a lean manufacturing system?	06
		OR	
4	a	What are the factors considered in understanding the inventory variation in a lean environment? Explain the components of total inventory.	08
	b	Explain the four tactics and skills for quantity control pillars (Jidoka pillar).	08
5		Discuss the need and importance of integrating lean strategy with culture and focus of the organization.	16
		OR	
6	a	Explain the techniques for reducing the lead times.	10
	b	Explain the four key implementing Lean Strategies adopted on the production line.	06
7		What are the 8 steps adopted as a prescription for the Lean implementation? Explain in detail.	16
8	a	Discuss the H-K planning model.	08
	b	To achieve the gains after implementation, the organizations need to reduce the variation. Explain the approach that can be used to reduce these variations.	08

16IM7G2

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Write the definition of Lean in the context of Operations Management.	02
	1.2	Differentiate between $KANBAN$ and $CONWIP$.	02
	1.3	List the tactics used by Jidoka.	02
	1.4	What are the seven wastes according to Ohno.	02
	1.5	List the 5s techniques.	02
	1.6	Write the formula to calculate work in process according to Little's law.	02
	1.7	Define takt time with suitable example.	02
	1.8	State the characteristic of Goal.	02
	1.9	Write the formula to calculate OEE with notations.	02
	1.10	Value stream mapping is part of which of the lean implementation strategies? And why?	02

PART-B

2	a	List and explain the two pillars of the Toyota Production System.	08
	b	Elaborate the key features of the important three revolutionary concepts in the TPS.	08
3	a	Explain the six rules of $KANBAN$ management.	08
	b	Evaluate Make to order and Make to stock production systems.	08
		OR	
4	a	Illustrate with an example the need for work in process (WIP) inventory. How do you compute WIP inventory?	08
	b	Explain the most commonly used two types of kanban.	08
5	a	Summarize the seven basic techniques that can be employed to reduce lead time and improve flow.	08
	b	List the eight important skills that a problem solver should possess.	08
		OR	
6	a	Explicate any one of the four lean implementation strategies. Also outline the tools used.	08

	b	What happens when Jidoka is not done well? Discuss with an appropriate case.	08
7	a	Enumerate the ten reasons for failure of lean initiatives in organizations.	08
	b	Examine the five precursors to implement the lean initiative.	08
8	a	With the help of a diagram, explain the $H - K$ planning model.	08
	b	To achieve the gain, we need to reduce the variation. State and explain the specific approach that can be made to reduce variation.	08

Quality Management

VII Semester SEE Question Papers

Course codes: 18IM7G1

- Feb/Mar 2023 (18IM7G1)

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18IM7G1

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Feb/Mar-2023

Industrial Engineering and Management

QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Block diagrams should be drawn wherever necessary to substantiate your answers.

PART-A

1	1.1	List any two ideas of Feigenbaum's systematic approach for quality.	02
	1.2	List any four QC tools that were evolved as an evolution of quality methodology.	02
	1.3	Define 'management by process'.	02
	1.4	Differentiate between proactive and reactive problem solving.	02
	1.5	List any two reasons for the importance of teamwork activities in quality improvement.	02
	1.6	List any two alignment purposes of Hoshin management.	02
	1.7	List the four costs of Six Sigma.	02
	1.8	Define rolled throughput yield.	02
	1.9	List any two objectives of Six Sigma.	02
	1.10	Define design for Six Sigma.	02

PART-B

2	a	Explain briefly the Juran's quality trilogy.	06
	b	Describe in detail the four fitness of quality.	10
3	a	Explain briefly the market-in concept that focuses on customer satisfaction.	10
	b	Describe briefly the WV model of continuous improvement.	06
		OR	
4	a	Explain briefly the four parts of the process for identifying the problem in reactive problem solving effort.	08
	b	Describe briefly the standard steps for proactive improvement problem.	08

5	a	Explain briefly the seven principles for activating teamwork based on Shoji Shiba's observation of QC circles in Japan.	08
	b	Explain briefly the importance of CEO involvement in TQM mobilization.	08
		OR	
6	a	Describe briefly the main components of Hoshin management.	08
	b	Explain briefly the six elements of infrastructure for networking.	08
7	a	Explain briefly the five different key players of Six Sigma.	08
	b	Describe briefly the different guidelines for metrics in Six Sigma.	08
8	a	Explain briefly the fundamental building blocks of 'growth management system'.	10
	b	Explain briefly the five step process of $DFSS$ model.	06

Additive Manufacturing Methods

VII Semester SEE Question Papers

Course codes: 16IM7G1

- Jan/Feb 2021 (16IM7G1)

16IM7G1

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

ADDITIVE MANUFACTURING METHODS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The machining process that adds rather than removes material is called_____.	01
	1.2	Additive manufacturing is generally a _____ process.	01
	1.3	A computer model of a part design on a CAD system is called _____ prototype.	01
	1.4	_____ is an additive-plus-subtractive process.	01
	1.5	Stereo lithography process is _____ prototyping.	01
	1.6	The act of turning a digital 3D model into thin layers used for 3D printing is called _____.	01
	1.7	The concept of extrusion is used in the process of _____.	01
	1.8	In selective laser sintering, the input material is in the form of_____.	01
	1.9	In a typical coordinate measuring machine (CMM), the interfacing hardware, control box, processing computer and software, are parts of _____unit.	01
	1.10	List the materials used for laminated object manufacturing.	01
	1.11	In the _____ process, one starts with a single block of solid material larger than final size of the desired object and material is removed until the desired object and material is removed until the desired shape is reached.	01
	1.12	Materials used to build the 3D prints is _____.	01
	1.13	Stereo lithography apparatus uses a pool of resin to create the _____ part.	01
	1.14	The most popular file format for 3D printing_____.	01
	1.15	Checking 3D CAD data is the process of the _____ stage.	01
	1.16	If the direction of the vertices list for a boundary is clockwise, it indicates that the solid material is _____the polyline.	01
	1.17	STL consists of an unordered list of _____ facets representing the outside skin of an object.	01
	1.18	STL file format is represented by interaction of _____.	01
	1.19	List the data interfacing formats.	01
	1.20	In STL files, if chord length decrease then accuracy_____.	01

PART-B

2	a	Enumerate the importance of rapid prototyping systems? Explain rapid prototyping systems liquid, solid and powder bases prototyping.	08
	b	List the steps in rapid prototyping process chain.	08
3	a	Explain the principles and applications of stereo lithography systems.	08
	b	Explain Stereo-lithography apparatus operations.	08
		OR	
4	a	As opposed to many of the liquid-based RP systems which use photosensitive polymer, water is used in the Rapid Freeze Prototyping (RFP). What are the pros and cons of using water?	08
	b	List advantages and disadvantages of rapid freeze modeling.	08
5	a	Describe the process flow of Stratasys' Fused Deposition Modeling.	08
	b	Explain the specification of Laminated Object Manufacturing of the type LOM-1015Plus and LOM-2030H for the following: i. Motion Control ii. Part accuracy XYZ directions, mm(in) iii. Material thickness, mm(in) iv. Material size.	08
		OR	
6	a	Describe the critical factors that will influence the performance and functions of 3D systems multi jet modeling system.	08
	b	Discuss at least four advantages of 3D systems multi jet modeling system.	08
7	a	Explain the purpose and the two principles of a selective laser sintering.	08
	b	Explain the working principle of Pro Metal's 3D printer and its applications.	08
8	a	Write a note on STL format with its different types of format.	08
	b	What causes missing facets or gaps and degenerate Facets to occur in STL format?	08

Big Data Analytics

VII Semester SEE Question Papers

Course codes: 16IM7F3

- Jan/Feb 2021 (16IM7F3)
- Nov/Dec 2019 (16IM7F3)

16IM7F3

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

BIG DATA ANALYTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	What is Big Data? Give examples.	02
	1.2	What is RDMBS?	02
	1.3	What are aggregate data models?	02
	1.4	What are unstructured data?	02
	1.5	What is Hadoop?	02
	1.6	List the Hadoop tools used for data gathering.	02
	1.7	What is HDFS?	02
	1.8	What is MapReduce in Hadoop?	02
	1.9	List the phases of MapReduce programs?	02
	1.10	Write a note on PIG?	02

PART-B

2	a	Identify and discuss the different types of Big data.	08
	b	With the help of neat sketch, explain the concept of Master and Slave replications.	08
3	a	Differentiate between SQL and NoSQL databases and identify typical characteristics of NoSQL databases.	08
	b	Identify and discuss different types of data modeling for NoSQL database management.	08
		OR	
4	a	What are the advantages of using non-relational models? Is there a future for Schema-less data?	08
	b	What are Graph Databases? “Graph database matter lot in big data”. Justify the statement.	08
5	a	Identify and discuss Apache Hadoop sub-projects.	08
	b	With neat sketch, explain the Hadoop architecture.	08

		OR	
6	a	Identify and discuss the features of 'Hadoop'.	08
	b	Discuss the role and importance of Network Topology in Hadoop.	08
7	a	Identify and discuss the components of HDFS architecture.	08
	b	Discuss briefly the process or phases of working of MapReduce.	08
8	a	Discuss HBase architecture with neat sketch.	08
	b	List down the points that differentiate the data models of Cassandra from that of an RDBMS.	08

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

BIG DATA ANALYTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Give any two examples for sources of big data.	02
	1.2	Differentiate between name node and data node.	02
	1.3	What is unstructured data?	02
	1.4	Give any four examples of structured data.	02
	1.5	Mention the characteristics of big data.	02
	1.6	Give any two examples of data theft.	02
	1.7	Define HiveQL data.	02
	1.8	Mention any two key features of Cassandra data model	02
	1.9	What is Hadoop streaming?	02
	1.10	Mention Mapreduce types.	02

PART-B

2	a	Explain "Big Data" and what are five 'V's of Big Data?	08
	b	With examples, describe the various types of data used in big data application?	08
3	a	Differentiate between Aggregate Data models and Distribution models with suitable examples.	08
	b	What are the different types of NoSQL databases?	08
		OR	
4	a	Why successful enterprises rely on NoSQL? Explain	08
	b	Explain with the help of suitable examples the concept of schemaless databases.	08
5	a	Illustrate and describe the anatomy of file read in *HDFS*?	08
	b	How is Data Integrity achieved in *HDFS*? Is Data Integrated in *HDFS*?	08
		OR	
6	a	Write a brief note on serialization in Hadoop.	08
	b	Illustrate and describe the anatomy of file write in *HDFS*?	08

7	a	What is the purpose of Apache Hadoop *YARN*? Briefly explain.	08
	b	Differentiate between shuffling and sorting in Hadoop MapReduce?	08
8	a	Explain four types of execution in pig Environment.	08
	b	With the help of suitable example explain HiveQL queries.	08

Energy Management

VII Semester SEE Question Papers

Course codes: 16IM7G4

- Jan/Feb 2021 (16IM7G4)
- Nov/Dec 2019 (16IM7G4)

16IM7G4

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

ENERGY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Define energy management.	02
	1.2	State the three primary phases of energy management program.	02
	1.3	What should be the objective of demand side of energy management?	02
	1.4	Mention any four mechanisms to transfer with the surroundings in a HCAC system.	02
	1.5	Mention any two main energy-using components of an Industrial HVAC system.	02
	1.6	Illumination is said to be dynamic when it involves a combination of _____ and _____ to provide the required level of illumination.	02
	1.7	List any four most energy-intensive industrial sectors, which are accounting for an estimated 77% of industrial direct CO_2 emission.	02
	1.8	Define Process Energy.	02
	1.9	List the five main elements of the building envelope which determines the heat gains and losses.	02
	1.10	If you are investing 100000 in a project that will save 2000 per month of energy, payback period is _____ years.	02

PART-B

2	a	Discuss the 3 basic approaches for minimizing the quantity of energy used.	08
	b	"The environment is directly and indirectly affected by energy production and use". Elucidate this statement with suitable examples.	08
3	a	How can a HVAC system be designed effectively for the best concern for human comfort and health?	08
	b	Discuss in brief, any five practical techniques for reducing energy use in new and existing facilities.	08
		OR	

4	a	Briefly discuss any six case examples of energy management opportunities for HVAC controls.	08
	b	Considering suitable case examples, explain the four general principles for energy management in lighting systems.	08
5	a	"The general principles of energy management can be applied to the management of process energy use in a variety of ways". Considering any suitable examples for five varieties of ways, justify the above statement.	08
	b	Explain the concept of energy storage for process industries.	08
		OR	
6	a	Explain the five key areas of process heat.	08
	b	Explain the concept of energy storage for agriculture and food processes.	08
7	a	List and explain the six principles of sustainable design.	08
	b	Examine the variety of parameters influencing building energy use.	08
8	a	The capital cost of the DG set is $Rs.9{,}00{,}000$, the annual output is $219MWh$ and the maintenance cost is $Rs.30000$ per annum. The cost of producing each unit of electricity is $3.50\ Rs./kWh$. Find the total cost of a diesel generator operating over a $5year$ period, taking into consideration both fixed and variable cost.	06
	b	A new small congestion plant installation is expected to reduce a company's annual energy bill by $Rs.4{,}86{,}000$. If the capital cost of the new boiler installation is $Rs.22{,}20{,}000$ and the annual maintenance and operating costs are $Rs.42{,}000$ calculate the expected payback period for the project.	03
	c	A company borrows $Rs.30{,}00{,}000$ to finance a new boiler installation. If the interest rate is $10\%\ per\ annum$ and the repayment period is $5years$, calculate the value of the total repayment and the monthly repayment value, assuming- i. Simple interest and ii. Compound interest.	07

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

ENERGY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	List any two basic approaches for minimizing the quantity of energy used.	02
	1.2	Mention the three phases of energy management program.	02
	1.3	List the four mechanisms of heat exchange with the surroundings.	02
	1.4	List the main energy- using components of a commercial or industrial $HVAC$ system.	02
	1.5	How do you reduce losses due to heat transfer to the environment?	02
	1.6	What is the purpose of process control.	02
	1.7	Mention the parameter which influence the building energy usage.	02
	1.8	What are Green buildings?	02
	1.9	Define life cycle costing.	02
	1.10	Draw Break Even chart.	02

PART-B

2	a	Briefly summarize some general principles that are applicable to wide variety of situations to manage energy.	10
	b	Identify few energy conversion devices used in your daily life and their uses.	06
3	a	Enumerate four general principles for energy management in lighting systems.	06
	b	Discuss the energy management opportunities in lighting systems.	10
		OR	
4	a	Explain general principles for energy management in $HVAC$ systems.	10
	b	Highlight two points of view considered by a building designer in lighting.	06
5	a	Provide an overview of energy use and management with respect to key areas of heating process.	10
	b	List few energy management opportunities encountered in manufacturing operations.	06
		OR	

6	a	List and briefly explain new types of controls and methods developed to control the process.	06
	b	Develop an energy management plan to reduce energy usage at your institution.	10
7	a	Mention the six principles of sustainable design.	06
	b	"Green buildings" are important new concepts that have emerged in last several decades and are finding growing acceptance among building designers, contractors and building owners. Elucidate this statement and provide reasons for growing acceptance of green building concept of designers, contractors and building owners.	10
8	a	Write a note on economic aspects of energy management.	08
	b	How management of energy can have an impact on environmental aspects, use case examples to support your explanation.	08

Financial Accounting and Costing

VII Semester SEE Question Papers

Course codes: 16IM72

- Jan/Feb 2021 (16IM72)
- Fast Track Oct 2020 (16IM72)
- Nov/Dec 2019 (16IM72)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Mention any one difference between Financial Accounting and Management Accounting.	01
	1.2	What is the meaning of going concern concept in accounting?	01
	1.3	Mention the need for accounting standards.	01
	1.4	What is the concept used in this accounting equation Capital + Liabilities = Asset?	01
	1.5	Define the term trial balance.	01
	1.6	Profit and Loss Account measures net income by matching revenues and expenses according to the accounting principles. Do you agree with this statement? Why or Why not?	01
	1.7	Give an example of intangible asset.	01
	1.8	What is the term used in accounting to denote debts which cannot be recovered or become irrecoverable?	01
	1.9	Mention any one purpose for which cost sheet is used in manufacturing plant.	01
	1.10	Which method of costing is applicable in the case of an organization providing printing service using offset printing services to clients?	01
	1.11	Which method of costing is applicable in the case of cosmetic company producing cosmetics in batches?	01
	1.12	Give an example of unavoidable losses in the context of process costing.	01
	1.13	Mention any one managerial use of Standard costing.	01
	1.14	Write down the formula for material yield variance.	01
	1.15	Mention any one method of absorbing overhead expenses.	01
	1.16	Define variance.	01
	1.17	What is a master budget?	01
	1.18	Mention any one assumption made while preparing a flexible budget.	01
	1.19	What is the purpose of preparing a purchase budget?	01
	1.20	What is the basis for fixing the opening balance of cash budget for the subsequent period determined?	01

PART-B

2 a Prepare a Journal in the books of *K.K.* Co. from the following transactions:

2018	*Rs.*	2018	*Rs.*
Dec. 1 *Started business with a capital of*	50000	*Dec.* 15 *Purchased goods from Ram*	4000
Dec. 8 *Purchased goods for cash*	4000	*Dec.* 18 *Purchased goods from Ram*	300
Dec. 6 *Paid into bank*	20000	*Dec.* 20 *Purchased goods from Ram*	1000
Dec. 9 *Paid to Ram*	1980	*Dec.* 22 *withdrawn from Bank*	3000
Dec. 9 *Discount allowed by him*	20	*Dec.* 25 *Paid Ram by cheque*	500
Dec. 10 *Cash sales*	3000	*Dec.* 31 *withdrawn for personal use*	200
Dec. 12 *Sold to Hari for cash*	2000		

10

b Discuss the matching principle and the accrual principle of accounting. 06

3 a From the following balances extracted from the books to Mr. Bansi Lal, prepare the Trading Account for the year ending 31st March 2006.

Purchases	42500	*Power*	300
Mfg. Expenses	1950	*Wages*	5000
Sales	67500	*Op. Stock*	10000
Carriage inwards	100	*Sales returns*	50
Freight and duty	5000	*Purchase returns*	200
Stores consumed	200	***Trade expenses***	

The value of stock unsold is *Rs.* 12,000. 12

b What are Personal, Nominal and Real Accounts? Give an example of transactions related to Personal, Nominal and Real Accounts. 04

OR

4 a The following balances are extracted from the books of Kautilya & Co. on 31st March, 2006. You are required to prepare the Trading and Profit and Loss Account and a Balance sheet as on that date.

Stock on April 1	500	*Commission (Cr)*	200
B/R	2250	*Returns Outwards*	250
Purchases	19500	*Trade Expenses*	100
Wages	1400	*Office Fixtures*	500
Insurance	550	*Cash in hand*	250
Sundry Debtors	15000	*Cash at Bank*	2375
Carriage Inwards	400	*Rent & Taxes*	550
Commission (Dr.)	400	*Carriage Outwards*	725
Stationary	225	*Sales*	25000
Interest on Capital	350	*Bills Payable*	1500
Return Inwards	650	*Creditors*	9825
		Capital	8950

The closing stock was valued at *Rs.* 12,500. 12

	b	Mention any two uses each for the balance sheet and Profit & Loss accounts that the stake holders can derive.	04
5	a	The following are the costing records for the year 2017 of a manufacturer: Production 10,000 units; Cost of Raw Materials $Rs.2{,}00{,}000$; Labour Cost $Rs.1{,}20{,}000$; Factory Overheads $Rs.80{,}0000$; Office Overheads $Rs.40{,}000$; Selling Expenses $Rs.10{,}000$, Rate of Profit 25% on the Selling Price. The manufacturer decided to produce 15,000 units in 2017. It is estimated that the cost of raw materials will increase by 20%, the labour cost will increase by 10%, 50% of the overhead charges are fixed and the other 50% are variable. The selling expenses per unit will be reduced by 20%. The rate of profit will remain the same. Prepare a Cost statement for the year 2017 showing the total Profit and Selling Price per unit.	12
	b	What is the prime cost? What are the elements of cost included in prime cost?	04
		OR	
6		Product-X is obtained after it passes through three distinct processes. You are required to prepare process account from the following information: $PARTICULARS$ / $PROCESSES$: $TOTAL$, I, II, III $Material$: 15084, 5200, 3960, 5924 $Daily\ Wages$: 18000, 4000, 6000, 8000 $Production\ Overhead$: 18000, , , 1,000 units @ Rs.6 per unit was introduced in Process I production overhead to be distributed at 100% on direct wages. $ACTUAL\ OUTPUT$ / $UNITS$ / $NORMAL\ LOSS$ / $VALUE\ OF\ SCRAP\ (Rs./\ UNIT)$: $PROCESS-I$: 950, 5%, 4 $PROCESS-II$: 840, 10%, 8 $PROCESS-III$: 750, 15%, 10 Prepare Process Accounts for I, II & III.	16
7	a	From the following you are required to calculate i) Material Usage Variance ii) Material Price Variance iii) Material Cost Variance Quantity of material purchased: 3,000 units Value of material purchased: $Rs.9{,}000$ Standard quantity of material required for one tonne of finished product: 25 units Standard rate of material: $Rs.2$ per unit Opening stock of material: NIL Closing stock of material: 500 units Finished production during the period: 80 tonnes	10
	b	Standard Cost is defined as "the predetermined cost that is calculated at the management's standards of efficient operations and the relevant necessary expenditure". Elaborate on this statement with examples to illustrate your points.	06

8	*ABC* Ltd., a newly started company wishes to prepare Cash Budget from January. Prepare a cash budget for the first six months from the following estimated revenue and expenses. Cash balance on 1st January was *Rs.*10,000. New machinery is to be installed at *Rs.*20,000 on credit, to be repaid by two equal installments in march and April; sales commission @ 5% on total sales is to be paid within a month following actual sales. *Rs.*10,000 being the amount of 2nd call may be received in March. Share premium amounting to *Rs.*2,000 is also obtained with the 2nd all. Period of credit allowed by supplier – 2 months; Period of credit allowed to customers- 1 month, delay in payment of overheads-1 month. Delay in payment of wages- $\frac{1}{2}$ month. Assume cash sales to be 50% of total sales.	16

Month	*Total Sales*	*Materials*	*Wages*	*Overheads*	
				Production	*Selling & Distribution*
January	20000	20000	4400	3200	800
February	22000	14000	4400	3300	900
March	28000	14000	4600	3400	900
April	36000	22000	4600	3500	1000
May	30000	20000	4000	3200	900
June	40000	25000	5000	3600	1200

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Fast track Examinations October-2020

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	_____ is a financial statement that displays the revenues and expenses of a company for a period of time.	01
	1.2	State the matching concept of accounting.	02
	1.3	In journal, transactions are recorded in _____ order.	01
	1.4	Indicate how assets, liabilities and capital are affected in the following transactions with an accounting equation: purchase of machinery for cash *Rs.* 3,00,000.	02
	1.5	Drawing account is _____ type of account.	01
	1.6	A person who owes money to the firm is known as _____.	01
	1.7	The act of transferring the information from the Journal to Ledger is called as _____.	01
	1.8	Loss leads to Reduction in _____.	01
	1.9	Cost of floppy disk used for office computer is part of _____ overhead.	01
	1.10	A product cost estimates derived from a competitive market price is called as _____ cost.	01
	1.11	Warehouse rent is a part of _____ overhead.	01
	1.12	Factory cost = _____ + factory overheads.	01
	1.13	Two methods used for calculation of equivalent production are _____ and _____.	01
	1.14	The _____ costing system is best suited for industries where homogeneous products are manufactured on continuous basis.	01
	1.15	Material usage variance is the sum of _____ and _____.	02
	1.16	Difference between the standard cost and actual cost is called as _____.	01
	1.17	A _____ is tool which helps the management in planning and control of all business activities.	01

PART-B

2	a	Kiran had the following transaction. Prepare an accounting equation to show their effect on assets, liabilities and capital. i) Commenced business with $Rs.1{,}00{,}000$. ii) Purchased furniture for cash $Rs.50{,}000$. iii) Purchased a building for $Rs.1{,}50{,}000$ giving $Rs.50{,}000$ in cash and balance through loan. iv) Sold goods costing $Rs.20{,}000$ for $Rs.25{,}000$. v) Purchased old vehicle (for office use) for cash $Rs.20{,}000$. vi) Paid cash $Rs.5{,}000$ for loan and $Rs.3{,}000$ for interest. vii) Withdrew cash for personal use $Rs.5{,}000$ viii) Furniture costing $Rs.10{,}000$ is valued at $Rs.9{,}000$	10
	b	"Accounting equation holds good under all circumstances". Justify this statement with help of six illustrations.	06
3		Following is the Trial balance of M/s.*BMW* & Associates for the year ended 31 *Jul* 2011. Prepare Profit and Loss A/c and Balance Sheet.	

Particulars	*Rs.*	*Particulars*	*Rs.*
Capital	813400	*Debtors*	42000
Sales	1090000	*Creditors*	39000
Plant & Machinery	356000	*Purchase Returns*	24500
Purchases	437000	*Stock*	150500
Freight Inward	35000	*Discount Allowed*	40000
Salaries	110000	*Insurance*	15000
Other administration Expenses	60000	*Advertisements*	56000
Loan from IDBI	350000	*Building*	650000
Furniture	75000	*Bad Debts*	2000
Wages	153900	*Investments in NSCs*	90000
Depreciation on machinery	25000	*Cash*	6500
		Bank	13000

Closing stock as on 31 *Jul* 2011 is $Rs.57000$. 16

OR

4 a From the data given below, prepare trading and Profit & Loss A/c of *B&D* Chemicals for quarter ended *Jun* 2011.

Particulars	*Amount*
Sales	500000
Salaries	46000
Opening stock	6000
Depreciation on Machinery	15000
Prepaid rent	7000
Audit fees	10000
Interest on Loan	25000
Outstanding salaries	5000
Consultancy Fees	5500
Closing stock	9000
Electricity charges	13000
Telephone expenses	3000
Rent	35000
Purchase	175000
Wages	60000

10

b Differentiate between:
i) Journal and ledger
ii) Trial balance and balance sheet. 06

5 a The modern manufacturing company submits the following information on *March* 31st, 2011.

	Rs.
Sales for the year	2,75,000
Finished goods inventories (beginning)	7,000
Work in progress inventories (beginning)	4,000
Purchase of raw material	1,10,000
Material at the beginning of the year	3,000
Material at the end of the year	4,000
Direct labour	65,000
Factory overheads	60% of labour cost
Finished goods inventories (ending)	8,000
Work in progress inventories (ending)	6,000
Selling expenses	10% of sales
Administrative expenses	5% of sales

Prepare a cost sheet for the year ending 31st *March* 2011. 10

b Highlight the use of costing information for the following factors. Use a case Example of a component manufacturing unit to illustrate your answer:-
i) Determining selling price
ii) Controlling cost
iii) Providing information for decision-making. 06

OR

6 From the following data, prepare process a/c of 1,2 and 3. The output of each process is treated as input for next process. 16

	Process 1	*Process* 2	*Process* 3
Material	70000	15000	30000
Wages	38000	40000	25000
Other expenses	11000	12000	15000

Factory overheads of $Rs.40000$ to be apportioned on the basis of wages paid. There was no opening or closing stock. 15000 units have been introduced in 1st process and the output of each process is as under

	Output	*Normal loss*
Process 1	14550	3%
Process 2	13500	6%
Process 3	13300	2%

7 To produce $500kgs$ of product "Hathi soap", following materials were consumed. 16

Raw material	$Qty(kgs)$	*Price per kg*
Oil	390	140
Caustic soda	240	75

As per standards, to produce $500kgs$ of product, following materials were to be consumed.

Raw material	$Qty(kgs)$	*Price per kg*
Oil	400	120
Caustic soda	200	70

Calculate the material variances from the following figures.

8 The cost of an article at the capacity of 5,000 units is given under A below. For variation of 25% in capacity above or below this level, the individual expenses vary as indicated under B below. 16

	A	$B\ (percent)$
Material cost	25,000	100 $(variable)$
Labour cost	15,000	100 $(variable)$
Power	1,250	80 $(semi-variable)$
Repairs	2,000	75 $(semi-variable)$
Stores	1,000	100 $(variable)$
Inspection	500	20 $(semi-variable)$
Administrative overheads	5,000	25 $(semi-variable)$
Selling overheads	3,000	50 $(semi-variable)$
Depreciation	10,000	100 $(fixed)$
Total	62,750	
Cost per unit	12.55	

Prepare the production cost budget at 4,000 units and 6,000 units.

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	_____ accounting concept states that transaction and events that can be expressed in money terms are recorded in the books of accounts.	01
	1.2	State the matching concept of accounting.	02
	1.3	$XYZ's$ trial balance as at 31st December $20X2$, contains the following information: 12% bank loan $Rs.40{,}000$, interest paid $Rs.3{,}800$. Interest debited to profit and loss account is _____.	02
	1.4	Indicate how assets, liabilities and capital are affected due to the following transactions with an accounting equation: Purchase of machinery for cash $Rs.3{,}00{,}000$.	02
	1.5	Mr. Vasu's trial balance sheet has the following entries: Opening stock $Rs.500$, Purchases $Rs.2{,}500$, Sales $Rs.3{,}600$ and Closing stock $Rs.300$. Calculate Mr. Vasu's gross profit _____.	02
	1.6	Cost of floppy disk used for office computer is a part of _____ overhead.	01
	1.7	_____ costs are directly/indirectly identified with product and include direct materials, direct labour and factory over heads.	01
	1.8	In _____ costing, jobs receive varying degrees of skill and attention from each production department or operation.	01
	1.9	Factory cost = _____ + factory overheads	01
	1.10	Two methods used for calculation of equivalent production are _____ and _____.	02
	1.11	Equivalent units are computed when some units remain _____ at the end of a particular period.	01
	1.12	Material usage variance is the sum of _____ and _____.	02
	1.13	Difference between the standard cost and actual cost is called as _____.	01
	1.14	_____ budget normally consist of (i) operating budget (ii) financial budget and (iii) special decision budget.	01

PART-B

2 a Journalise the following transactions:

2014		*RS.*
Jan. 1	*Started business with cash*	50,000
Jan. 3	*Paid into Bank*	40,000
Jan. 5	*Sold goods to Mohan*	22,000
Jan. 9	*Goods returned by Mohan*	2,000
Jan. 11	*Goods purchased from Shyam*	31,500
Jan. 15	*Goods returned to Shyam*	1,500
Jan. 18	*Bought furniture for office use by cheque*	9,000
Jan. 22	*Purchased goods for cash*	1,000
Jan. 22	*Paid cartage*	50
Jan. 30	*Paid interest on loan*	500

10

b "Accounting equation holds good under all circumstances", justify this statement with the help of six illustrations. 06

3 a From the following balances extracted from the books of Mythri & Co., prepare a trading and profit & loss account and balance sheet as on 31st December, 2012.

Particulars	*Rs.*	*Particulars*	*Rs.*
Opening stock	11,000	*Return outwards*	500
Bills receivable	4,500	*Trade expenses*	200
Purchases	39,000	*Office expenses*	1,000
Wages	2,800	*Cash in hand*	500
Insurance	700	*Cash at bank*	4,750
Sundary debtors	30,000	*Rent and taxes*	1,100
Carriage inwards	800	*Carriage outwards*	1,450
Commision (Dr.)	800	*Sales*	60,000
Interest on capital	700	*Bills payable*	3,000
Stationary	450	*Creditors*	19,650
Return inwards	1,300	*Capital*	17,900

The stock on 31st December 2012 was valued at *Rs.* 25000 13

b State the debit and credit rules for Personal, Real and Nominal accounts. 03

OR

4 a A book beeper has submitted to you the following trail balance wherein the totals of the debit and credit balance are not equal. Redraft the trial balance sheet sheet correctly as at 31st March $20X2$

Particulars	*Dr.* (*Rs*)	*Cr.* (*Rs*)
Capital	--	7,670
Cash in hand	– –	30
Purchases	8,990	– –
Sales	– –	11,060
Cash at bank	885	– –
Fixtures and fittings	225	– –
Free hold premises	1,500	– –
Water and electricity	65	– –
Bills receivable	– –	825
Return inward	– –	30
Salaries	1,075	– –
Creditors	– –	1,890
Debtors	5,700	– –
Stock at 1 *April* 20*X*1	3,000	– –
Printing	225	– –
Bills payable	1,875	– –
Rates, taxes & insurance	190	– –
Discount received	445	– –
Discount allowed	– –	200
	24,175	21,705

10

b Differentiate between (i) Journal and ledger (ii) Trial balance and Balance sheet. 06

5 a The following data relates to a manufacturing firm *ABC* Ltd. Calculate the cost of goods sold.

Particulars	*Rs.*
Raw material purchased	30,450
Wages	6,500
Power and fuel	675
Repairs	385
Maintenance	250
Consumables	320
Depreciation	1,500
Factory rent	20

Stock:	*Rs*
Raw material	
Opening	3,900
Closing	3,750
Work in progress	
Opening	10,950
Closing	10,700
Finished goods	
Opening	8,000
Closing	8,300

10

b Highlight the use of costing information for the following. Use a case example of a component manufacturing unit to illustrate your answer:-

i) Determining selling price,
ii) Controlling cost,
iii) Providing information for decision-making. 06

OR

6 Laxmi chemical Ltd processes a range of products including a detergent, "Mysore", which passes through three processes before completion and transfer to the finished goods warehouse. During September, data relating to this product were as follows:

	Process I	***Process II***	***Process III***	***Total***
Basic Raw Material (10,000 *units*)	*Rs.* 6,000	– –	– –	*Rs.* 6,000
Direct Raw Material added in process(*Rs*)	8,500	9,500	5,500	23,500
Direct Wages(*Rs*)	4,000	6,000	12,000	22,000
Direct expenses(*Rs*)	1,200	930	1,340	3,470
Production Overhead(*Rs*)				16,500
Output (*units*)	9,200	8,700	7,900	
Normal loss in process of input (%)	10	5	10	
Scrap value loss per unit	0.2	0.5	1	

The production overhead is absorbed as a percentage of direct wages. There was no stock at the start or at the end of any process.

You are required to prepare the following accounts: (a) Process I; (b) Process II; (c) Process III; (d) Abnormal loss; (e) Abnormal gain. 16

7 Calculate price, usage and mix variance from the following data.

	Standard			*Actual*		
Material	*Quantity* (*kgs*)	*Price* (*Rs*)	*Total* (*Rs*)	*Quantity* (*kgs*)	*Price* (*Rs*)	*Total* (*Rs*)
A	4	1	4	2	3.5	7
B	2	2	4	1	2.0	2
C	2	4	8	3	3.0	9

16

8 The cost of an article at the capacity of 5,000 units is given under *A* below. For variation of 25% in capacity above or below this level, the individual expenses vary as indicated under *B* below

	***A*(*Rs*)**	***B* (*percent*)**
Material cost	25,000	100 (*variable*)
Labour cost	15,000	100 (*variable*)
Power	1,250	80 (*semi – variable*)
Repairs	2,000	75 (*semi – variable*)
Stores	1,000	100 (*variable*)
Inspection	500	20 (*semi – variable*)
Administrative overheads	5,000	25 (*semi – variable*)
Selling overheads	3,000	50 (*semi – variable*)
Depreciation	10,000	100 (*fixed*)
Total	62,750	
Cost per unit	12.55	

Prepare the production cost budget at 4,000 units and 6,000 units 16

Foundations of Business Analytics

VII Semester SEE Question Papers

Course codes: 16IM74

- Jan/Feb 2021 (16IM74)
- Fast Track Oct 2020 (16IM74)
- Nov/Dec 2019 (16IM74)

16IM74

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

FOUNDATIONS OF BUSINESS ANALYTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Mention any ways in which semi-structured data is managed and stored.	02
	1.2	Expand the acronym DTD's.	02
	1.3	Arrange the following features into their respective categories: OLTP or OLAP. Subject centric, Complex queries, Historical data, Frequent Updates and Inserts.	02
	1.4	Mention few advantages of OLTP system.	02
	1.5	What is subject oriented, integrated, time variant and non-volatile data warehouse?	02
	1.6	Identify any two characteristics of dimension table.	02
	1.7	What are the components of metric data?	02
	1.8	Identify four perspectives of balanced score card.	02
	1.9	List three levels of data security concerns for mobile BI.	02
	1.10	What is cloud computing?	02

PART-B

2	a	Compare and contrast structured, semi-structured and unstructured data.	08
	b	Identify challenges faced while extracting information from stored semi-structured data.	08
3	a	With the help of suitable block diagram, identify role of OLAP tools in the BI architecture.	08
	b	What is data mining? Briefly explain techniques used by data mining to discover new patterns in data.	08

OR

4	a	Bring out the difference between Business Intelligence and Business analytics with the help of suitable examples.	08
	b	Highlight the various skills required to transform into a data scientist.	08
5	a	What are the goals that a data warehouse is meant to achieve?	08
	b	How is data integrity different from data quality? Explain.	08
		OR	
6	a	Compare the three types of handling of slowly changing dimensions.	08
	b	Enumerate different steps used to arrive at the ER diagram with help of an example.	08
7	a	Identify few sample KPI's used by the "Human capital and training management" division of GoodFood Restaurants Inc.	08
	b	Assume that you are holding a post of HR manager in XYZ company. Identify the various KPIs you will consider for your organization.	08
8	a	"Business Intelligence on the cloud" – why do you think this is important? Explain.	08
	b	What are the benefits and drawbacks of integrating social media information into Business Intelligence applications?	08

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

FOUNDATIONS OF BUSINESS ANALYTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Two of the challenges associated with unstructured data are _____ and _____.	02
	1.2	_____ data format uses atomic value for its characters.	01
	1.3	A SQL table can be classified as an example of _____ data.	01
	1.4	Two of the ways in which semi-structured data is managed and stored are _____ and _____.	02
	1.5	Systems managing transaction oriented applications like mainly entry, storage and retrieval of data are known as _____ systems.	01
	1.6	Compared to $OLAP$, the $OLTP$ systems are usually _____ in processing speed and _____ in space requirements.	02
	1.7	Attention towards relevance, accuracy, coherence in a data set is meant to improve its _____.	01`
	1.8	Any business intelligence component framework can be divided into 3 layers: _____, _____ and _____.	02
	1.9	Business intelligence applications like $DSS, OLAP$, various analytics etc. can broadly be divided into two categories: _____ solutions and _____ solutions.	02
	1.10	What are the similarities between an operational data store and a data warehouse?	02
	1.11	Overall view of an enterprise including key performance indicators are made available in a _____.	01
	1.12	Identify quantum and stratum in the following sentence on $KPI's$: "FBA course results in the year 2019 is 50% S grades, against 30% S grades in the year 2018".	02
	1.13	Software as subscription, increased scalability, reduced cost of maintenance of software are benefits associated with _____ and BI.	01

PART-B

2	a	What are the characteristics of Internet ready applications? Relate any four of these to a large home-appliance manufacturing organization.	10
	b	How is unstructured data set managed to enable storage and search?	06

3	a	What are the different *OLAP* architectures? Briefly mention their advantages and disadvantages.	09
	b	Transaction data of a toy-store is provided as follows for a month (sales data in Rs).	

Week	*Boys*	*Girls*
1	7041	9883
2	9680	2366
3	9660	7003
4	7456	8359

i) Illustrate the *OLAP* operations drill-up and pivot.
ii) List the other *OLAP* operations which can be performed on a larger data set — 07

OR

4		Briefly illustrate data models for *OLTP* and *OLAP* using a data base of Human Resource department of a large organization. Clearly show the keys, facts and dimensions in the model for the database.	16
5		What is meant by data integration? What are the two main approaches to data integration? Why is it needed? Briefly explain with the help of student performance databases present in different departments of a college.	16

OR

6		How is data quality different from data integrity? Provide measures of data quality. Provide examples for each measure.	16
7	a	With the help of an example of a managers responsibility for the customer service function of a firm, differentiate between balanced score cards and dashboard for viewing his/her performance.	08
	b	With the help of examples, explain any two kinds of analysis.	08
8		What more can social *CRM* do as compared to traditional *CRM* which has been in use for about 25 years already? Explain with the help of operations at a car dealership. Include a minimum of 4 points which is not possible with traditional *CRM*.	16

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

FOUNDATIONS OF BUSINESS ANALYTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What are the biggest sources of unstructured data in an enterprise? (List any two).	01
	1.2	Identify the key challenge in case of semi structured data.	01
	1.3	Do you think structured data stores be confined to contain/hold only structured data? Give reasons.	02
	1.4	What is content addressable storage?	02
	1.5	Classify the following features into respective categories of *OLTP* and *OLAP*:- Historical data, Normalized storage, Frequent updates and inserts, Adhoc queries.	02
	1.6	_____ are the measures that help an organization measure its performance.	01
	1.7	_____ helps unravel hidden patterns and spot trends in historical data.	01
	1.8	Identify the four distinct components of data warehouse.	02
	1.9	Illustrate the various data integration technologies used in a business organization.	02
	1.10	State any four benefits of using cloud computing.	02
	1.11	What are the four perspectives of balanced scorecard?	02
	1.12	Distinguish between tabular reports and matrix reports.	02

PART-B

2	a	Think of a retail organization of your choice, identify and explain the core business processes for the chosen organization.	08
	b	Classify the given data into structured, unstructured and semi structured data: video, *HTML*, e-mail, research paper, A *SQL* Table, Webpage, zipped file and an *OLTP* interface.	04
	c	Enumerate the advanced common capabilities of Internet ready applications (any 8).	04
3	a	Identify the common *OLAP* operations on multi-dimensional data and explain any four in detail citing suitable examples.	10
	b	How can business intelligence lead to performance enhancement? Explain with suitable examples.	06
		OR	

4	a	Citing suitable examples list and explain various types of metadata.	10
	b	Assuming suitable examples differentiate between snowflake schema and star schema.	06
5	a	Compare and contrast the various approaches for handling slowly changing dimensions. Use an example to clearly explain your answer.	09
	b	What is data profiling? When and how is data profiling conducted? Explain.	07
		OR	
6	a	Enumerate the goals that a data warehouse is meant to achieve.	06
	b	Differentiate between top-down approach and the bottom up approach to building a data warehouse? (Any five differences).	05
	c	What constitutes a fact table? What are the various types of facts? Explain using examples.	05
7	a	"You cannot manage what you cannot measure and you cannot measure what you cannot define". Comment on the above statement citing suitable example.	06
	b	Create a balanced score card for a fictitious enterprise. Explain the rationale behind it.	10
8	a	"$BI + ERP =$ increased value". Comment on the statement giving relevant examples.	06
	b	Highlight the various benefits of business intelligence in ERP.	06
	c	Differentiate between push and pull reporting as applicable to Mobile Business Intelligence.	04

Industry 4.0 - A Growth in Manufacturing

VII Semester SEE Question Papers

Course codes: 16IM7F1

- Jan/Feb 2021 (16IM7F1)
- Fast Track Oct 2020 (16IM7F1)
- Nov/Dec 2019 (16IM7F1)

16IM7F1

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

INDUSTRY 4.0 – A GROWTH IN MANUFACTURING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What is Industry 4.0?	02
	1.2	What is IoT?	02
	1.3	What are the components of ROBOT?	02
	1.4	What is cloud robotics?	02
	1.5	What is stereo lithography?	02
	1.6	What is maturity model?	02
	1.7	What is lean manufacturing system?	02
	1.8	List the technologies for successful implementation of Industry 4.0	02
	1.9	What is technology road map in the context of Industry 4.0?	02
	1.10	Distinguish between Industry 3.0 and Industry 4.0	02

PART-B

2	a	Explain the concepts and components of Industry 4.0	08
	b	Explain the IoT value creations layers and technologies.	08
3	a	What is Lean production system? Explain the features of Lean production system along with a focus on wastes.	08
	b	What are the various advanced industrial 4.0 technologies and cyber-physical systems that can be employed for waste elimination in advanced manufacturing systems? Explain in detail any two.	08
		OR	
4	a	Review the existing Industry 4.0 maturity and readiness models.	08
	b	Explain the problems encountered while implementing Industry 4.0 strategies.	08
5	a	Explain the technology road map for Industry 4.0 along with an illustration.	08
	b	Explain the new product and process development phase in Industry 4.0	08

		OR	
6	a	Explain the features of project portfolio optimization model.	08
	b	Review the talent development practices for Industry 4.0 environment.	08
7	a	Explain the various industrial robots used to bring intelligent and sensory capabilities in the manufacturing process.	08
	b	Explain the potential industrial robotic applications and the next generation of robotics planned to be utilized in the Industry 4.0	08
8	a	Explain the application areas of additive manufacturing with suitable examples.	08
	b	Discuss the advantages and disadvantages of additive manufacturing.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

INDUSTRY 4.0- A GROWTH IN MANUFACTURING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	In European Union, almost _____ of the GDP is explicated for by industry, which also effectuated approximately _____ job opportunities.	02
	1.2	In successful system adaption to Industry 4.0, horizontal integration is through _____.	01
	1.3	IP stands for _____ in communication and Networking.	01
	1.4	Lean Producers adopt the philosophy of doing more with less by eliminating _____ activites from _____.	02
	1.5	The waste in Japanese is called _____.	01
	1.6	The maturity model is formed in four stages and seven dimensions. Stages are: a. b. c. d.	02
	1.7	A _____ is defined in a time-based plan and portrays where an industry is, where it needs to go and how to get it there.	01
	1.8	The new product and process development phase allows _____ goals and projects considering different principles and constraints on separate _____ against a shared vision.	02
	1.9	ICT and other 'hard skills' required for Industry 4.0 have a more _____ character than basic literacy and professional know-how.	01
	1.10	Talent = Competence * _____.	02
	1.11	Shoe production steps: Cutting, Skiving, _____ and _____.	02
	1.12	In an Industry 4.0 factory, the robots endowed with the advanced capabilities owing to the information, _____ and _____ are able to collaboratively work with human workers and collaboratively with the other robots in an assembly line.	02
	1.13	$3DP$ stands for _____.	01

PART B

2	a	Discuss the components of Industry 4.0	08
	b	Justify Adaptive robotics, Additive manufacturing are the supportive technologies for successful implementation of Industry 4.0	08

3	a	How lean tools or techniques can handle seven wastes. Explain with a table. Table: The Seven Wastes versus Lean Tools/Techniques	08
	b	Automation is one of the Lean Production Application. Explain E-Kanban systems and automation in error-proofing in order to achieve the same.	08
		OR	
4	a	Explain the *IMPULS* with reference to Industry 4.0 readiness.	08
	b	Illustrate the strategy and organization maturity levels requirements in a proposed Industry 4.0 maturity model.	08
5	a	Write the Industry 4.0 transformation challenges.	08
	b	Predict the strategy phase and the New Product and Process Development phase.	08
		OR	
6	a	Identify the skill requirements in the Digital World.	08
	b	Explain Talent Development Practices for Industry 4.0.	08
7	a	Examine the power consumption in manufacturing using a structured neural network.	08
	b	Explain any three techniques used for predictive analytics.	08
8	a	Evaluate the recent technological components of Robots.	08
	b	Write six advantages of Additive manufacturing.	08

16IM7F1

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

INDUSTRY 4.0 – A GROWTH IN MANUFACTURING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	How Industry 4.0 is different from Industry 3.0.	02
	1.2	How do you justify the use of lean concept in automation system?	02
	1.3	List any two advanced components of robots in the era of Industry 4.0	02
	1.4	Mention any two dis-advantages of additive manufacturing.	02
	1.5	What is the form of material used in *FDM* and Stereo lithography technology?	02
	1.6	What are the objectives of Industry 4.0?	02
	1.7	What do you mean by *IMPULS* with respect industry 4.0?	02
	1.8	How industry 4.0 is different from IIoT	02
	1.9	What is Big data and Analytics with respect to IoT?	02
	1.10	What are the risks of an Industry 4.0 solution?	02

PART-B

2	a	With supporting technology, explain industry 4.0 framework.	08
	b	Explain Key Business Model Components of Smart and Connected Products.	08
3	a	Explain the role of lean tool for industry 4.0.	08
	b	Explain the following terms with respect to Automation Based Lean Production Application i) *E*-Kanban System ii) Automation in error proofing.	08
		OR	
4	a	Briefly explain Existing Industry 4.0 Maturity and Readiness Models.	08
	b	Discuss the strategies of proposed Industry 4.0 Maturity Model.	08
5	a	List the key benefits of technology road mapping.	08
	b	Describe the various challenges involved in Industry 4.0 transformation.	08
		OR	

6	a	Explain proposed Framework for Technology Roadmap of Industry 4.0.	08
	b	List and explain the assumptions made in project Portfolio Optimization Model.	08
7	a	How is multivariate analysis performed in big data? Show with an illustration.	08
	b	Write a short note on Cloud Robotics.	08
8	a	Briefly explain about Pre-build, Part-Build and Post-build process of SLA.	08
	b	Explain the principle, process, strengths and weaknesses of FDM.	08

Principles of Soft Computing

VII Semester SEE Question Papers

Course codes: 16IM71

- Jan/Feb 2021 (16IM71)
- Fast Track Oct 2020 (16IM71)
- Nov/Dec 2019 (16IM71)

16IM71

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RV COLLEGE OF ENGINEERING®

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VII Semester B. E. Examinations Jan/Feb- 2021

Industrial Engineering and Management

PRINCIPLES OF SOFT COMPUTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

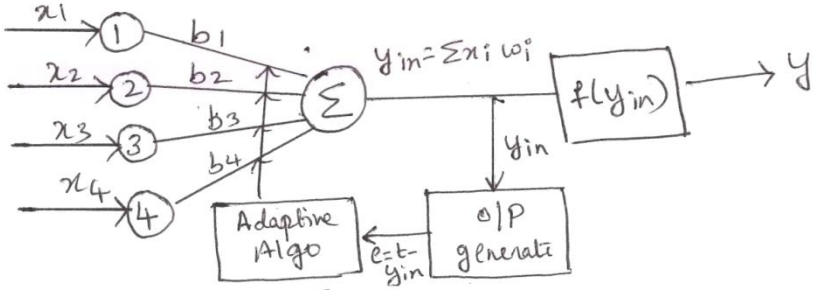
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Identify relevant or specific model in which chain rule differentiation is used.	01
	1.2	In GA, _____ operators usually work on a population, and may serve to remove weaklings, or to select strong individuals for reproduction.	01
	1.3	In Genetic Algorithm, an objective function takes a _____ as its parameter and typically gives a real-valued result that represents the fitness of goodness of the solution.	01
	1.4	_____ represents every individual of the potential solution as an array of sequences of strings.	01
	1.5	_____ and _____ approaches are used to investigate the layout of temporary facilities in relation to the planned buildings in a construction site.	01
	1.6	Sensing, sensory perception, actuation, control and knowledge based decision-making are crucial for the proper performance of an _____.	01
	1.7	A knowledge-based system may be organized and developed according to one or a combination of several architectures. List the three commonly used architectures.	01
	1.8	Consider a 4-input neuron which has weights 1, 2, 3 and 4 respectively. The transfer function is linear with the constant of proportionality being equal to 2. Now the inputs are 2, 3, 4 and 0 respectively. The output will be _____.	01
	1.9	Evolutionary computing techniques use different combination of operations, the _____ operation intends to reduce diversity of the sample population. On the other hand, the _____ operators try to increase the diversity of the population.	02
	1.10	Identify the features of Population-based incremental learning (PBIL). (Write any three features)	02
	1.11	List any 4 applications of Artificial Neural Network.	02
	1.12	The basic idea of Genetic Algorithm is to represent every individual of the potential solution as an array of sequencing of strings known as _____ and each string in it is called _____.	02

	1.13	List techniques to solve optimization problem.	02
	1.14	Identify the limitation of the Hopfield network model.	02

PART-B

2	a	Exemplify the complete structure of an intelligent process with necessary diagram.	08
	b	With example, explain Frame based system architecture for a knowledge based system, with neat diagram.	08
3	a	Consider two propositions A and B. Form the truth table of the combined proposition $(NOT\ A)\ OR\ B$. Show that, in accordance with two-valued crisp logic, this proposition is identical to the statement "$IF\ A\ THEN\ B$".	04
	b	Differentiate between the following with relevant representation: i. Height of a fuzzy set and support set ii. Set equality and grade of equality iii. Set inclusion and grade of inclusion.	04
	c	With the necessary fuzzy logic sets and membership function, arrive at Larsen ⊆ Mamdani ⊆ Zadeh ⊆ Dienes-Rescher ⊆ Lukasiewicz	08
		OR	
4	a	If "$A \rightarrow B$" is F and "$B \rightarrow A$" is T, what is the truth value of "$(A \rightarrow B)\ AND\ (B \rightarrow A)$" ? Using this result and the truth table of "$A \rightarrow B$", determine the truth table of "$A \leftrightarrow B$".	04
	b	Identify the following membership function and analyze with the help of diagram (if needed) and explain. i. $\mu_{A-B}(x) = \begin{cases} 1 & if\ \mu_A(x) = \mu_B(x) \\ \mu_A(x) T \mu_B(x) & otherwise \end{cases}$ ii. $\mu_{A \subset B}(x) = \sup[c \in [0,1] \mid \mu_A(x)\ Tc \leq \mu_B(x)]$ iii. $\mu_{A \rightarrow B}(x,y) = \mu_A(x)\mu_B(y) \quad \forall x \in X,\ \forall y \in Y$	06
	c	Differentiate the following with respect to fuzzy logic specifying relevant equations. i. Projection and Cylindrical extension ii. Join and Composition	06
5	a	Write the topology of Kohonen's self-organizing network and explain.	08
	b	Consider an ADALINE network shown in Fig 5b. Train the network for XOR function with bipolar inputs and target perform 1 Epoch of training. Initial weights are $w_1 = w_2 = b = 0.2$ and learning rate $h = 0.2$ Write the steps while calculating; find the MSE (mean square error). NOTE: Make a note if any assumptions are made.  Fig 5b	08

OR

6 a The network shown in Fig 6b is feed-forward, the weights are as given.

$w_{13} = -2, w_{23} = 3, w_{14} = 4, w_{24} = -1, w_{35} = 1, w_{45} = -1, w_{36} = -1, w_{46} = 1$

Each of the nodes $3, 4, 5$ and 6 uses the following activation function.

$$\Phi(v) = \begin{cases} 1 & if\ v \geq 0 \\ 0 & otherwise \end{cases}$$

Where v denotes the weighted sum of a node. Each of the input nodes $(1\ \&\ 2)$ can only receive binary values (either $0\ or\ 1$). Calculate the output of the network $(y_5\ \&\ y_6)$ for each of the input patterns (record all the calculations).

Pattern	P_1	P_2	P_3	P_4
Node 1	0	1	0	1
Node 2	0	0	1	1

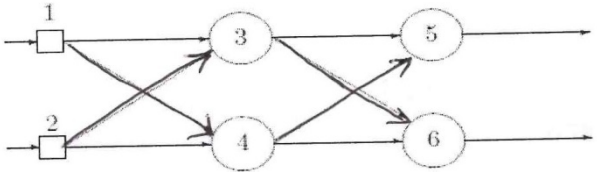


Fig 6a

08

b For the Multilayer Perceptron (MLP) shown in Fig 6b, initialize the initial weight as

0.3, set learning rate to 0.2

$x(1) = (0.2, 0.6), \quad t(1) = (0.88)$

$x(2) = (0.9, 0.4), \quad t(2) = (0.82)$

With $x\ \&\ t$ being the input and the output data respectively. Biases are treated here as connection weights that are always multiplied by -1 through a neuron to avoid special case calculation for biases. Each neuron uses a unipolar sigmoid activation function.

i. Make a note of all initializations and assumptions done if any

ii. Perform training only for loop 1 first pattern and make note on – forward propagation, output error measure and error propagation, update weights.

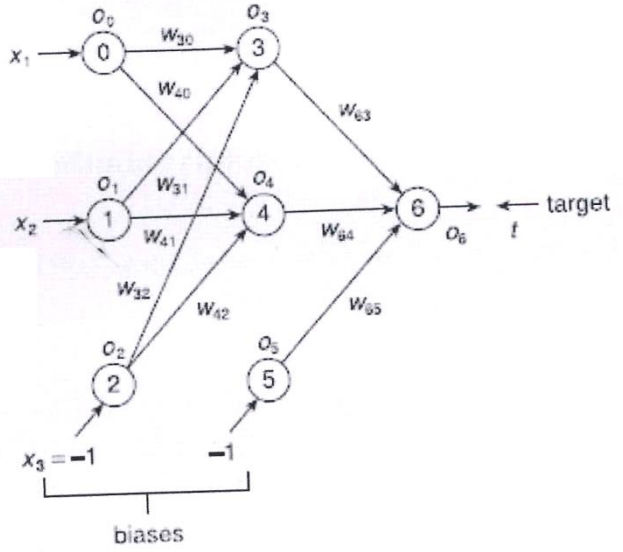


Fig 6b

08

7	a	How Integration of genetic algorithms with fuzzy logic is performed? Explain.	04
	b	With the aid of an example, explain how the population is generated in population based incremental?	04
	c	Explain the following applications of evolution strategies. i. Image processing and computer vision systems ii. Task scheduling by ES.	08
8	a	Identify the steps involved in proposing layout planning. With a brief illustration, explain the weight factors in generating layout planning.	10
	b	Write code division multiple access (CDMA) cellular network topology.	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

PRINCIPLES OF SOFT COMPUTING

Time: 03 Hours ***Maximum Marks: 100***

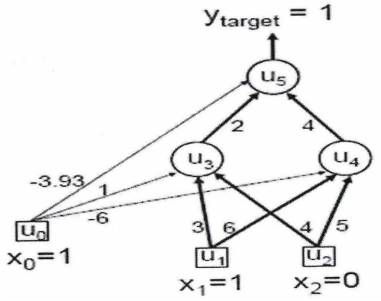
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	An artificial neural network (ANN) is typically composed of a set of parallel and distributed processing units, called _____.	01
	1.2	List the five mutation operators	01
	1.3	If a Genetic Algorithm converges to a set of solutions within a local minimum, which is not a global minimum, then the GA is said to have _____.	01
	1.4	_____ refers to a condition in the genotype structure where genes are highly interdependent	01
	1.5	Any database contains the _____, _____ and _____.	01
	1.6	Sensing, sensory perception, actuation, control and knowledge based decision-making are crucial for the proper performance of an _____.	01
	1.7	An _____ is a special type of knowledge based system, it contains the traditional constituents of the latter such as a knowledge base, an inference engine and a human/machine interface.	01
	1.8	_____ and _____ are the two strategies available to perform reasoning and make inferences in a production system	01
	1.9	The basic idea of genetic algorithm is to represent every individual of the potential solution as an array of sequences of strings known as _____ and each string in it is called _____.	02
	1.10	List the conflict resolution methods in reasoning strategies of production system	02
	1.11	List the categories of Numerical learning, supervised and the unsupervised algorithms.	02
	1.12	Give the two definitions of fuzzy implication.	02
	1.13	List any four applications of Artificial Neural Network.	02
	1.14	Identify the features of Population based Incremental learning ($PBIL$) (Write any three features).	02

PART B

2	a	With the help of a schematic model explain the dynamics of intelligence	08
	b	An aircraft emergency locator transmitter (ELT) is a device designed to transmit a signal in the case of a crash. The Alitiguage Manufacturing company makes 80% of the $ELTs$, the Bryant company makes 15% of them and the Chartair company makes the other 5%. The $ELTs$ made by Altigauge have 4% rate of defects, the Bryant $LLTs$ have a 6% rate of defects, and the Chartair $ELTs$ have a 9% rate of defects(whch helps to explain why Chartair has the lowest market share) (Hint: three events corresponding to the selection of ELT use $P(A/D) = \frac{P(A).P(D/A)}{[(P(A).P(D/A))]+[(P(B).P(D/B))]+[(P(C).P(D/C))]}$ i) ii) If an ELT is randomly selected from the general population of all the $ELT's$, find the probability that it was made by Altigauge Manufacturing Company. iii) It a randomly selected ELT is then tested and is found to be defective, find the probability that it was made by the Altigauge Manufacturing company.	08
3	a	Prove that the "minimum" operator is largest T-norm and the "maximum" operator is the smallest s-norm. Specifically show that $\min(x,y) \geq xTy$ and $\max(x,y) \leq xSy$. From the result prove that $xTy \leq min(x,y) \leq max(x,y) \leq xSy$	08
	b	With necessary symbolic representation discuss the fuzzy complement operation, due to Yager and Sugeno.	08
		OR	
4	a	Consider the fuzzy set A in the universe, R, the real line whose membership function is given by $\mu_A(x) = \begin{cases} 1-2\lvert x-1\rvert \; for \; \lvert x-1\rvert \leq 0.5 \\ = 0, otherwise \end{cases}$ i) Sketch this membership function ii) What is the support set of A? iii) What is the ½ cut of A?	08
	b	Identify with an example Cartesian product of fuzzy sets. Explain with two crisp set and two fuzzy set.	08
5	a	Summarize the different learning stages of a Hopfield network.	08
	b	For the multilayer perceptron (MLP) training for logical OR is shown in the fig 5b, weights on the arrows are $W_{13} = 3, W_{35} = 2, W_{24} = 5$, bias weight $W_{04} = -6$, Input pattern $X_1 = 1, X_2 = 0$ target output is Target = 1. Find the output of each neuron u_3, u_4, u_5 and calculate δ_5 to u_5  Fig 5b	08

		OR	
6	a	Differentiate Adaptive Linear Neuron($ADALINE$) and $MADLINE$ with the structure.	08
	b	How is the momentum effect with the rate of convergences in MLP?	08
7	a	Genetic algorithms have a number of drawbacks, as is the case of other optimization techniques.	08
	b	Write the steps in implementing Genetic Algorithms. With a neat schematic representation explain the same.	08
8	a	Exemplify overview of conventional bandwidth allocation techniques for learning based resource optimization in ATM networks.	08
	b	Write how mobile location is obtained using signal strength.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

PRINCIPLES OF SOFT COMPUTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

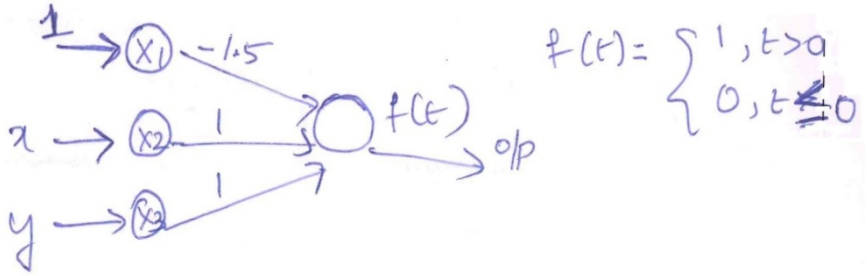
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

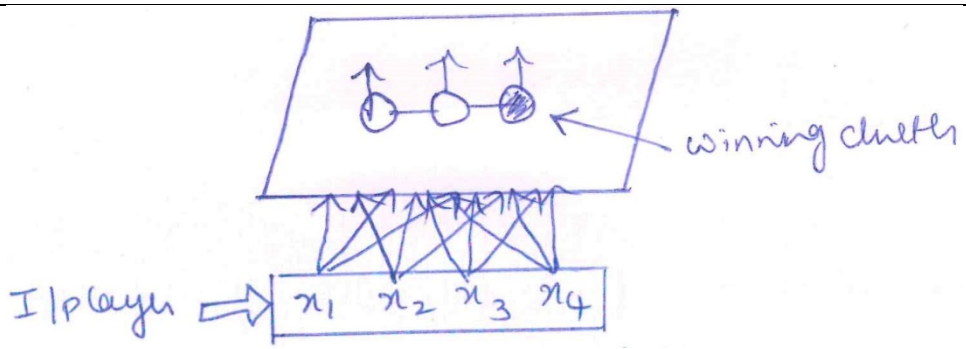
PART-A

1	1.1	List conflict resolution methods in the context of data matching.	02
	1.2	Tabulate the following properties of $T-norm$ (Triangular norm) and $S-norm$ ($T-conorm$) • Commutative • Associative • Boundary condition • Demorgan's law.	02
	1.3	List the four steps of genetic programming implementation.	02
	1.4	Give the two definitions of fuzzy implication.	02
	1.5	List the 4 unsupervised training procedures to determine the centers of the radial basis function.	02
	1.6	Write the mathematical representation of fuzzy complement operation for Yager and Sugeon.	02
	1.7	Write the 5 mutation operators.	02
	1.8	Training of the radial basis function network contains two separate steps. List them.	02
	1.9	Write the structure of feed forward and recurrent topology in neural network.	02
	1.10	List four components of expert system.	01
	1.11	Write any two algorithms to solve optimization problems.	01

PART-B

2	a	With the help of neat diagram, explain general purpose structure of an Intelligent Machine (each block explanation to be given).	08
	b	Define Expert System? With neat diagram, explain both components and operational schematics of an Expert System.	08
3	a	Recommend a room comfort control system with Temperature (T) and Humidity (H) process variables, sensor signals provides fuzzy logic controllers and determines the cooling rate (C) that generates air conditioning. Now write the basic composition on fuzzy rules (4), membership functions and also comfort control system diagram and explain.	10

	b	Prove that the "minimum" operator in the largest $T-norm$ and the "maximum" operator in the smallest $S-norm$, specifically show that $min(x,y) \geq xTy$ and $max(x,y) \leq xSy$. From this result, prove that $xTy \leq min(x,y) \leq max(x,y) \leq xSy$.	06
		OR	
4	a	Consider a fuzzy set $'A'$ in the universe $'R'$, the real line, whose membership function is given by $\mu_A(x) = 1-2\|x-1\|$ for $\|x-1\| \leq 0.5$ $= \theta$ $otherwise$ i) Sketch this membership function ii) What is the support set of A ? iii) What is the $\frac{1}{2}-cut\ of\ A$? iv) Another fuzzy set, B, is given by the membership function $\mu_B(x) = 1-\|x-2\|$ for $\|x-2\| \leq 1$ $= \theta$ $otherwise$ Sketch this membership function. Which one of the sets A and B is fuzzier? Explain.	08
	b	Identify the definition of the following. i) Set inclusion $(A \subset B)$ ii) Grade of inclusion iii) Set equality $(A = B)$ iv) Grade of equality Where A and B are fuzzy sets?	08
5	a	Consider a single perceptron with weights given in the Fig 5a and identify whether the network satisfy AND, OR, XOR logical operation.  Fig 5a	08
	b	Summarize multilayered perceptron network using Back propagation learning algorithm. (Represent the steps with clear mathematical formulas).	08
		OR	
6	a	A Kohenen self-organizing map is used to cluster four vertex given by(1,1,1,0), (0,0,0,1), (1,1,0,0) & (0,0,1,1); maximum number of cluster to be formed is $n = 3$ as shown below in Fig 6a. Suppose learning rate is given by $\alpha(0) = 0.3$ $\alpha(t+1) = 0.2\,\alpha(t)$ Each step updates only one cluster. Find the weight matrix for first two vectors only and update the matrix.	

			
		Fig 6a : structuring of neural network	08
	b	Write a note on the following: i) Neural network in Robotics ii) Neural networks for process monitoring and optimal control.	08
7	a	Summarize the steps for implementing genetic algorithms; write the block diagram to depict the steps.	08
	b	Write a brief note on the four issues in genetic algorithms.	08
8	a	Give the introduction, technique or algorithm and conclusion or results for the following. i) Learning based resource optimization in ATM ii) Mobile position estimation using an RFB.	12
	b	Write the main feature and the drawback when software tools are used for solving layout planning problem.	04

Technology Management

VII Semester SEE Question Papers

Course codes: 16IM7F4

- Jan/Feb 2021 (16IM7F4)
- Nov/Dec 2019 (16IM7F4)

16IM7F4

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

TECHNOLOGY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Illustrate the continuum of knowledge and highlight the position of technology in the continuum.	02
	1.2	Distinguish between Technologist and Technician.	02
	1.3	Identify any two conditions that need to be satisfied for an invention to be granted patent status.	02
	1.4	Indicate the three levels of diffusion in case of new or improved technologies as advocated by Stoneman.	02
	1.5	The neoclassical theory of economic growth, explained growth in terms of two factors of production. What are the two factors?	02
	1.6	Write the equation representing Cobb Dongles function.	02
	1.7	What is the basic purpose of Technological Strategy as advocated by Twiss?	02
	1.8	What is the purpose of Benchmarking process?	02
	1.9	What is meant by product differentiation? Give an example.	02
	1.10	List the five key elements of concurrent engineering approach.	02

PART-B

2	a	Illustrate the key constituents of a specific technology. Provide an overview on any two of the constituents.	08
	b	Tabulate the five levels of technology along with the characteristics of each level. Also indicate the pivotal staff for each level.	08

3	a	Illustrate the life cycle characteristics of a technology. Also explain the elements involved.	08
	b	Illustrate the relationship between technological stages and product life cycle along with corresponding features in a tabular form.	08
		OR	
4	a	Review the criteria for assessing the progress of technology.	08
	b	Explain the pattern of diffusion with the aid of 'S' curve.	08

5	a	Discuss the constituents of technological economics as a subject for study.	08
	b	What is engineering economics? Bring out the distinction among technological and engineering economics.	08
		OR	
6	a	Technology is marketable in several forms. List and explain any four such forms in detail.	08
	b	Economy of scale is an important feature of industrial operations, particularly with capital intensive manufacture. In the above context, explain different activities to which economy of scale can be applied.	08

7	a	Summarize the characteristics of initiative strategies and responding strategies.	08
	b	What is meant by competitive technology for a firm? Review the main features of a competitive technology.	08

8	a	Provide a brief note on the nine stages of evaluation, which are the basis for structuring of TCA (Technological Completion Analysis) technique.	08
	b	Moenaert et.al lists four factors which a firm needs to consider before committing itself to a new technology. Explain the four factors in detail.	08

16IM7F4

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

TECHNOLOGY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	List the importance of Technology in Educational Institutions.	02
	1.2	Justify the impact of Technology on environment.	02
	1.3	Define: a) Technological Progress b) Innovation.	02
	1.4	List the two categories of motivation for Technological change.	02
	1.5	List the two examples for Technological discontinuities.	02
	1.6	Define: a) Technological Economics b) Technological Convergence.	02
	1.7	Differentiate between the initiative strategy and responding strategies.	02
	1.8	Define Technology Cluster strategy.	02
	1.9	List the three stages for adoption of new technology.	02
	1.10	Define concurrent Engineering.	02

PART-B

2	a	Why has technology management emerged as an important area for study and research? Discuss the status of technology management in India?	08
	b	"For all countries, the most practical strategy for technology development is to make some and buy some." Explain with examples. What general aspects of classification would you like to suggest for planning fusion of technologies within a product technology portfolio.	08
3	a	"Every technology eventually reaches a decline phase." Discuss the reasons for such decline and suggest measures that could be helpful in arresting or postponing such decline.	08
	b	"Indian firms must learn to absorb imported technologies if they want to become self-reliant." Explain the statement, highlighting the need for technology diffusion. What should be the important elements of management of technology diffusion at the enterprise level?	08

		OR	
4	a	"The addition of new technology significantly improves product or process performance" Explain the statement, highlighting the need for Mature technology.	08
	b	"New Technologies are the result of a complex interaction between scientific advances, institutional factors and economic Mechanisms" Justify the statement.	08
5	a	Enumerate on the concepts of Technological and Engineering Economics.	08
	b	Discuss the role of technology strategy at the corporate level. What steps would you consider in formulating a viable technology strategy for a business firm?	08
		OR	
6	a	Enumerate on the concepts of production functions.	08
	b	'Economics technology presumes an environment, set of given technologies and input factors' Justify the statement considering the technology at macroeconomic level.	08
7	a	Explain briefly the concepts of Technology Gap Analysis.	08
	b	Describe briefly the concepts of technological alliances and technology crisis.	08
8	a	"Technology is an important enabler of competitiveness" Describe briefly the analysis of technology competition with suitable examples.	08
	b	With the help of an example, explain the implementation of Concurrent Engineering in product development.	08

Data Warehousing and Data Mining

VII Semester SEE Question Papers

Course codes: 12IM7E5

- Nov/Dec 2018 (12IM7E5)
- Nov/Dec 2017 (12IM7E5)
- Nov/Dec 2016 (12IM7E5)

12IM7E5

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

DATA WAREHOUSING AND DATA MINING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	_____ is the repository of data used for management decision support system.	01
	1.2	_____ contains the measurement of business processes, and it contains foreign keys for the dimension tables.	01
	1.3	State the concept of normalization.	02
	1.4	List three types of data warehouse users.	02
	1.5	The first step in data cleaning process is _____.	01
	1.6	A _____ says that each value of a given attribute must be different form all other values for that attribute.	01
	1.7	_____ find discrepancies by analyzing that data to discover rules and relationships and detecting data that violate such conditions.	01
	1.8	Kashunen – Lorve method is also called as _____.	01
	1.9	_____, _____ and _____ are three elements defining data quality.	02
	1.10	_____ investigates how computers can learn based on data.	01
	1.11	_____ techniques can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintains the integrity of the original data.	01
	1.12	_____ routines work to "clean" the data by filling in missing values, smoothing noisy data, identifying as removing outlines and resolving inconsistencies.	01
	1.13	_____ is a two-step process, consisting of learning step and a classification step.	01
	1.14	_____ is the learning of decision trees from class labeled training tuples.	01
	1.15	Classify different types of data mining systems according to the kinds of databases mined.	02
	1.16	Expand $DBSCAN$.	01

PART-B

2	a	Illustrate with the help of neat diagram the concepts of star schema and snow flake schema.	08
	b	Bring out the difference between online transactional and online analytical processing systems.	08

		OR	
3	a	With the help of data modeling diagrams explain the essential components of data warehouse architecture.	10
	b	Briefly explain common risks identified in data warehousing.	06
4	a	Distinguish between internal and external meta data. What are the elements of meta data strategy?	10
	b	List some of the common sources of meta data.	06
		OR	
5	a	Sketch a neat block diagram showcasing how to locate meta data within the physical data warehouse architecture.	10
	b	Explain how data warehousing and data mining are related.	06
6	a	In real world data, tuples with mixing values for some attributes are common occurrences. Describe various methods for handling this problem.	08
	b	Data quality can be assessed in terms of several issues, including accuracy, completeness, and consistency. For each of the above three issues, discuss how data quality assessment can depend on the intended use of the data, giving examples. Propose two other dimensions of data quality.	08
		OR	
7	a	What is data transformation? Explain various strategies associated with data transformation.	08
	b	Briefly highlight various techniques used to improve the efficiency of Apriori based mining.	08
8	a	With the help of an example explain about learning and classification involved in the data classification process.	10
	b	Differentiate between preprocessing and post-processing.	06
		OR	
9	a	What is classification by back propagation?	08
	b	Briefly describe the classification process using: i) Genetic algorithms ii) Rough sets iii) Fuzzy sets.	08
10	a	What are the typical requirements of clustering in data mining?	10
	b	Bring out the difference between partitioning and hierarchical methods of clustering.	06
		OR	
11	a	Discuss the applications of data mining in health care industry.	08
	b	Explain the concept of $DBSCAN$.	08

12IM7E5

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

DATA WAREHOUSING AND DATA MINING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	What is data warehousing?	02
	1.2	What is data mining?	02
	1.3	Differentiate between internal and external metadata?	02
	1.4	What are Snow Flake Schema?	02
	1.5	Differentiate between data integration and data reduction.	02
	1.6	List the association rules.	02
	1.7	What is cluster analysis?	02
	1.8	Write the acronym for DBSCAN.	02
	1.9	What are K-means?	02
	1.10	Define CRM.	02

PART-B

2	a	Explain the different ways of converting data into information.	08
	b	Explain the architectural challenges in data warehousing.	08
		OR	
3	a	Explain the phenomenon in data warehousing.	08
	b	Explain the database design in data warehousing systems.	08
4	a	Explain the types of metadata users in data management.	08
	b	Explain the classification of data mining systems.	08
		OR	
5	a	Explain the sources of metadata in data management.	08
	b	Explain the functionalities in data mining.	08
6	a	Explain discretization and concept hierarchy generation in data preprocessing.	08
	b	Explain FP-growth algorithm in data mining.	08
		OR	

7	a	Explain concept hierarchy generation in data processing.	08
	b	Briefly explain constraint based association in data mining.	08
8	a	Explain classification by decision tree induction.	08
	b	What is prediction? Explain classification by back propagation.	08
		OR	
9	a	Briefly explain Bayesian classification and its algorithm.	08
	b	Explain genetic algorithms in classification.	08
10	a	Explain Hierarchical clustering method in cluster analysts.	08
	b	Explain the application of industrial engineering in Health care system.	08
		OR	
11	a	What are outliers? Explain how they differ from noisy data.	08
	b	Explain the application of industrial engineering in e-commerce.	08

12IM7E5

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

DATA WAREHOUSING AND DATA MINING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

4. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
5. Answer FIVE full questions from Part B.

PART-A

1	1.1	Differentiate between a Part table and a dimension table.	02
	1.2	List any two termination conditions for decision tree algorithm.	02
	1.3	List any four strategies for data reduction.	02
	1.4	Define data mining and list any two industrial applications.	02
	1.5	Distinguish between star schema and snowflake schema.	02
	1.6	What is meant by a Metadata?	02
	1.7	State the concept of Normalization.	02
	1.8	Expand the acronym for: a) $DBSCAN$ b) RAD.	02
	1.9	Distinguish between classification and prediction.	02
	1.10	What are the two basic steps involved in process of data classification?	02

PART-B

2	a	Describe with neat sketch the three tier architecture of data warehouse.	08
	b	Describe the common risks identified in data warehousing.	08
		OR	
3	a	Explain star schema with an appropriate example.	06
	b	Describe the different ways of converting data into information.	10
4	a	Distinguish between external and internal metadata.	06
	b	Briefly describe the classification of data mining system.	10
		OR	
5	a	Describe various data mining functionalities.	12
	b	What do you mean by working with user meta data? How do you manage user meta data?	04

6	a	Describe various methods for handling data cleaning as part of data pre processing.	08
	b	Explain the working of Apriori algorithm for mining frequent patterns.	08
		OR	
7	a	Why we need to go for pre processing of data? Describe the major steps involved in data pre-processing?	10
	b	List and briefly explain the various data transformation methods.	06
8	a	Explain how back propagation performs learning on a multi layer feed forward neural network.	08
	b	The following table shows the midterm and final exam grades obtained for students in the data mining course	

Mid Term Exam	72	50	81	74	94	86	59	83	65	33	88	81
Final Exam	84	63	77	78	90	75	49	79	77	52	74	90

		Use the least square method to find an equation for the prediction of student's final exam grade based on the students midterm grade in the course.	08
		OR	
9	a	Briefly outline the major steps of decision tree algorithm.	08
	b	What is Genetic Algorithm? Briefly discuss the steps involved in GA.	08
10	a	What is an outlier? Briefly describe the different approaches behind statistical based outlier detection.	08
	b	Describe working of agglomerative and divisive hierarchical clustering.	08
		OR	
11	a	Describe the working of K-means clustering method.	10
	b	Write a brief note on various application of data mining in industrial engineering domain.	06

Retail Supply Chain Management

VII Semester SEE Question Papers

Course codes: 12IM7E4

- Nov/Dec 2018 (12IM7E4)
- Nov/Dec 2017 (12IM7E4)

12IM7E4

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

RETAIL SUPPLY CHAIN MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical table is permitted.

PART-A

1	1.1	Write four elements that are commonly considered as part of logistics.	02
	1.2	Give two examples each for qualitative and quantitative methods in merchandize forecasting.	02
	1.3	What is a planogram?	02
	1.4	Write the formula to calculate price elasticity.	02
	1.5	Distinguish between VMI and $CPFR$.	02
	1.6	In buyback contract the _____ can return back a portion of _____ inventory at predefined price.	02
	1.7	Three levels of customer service pyramid are _____, _____ and _____.	02
	1.8	$GCMMF$ stands for _____.	02
	1.9	List any four jewelry retailing supply chain characteristics/parameters.	02
	1.10	List two main components of $RFID$.	02

PART-B

2	a	What are the common elements between manufacturing and retail supply chains?	08
	b	List and explain the five inputs required for the merchandize budgeting process.	08
		OR	
3	a	Give at least four examples of strategies, tactical and operational decisions of a retail supply chain.	08
	b	What are the advantages and disadvantages of univariate models used in merchandize forecasting?	08
4	a	With the help of a block diagram, explain the components of assortment management.	10
	b	How can IT be used for pricing promotion and markdown?	06
		OR	

5	a	What are the objectives of retail promotion? (state any four).	08
	b	Identify the three different stages of space planning. Elaborate.	08
6	a	With a block diagram, explain the retailer driven replenishment/computer based ordering.	10
	b	List the pros and cons for global sourcing.	06
		OR	
7	a	Distinguish between purchasing, procurement and strategic sourcing.	08
	b	Discuss the different advanced payment technologies (any four).	08
8	a	How *RFID* can help in a food supply chain?	06
	b	List the different supply chain challenges faced by a footwear retailer.	10
		OR	
9	a	Identify the emerging trends in book and music retailing.	08
	b	Investigate the challenges involved in a food supply chain.	08
10	a	How cash and carry retailing benefits both supplier and customer?	08
	b	What is web 2.0? What are its advantages?	08
		OR	
11	a	Discuss the components of a retail *ERP* system.	08
	b	Discuss critical success factors for online retailing.	08

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VII Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

RETAIL SUPPLY CHAIN MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical table is permitted.

PART-A

1	1.1	Write four common processes/functions that are common for both manufacturing and retail supply chains.	02
	1.2	Write full form of the acronym $GMROI$.	01
	1.3	Sketch the customer decision tree/hierarchy for a toilet soap.	02
	1.4	Identify the phases in promotion management maturity model.	01
	1.5	Bio-degradable milk bottles are called _____ bottle.	01
	1.6	Write about $JIT2$ in one line.	01
	1.7	List two pros and cons of global sourcing.	02
	1.8	Sketch customer service pyramid and indicate the three tiers.	02
	1.9	$HACCP$ is increasingly recognized as the _____ and _____ vehicle for ensuring food safety.	01
	1.10	What are the two parts of prepack configuration decision?	02
	1.11	In home furnishing retailing location of the store is _____ factor.	01
	1.12	Name two global retail chains operating in organized $B2B$ cash and carry format in India.	01
	1.13	DCM sriram consolidated ltd. has successfully pioneered the concept of _____ rural departmental stores.	01
	1.14	A bar code contains _____ information encoded in the form of bars and spaces using international symbologies.	01
	1.15	_____ $RFID$ tag has no power source to transmit a signal.	01

PART-B

2	a	Enumerate the scope of retail logistics. Elaborate the elements.	08
	b	Explicate the assortment management processes of retail supply chains.	08
		OR	
3	a	What are the enabling components of category management and how do they facilitate category management decisions?	08
	b	You are the manager of a retail chain. You are required to prepare a six month merchandise budget. Make suitable assumptions.	08

4	a	Develop the assortment management framework. Sketch the block diagram. Provide brief summary of the framework.	08
	b	Provide a planogram design for a supermarket apparel section. Write a small note of processes for planogram management.	08
		OR	
5	a	How can information technology be used for pricing, promotion and markdown? Discuss.	08
	b	List any eight ECR best practices for the critical functional requirements of shelf ready packaging.	08
6	a	Write a detailed note on vendor managed inventory (VMI)/ continuous replenishment program (CRP), giving salient features. Advantages and cautions.	10
	b	Compare the home delivery policy adopted by two organized retail chains in your locality. What kind of supply chain infrastructure have they built to support the policy?	06
		OR	
7	a	What is meant by green sourcing? Explain some examples of the initiatives by the retailers on green sourcing.	08
	b	Illustrate the multichannel retailing challenges and required logistics capabilities with a block diagram and discuss.	08
8	a	What are the supply chain characteristics and challenges for fresh food retailing in India? Elaborate.	08
	b	Document the recent trends in dairy industry in India.	08
		OR	
9	a	Explain the concept of fast response. What steps zara had taken to make its supply chain responsive?	08
	b	Enumerate the characteristics of jewelry retailing supply chain in India.	08
10	a	Critically analyze the challenges and complexities of rural retailing supply chains in India. Cite examples.	08
	b	Identify critical success factors for online retailing and discuss.	08
		OR	
11	a	Discuss three important layers of retail analytic application. Illustrate with a sketch.	08
	b	Mobile applications are the next generation of technology enablers that leading retailers are looking at for bringing efficiency in their supply chain and customer service process. Provide your comments.	08

Supply Chain and Logistics Management

VII Semester SEE Question Papers

Course codes: 12IM72

- Nov/Dec 2018 (12IM72)
- Fast Track Jul 2018 (12IM72)
- Nov/Dec 2017 (12IM72)
- Fast Track Jul 2017 (12IM72)
- Nov/Dec 2016 (12IM72)
- Fast Track Jul 2016 (12IM72)

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Cisco's decisions regarding its choice of supply sources for components, contract manufacturing and the location and capacity of its warehouses, are all SC- _____ decisions.	01
	1.2	The most limited scope over which strategic fit is considered is one operation within a functional area within a company. This is referred to as _____ scope.	01
	1.3	List the supply chain macro processes.	02
	1.4	Inventory is not held by manufacturers at the factories but is held by distributors/ retailers in intermediate warehouses and package carriers are used to transport products from the intermediate location to the final customer. This network design option is called _____.	01
	1.5	One of the constraints in Capacitated Plant Location Model enforces that no plant can supply more than its capacity. This is mathematically represented as _____.	01
	1.6	Graphically depict variation of transportation cost and response time with number of facilities.	02
	1.7	_____ inventory is held, in case demand exceeds expectations to counter uncertainty.	01
	1.8	If the manager at B and M manages inventory such that the store does not run out of inventory in 6 out of 10 replenishment cycles, then store achieves a CSL of _____%.	01
	1.9	Define inventory turnover ratio.	02
	1.10	A _____ is a transportation route in which a truck either delivers product from a single supplier to multiple retailers or goes from multiple suppliers to a single retailer.	01
	1.11	_____ contract allows the buyer to modify the order (within limits) as demand visibility increases closer to the point of sale.	01
	1.12	Identify any two benefits of effective sourcing decisions.	02
	1.13	In the context of "Managerial Levers to achieve co-ordination', sharing point of sale data improves _____ accuracy.	01
	1.14	_____ is a SCM collaborative initiative that aims to improve the design of products through collaboration between manufacturers and suppliers.	01
	1.15	With one example clarify which are the parties involved in $B2C$ e- business.	02

PART-B

2	a	Describe the cycle and push/pull views of a supply chain. Which of the two views is very useful when considering strategic and operational decisions?	10
	b	'It is necessary to achieve a fit between competitive and functional strategies'. Justify this statement with an example.	06
		OR	
3	a	With example, illustrate the impact of [product life cycle on strategic fit between implied demand uncertainty and supply chain responsiveness.	10
	b	Write the block diagram to depict framework for structuring the drivers. How does Walmart use four drivers to strike balance between efficiency and responsiveness?	06
4	a	In the context of designing distribution network, elaborate on impact of e-business on customer service.	10
	b	Identify and describe the factors influencing Network Design Decisions.	06
		OR	
5	a	Draw the network of following distribution design options. Also bring out the advantages and limitations. i) Manufacturer storage with direct shipping ii) Distributor storage with last Mile Delivery.	10
	b	What are the key factors to be considered when designing the distribution network? Explain.	06
6	a	Weekly demand for Legos at a Walmart store is normally distributed, with a mean of 2500 boxes and a standard deviation of 500. The replenishment lead time is two weeks. i) Assuming a continuous –review replenishment policy, evaluate the safety inventory that the store should carry to achieve a CSL of 90 percent. ii) If the store manager decides to follow a periodic review policy to manage inventory of Legos and plans to review inventory after every four weeks, for a CSL of 90%, how much safety inventory should the store carry? What should its OUL be? iii) Compare above 2 review policies and comment on requirement of safety inventory.	12
	b	In the context of Multi-Echelon supply chains, elaborate on 'integer Replenishment Policy'.	04
		OR	

7	a	A company uses 12000 items per year. Carrying costs are 16% of the value of the item and the ordering costs are $Rs.20$ per order. The supplier however offers discounts as per the table 7.1 below. $Order\ Quantity$ | $Unit\ Price$ $0-2000$ | $Rs.3.00$ $2000-4000$ | $Rs.2.90$ $4000\ or\ more$ | $Rs.2.85$ Compute the economic order size.	11
	b	What is the role of cycle inventory in supply chain? Identify managerial levers that reduce lot size and cycle inventory in a supply chain without increasing cost.	05
8	a	Identify trade-offs that managers should consider when making transportation decisions. Explain.	10
	b	Justify in the context of tailored transportation: "a firm can meet customer needs at a lower cost by using tailored transportation to provide the appropriate transportation choice based on customer characteristics".	06
		OR	
9	a	Following table provides details of old supplier and a new probable supplier. If the weekly demand for a product is 1000 units, standard deviation per period is 300 units and holding cost at the company is 26%, evaluate the suppliers based on the impact on total cost and suggest who is better. Assume continuous review policy with required cycle service level of 95%. $Old\ Supplier$ | $New\ Supplier$ $Cost\ per\ unit\ = Rs.1/-$ | $Cost\ per\ unit\ = Rs.0.97/-$ $Lead\ time = 2\ Weeks$ | $Lead\ time = 6\ weeks$ $Standard\ deviation\ of\ Lead\ time = 1\ week$ | $Standard\ deviation\ of\ lead\ time = 4\ Weeks$ $Batch\ size = 2000\ units$ | $Batch\ size = 8000\ units$	16
10	a	Describe supply chain coordination, the bullwhip effect and their impact on SC performance.	10
	b	Identify and explain the key processes involved in customer Relationship Management.	06
		OR	
11	a	What is Vendor Managed Inventory? How it is different from continuous replenishment strategy? How manufacturer is benefitted out of VMI?	10
	b	Explain the importance of key metrics in building initiatives towards sustainable SC.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Fast Track Examinations July - 18

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	The customer order cycle occurs at the _____ interface and procurement cycle occurs at _____ interface.	02
	1.2	Write full form of the acronyms $ISCM, SRM, CRM$.	02
	1.3	Sate any two activities involved in the SRM macro process.	02
	1.4	A company's competitive strategy defines the set of _____, _____ that it seeks to satisfy through its _____ and services.	02
	1.5	Illustrate the supply chain responsiveness spectrum.	02
	1.6	The two major types of facilities are _____ sites and _____ sites.	02
	1.7	List two obstacles to achieving strategic fit.	02
	1.8	List two factors affecting optimum level of product availability.	02
	1.9	List the tradeoffs in transportation design.	02
	1.10	Suggest a suitable distribution network design for an organization that manufactures earth moving equipment. Justify in one line.	02

PART-B

2	a	Elaborate the three decision phases in a supply chain with examples.	08
	b	Develop the value chain for a typical manufacturing organization. Illustrate with a diagram.	08
		OR	
3	a	Write a detailed note on implied demand uncertainty. Explain the impact of customer needs on implied demand uncertainty. Sketch implied uncertainty spectrum and cite examples.	08
	b	Explain the components of inventory and transportation decisions in a supply chain.	08
4	a	Enumerate elements of customer service influenced by network structure and discuss.	08
	b	Elucidate the characteristics of manufacturer storage with direct shipping type of distribution network with emphasis on elements of customer service.	08
		OR	

5	a	Develop a framework for global site selection. Illustrate the four stages and discuss.	12
	b	Write a note on evaluating network design decisions using decision trees.	04
6	a	The Donimick's super market chain sells nut flakes, a popular cereal manufacturing by the Tastee Cereal Company. Demand for nut flakes is 1000 boxes per week. Dominick's has a holding cost of 25% and incurs a fixed trucking cost of $Rs.200$ for each replenishment order it places with Tastee. Given that Tastee normally charges $Rs.2$ per box of nut flakes, how much should Dominick's order in each replenishment lot? Tastee runs a trade promotion, lowering the price of Nut Flakes price reduction? Evaluate cycle inventory and average flow time for both the scenarios and calculate the amount forward buy.	10
	b	Bring out the importance of managing multi-echelon cycle inventory.	06
		OR	
7	a	Weekly demand for HP printers at a Sam's club store is normally distributed with a mean of 250 and a standard deviation of 150. The store manager continuously monitors inventory and currently orders 1000 printers each time the inventory drops to 600 printers. HP currently takes two weeks to fill an order. How much safety inventory does the store carry? What CSL does Sam's club achieve as a result of this policy? What fill rate does the store achieve?	12
	b	List and explain with examples the managerial levers to improve supply chain profitability.	04
8	a	Enumerate the Pros and Cons of different transportation networks.	10
	b	Explain tailored transportation.	06
		OR	
9	a	List the various factors used for supplier scoring and assessment. Explain any four.	08
	b	Discuss in detail with examples the impact of design collaboration on the supply chain performance.	08
10	a	One outcome of the lack of supply chain coordination is the bullwhip effect. Examine the statement with respect to various performance measures of a supply chain.	08
	b	Write a note on the supply chain macro processes and the transaction management foundation.	08
		OR	
11	a	High light the significance of collaborative planning forecasting and replenishment.	06
	b	Discuss the role of reverse logistics in the closed loop supply chain.	10

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R. V. COLLEGE OF ENGINEERING

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VII Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

4. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
5. Answer FIVE full questions from Part B.

PART-A

1	1.1	Represent using block diagram the typical stages of a Generalized components of a Supply chain.	01
	1.2	Consider the objective of a supply chain to maximize the overall value generated. Use the following identity: Supply chain surplus = Customer value – Supply chain cost, to define supply chain profitability.	01
	1.3	Mention the time frame considered typically for decisions made during the supply chain planning phase of a typical supply chain management system.	01
	1.4	Graphically represent the cost-responsiveness efficient frontier curve. How do you interpret this frontier in the context of supply chain to understand the supply chain capabilities?	01
	1.5	Define safety inventory.	01
	1.6	Mention the major category of factors that influence network design decisions in supply chain.	01
	1.7	Mention the typical inputs required for formulating and solving the capacitated plant location network optimization model.	01
	1.8	List out the categories of supply chain risks to be considered during supply chain network design.	01
	1.9	Diagramatically represent the configuration of a Dedicated network and a Fully Flexible network.	01
	1.10	Give reason for the following guideline for decision making under uncertainity for the Global supply chain design. “Use Financial Analysis as an input to decision making, not as the decision making process.”	01
	1.11	Define Holding Cost. What is the unit of measurement?	01
	1.12	If the demand increases by a factor of k, what will be the impact of this on the optimal lot size?	01
	1.13	What will be the impact of trade promotions on the lot size and Cycle inventory? Give reasons.	01
	1.14	List out the three basic questions that need to be considered when designing a transportation network between two stages of a Supply chain.	01
	1.15	What are the pros and cons of the option of Direct Shipping with milk runs in the design of transportation network?	01

	1.16	List the three types of risks to be considered when transporting a Shipment between two nodes in a transportation network.	01
	1.17	What modes of transportation are best suited for large, low value shipments? Why?	01
	1.18	Mention the key macro processes under Customer Relationship Management.	01
	1.19	Justify this statement "Use Information Technology (*IT*) systems to support decision making, not to make decisions."	01
	1.20	What do you understand by the term Sustainability as applicable to Supply Chain Management?	01

PART-B

2	a	"Successful Supply Chain Management requires many decisions relating to the flow of information product and funds." Discuss the three categories or phases of decisions related to decisions in a supply chain. Take a case example to highlight the importance of these decision phases.	08
	b	Compare and Contrast "Cycle view" versus the "push/pull view" of a Supply chain. Give illustrations for each of these.	08
		OR	
3	a	Justify as to why "There must be consistency between the customer priorities that the Competitive Strategy hopes to satisfy and Supply Chain capabilities that the supply chain strategy aims to build." Discuss the importance of achieving strategic fit with any one case study as an example.	08
	b	How could a bicycle manufacturer increase the responsiveness through decisions related to building Supply chain capabilities and design of facilities? Take the case study example of a bicycle manufacturer and Justify why this is a case of a responsive Supply chain, given the trend of environment friendly mobility solution in today's society.	08
4	a	Discuss the performance characteristics of manufacturer storage with direct shipping network along the dimensions of * Inventory, * transportation, * facilities and handling, * information and also * response time, * product variety, * product availability and * customer experience.	10
	b	Discuss in the Indian context the problems and issues related to last mile delivery and the implications of these in the designing of supply chain networks.	06
		OR	
5	a	Why has the online channel been more successful in the *PC* industry (personal computers) compared to the grocery industry? In the future how valuable is the online channel likely to be in the *PC* industry?	10

	b	Discuss the impact of the following on the location decisions in a supply chain • Import duties and exchange rates • Rise in transportation costs.	06
6	a	Discuss impact of following approaches for managing inventory in a supply chain with case examples. • Using common components across multiple products • Building inventory of high demand or predictable demand products.	08
	b	Discuss the following with examples for implementing inventory module in supply chain management systems. i) Coordinating planning across enterprise in the supply chain, ii) Take predictable variability into account when making strategic decisions.	08
		OR	
7	a	Discuss why subcontractors can offer products and services to a company more cheaply than if the company produced them itself. Take one case example of a product based organization and one case example of a service oriented organization and discuss.	10
	b	What is the difference between lot sized based and volume based quantity discounts?	06
8	a	Discuss the various modes of transportation used in typical supply chain and the performance characteristic of each of the modes in the transfer of materials in the supply chain.	10
	b	What is the meaning of Tailored transportation? Discuss the following forms of tailored transportation i) Tailored transportation by customer density and distance, ii) Tailored transportation by size of customer.	06
		OR	
9	a	An auto manufacturer sources both office supplies and sub systems such as seats. What, if any difference in sourcing strategy would you recommend for the two types of products? Discuss in detail.	08
	b	Discuss the importance of design collaboration on the supply chain performance.	08
10	a	Bring out the importance of information and information technology in a supply chain. Give a case example to highlight successful *IT* deployment in supply chain. How does *IT* enable the managing of supplier relationship management? Explain.	08
	b	Give a few examples to justify how the availability of real time information and its use can help improve Supply chain performance. One of the case example could be a carrier service.	08
		OR	

11	a	Discuss the role of sustainability in a supply chain. Take the case example of an OEM(Original Equipment Manufacturer) involved in assembly of light commercial vehicles and outline the sustainability initiatives that can be taken up.	08
	b	Elaborate key features of closed-loop supply chains.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical table is permitted.

PART-A

1	1.1	The phase in which company decides how to structure the supply chain is called _____ phase.	01
	1.2	_____ processes are initiated and performed in anticipation of customer orders.	01
	1.3	Draw cost – responsiveness efficiency frontier.	01
	1.4	A company's _____ strategy defines a set of customer needs that it seeks to satisfy through its products and services.	01
	1.5	Transportation costs are least in _____ type of distribution network design.	01
	1.6	Graphically show the relationship between number of facilities and inventory costs.	01
	1.7	_____ analysis evaluates the present value of any stream of future cash flows and allows management to compare two streams of cash flows in terms of their financial value.	01
	1.8	A mode with higher transportation costs can be justified if it results in significantly lower _____.	01
	1.9	If optimal lot size is 980 units annual demand is 12,000 units, then average flow time is _____ years.	01
	1.10	If demand is steady, then cycle inventory is given by _____.	01
	1.11	The store manager at B & M is responsible for inventory and purchase of films. He does not continuously track film inventory. Every Saturday he checks film inventory and the manager orders enough such that available inventory and size of the order total 1000 rolls of films.	01
	1.12	The cost of _____ is the loss incurred by a firm for each unsold unit at the end of the selling season.	01
	1.13	_____ mode of transportation is used primarily for the transport of crude petroleum, refined petroleum products and natural gas.	01
	1.14	_____ transportation is the use of more than one mode of transportation to move a shipment to its destination.	01
	1.15	With the _____ option, the buyer structures his transportation network so that all shipments come directly from each supplier to each buyer location.	01

	1.16	_____ is a SCM collaborative initiative that aims to improve the design of products through collaboration between manufacturers and suppliers.	01
	1.17	Fluctuations in orders increase as they move up the supply chain from retailers to wholesalers to manufacturers to suppliers is called _____.	01
	1.18	What are the key processes of SRM?	01
	1.19	_____ obstacles occur in situations when incentives offered to different stages or participants in a supply chain lead to actions that increase variability and reduce total supply chain profits.	01
	1.20	Define inventory turnover ratio.	01

PART-B

2	a	What is supply chain management? Explain with the neat sketch the supply chain network and flows.	06
	b	Bring out the comparison between efficient SC and responsive SC. It is necessary to achieve a fit between competitive and functional strategies. Justify this statement with an example.	10
		OR	
3	a	List key supply chain decision phases and explain the significance of each one.	09
	b	How does WalMart structure its drivers so that its competitive strategy and supply chain strategy are in harmony?	07
4	a	Explain decision tree methodology for evaluating network design decisions under uncertainty.	06
	b	Demand allocation decisions have to be made for plants in a supply chain in which many manufacturing plants serve different markets. Multiple plants may be used to satisfy demand at a market. Perform the following to formulate a mathematical model that minimizes total cost of production, inventory and transportation. i) Identify the inputs required for mathematical model ii) Identify objective function iii) Formulate the constraint equations for the model. Also draw supply chain network for above data.	10
		OR	
5	a	Elucidate with an example the role of network design decisions in supply chain.	08
	b	What are the key factors to be considered when designing the distribution network? Explain.	08
6	a	Daily demand for PCs at Dell is normally distributed, with a mean of 2,500 and a standard deviation of 500. A key component in PC assembly is the hard drive. The hard drive supplier takes an average of 7 days to replenish inventory at Dell. Dell is targeting a CSL of 90% (providing a fill rate close to 100%) for its hard drive inventory. Evaluate the safety inventory of hard drives that Dell must carry if the standard deviation of the lead time is seven days.	10

	b	Identify managerial levers that reduce lot size and cycle inventory in a supply chain without increasing cost.	06
		OR	
7	a	Demand for vitamins is 10000 bottles per month in Drugs Online (DO). DO incurs a fixed order placement, transportation and receiving cost of $Rs.100$ each time an order for vitamins is placed with the manufacturer. DO incurs a holding cost 20%. The price charged by the manufacturer varies according to the all unit discount pricing schedule shown. Evaluate the number of bottles that DO manager should order in each lot. *Order quantity* | *Unit price* 0 – 5000 | $Rs.3.0$ 5000 – 10000 | $Rs.2.96$ 10000 *or above* | Rs 2.92	11
	b	Why do manufacturers offer trade promotions? What impact do trade promotions have on supply chain performance?	05
8	a	The manager at Albertson's a grocery store selling online, has 5 orders that are to be delivered to customers. The location and order size for each customer are shown below *X coordinate* | *Y coordinate* | *Order size* *Warehouse* | 0 | 0 | *Customer* 1 | 7 | 7 | 70 *Customer* 2 | 12 | 5 | 90 *Customer* 3 | 7 | 4 | 50 *Customer* 4 | 3 | 4 | 60 *Customer* 5 | 12 | 0 | 80 The Albertson's fulfillment store has two trucks, each capable of carrying up to 200 units. Using the savings matrix and nearest neighbor methods devise a suitable delivery schedule.	10
	b	Given the customer density and distance from warehouse how should a firm like $W.W$. Grainger should tailor its transportation network?	06
		OR	
9	a	What are some ways that a firm like WalMart benefits from good sourcing decisions?	05
	b	What is the role of supply contracts? Elaborate on types of contracts and their impact on supplier performance and information distortion.	11
10	a	Evaluate the impact of bullwhip effect on supply chain performance.	10
	b	Explicate the impact of supply chain drivers on performance of sustainable supply chain.	06
		OR	
11	a	What is Vendor Managed Inventory (VMI)? How it is different from continuous replenishment strategy? How manufacturer is benefited out of VMI?	09
	b	How does business practice such as $CPFR$ improve supply chain performance? Discuss.	07

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

9. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
10. Answer FIVE full questions from Part B.
11. Use of statistical table is permitted.

PART-A

1	1.1	A _____consists of all parties involved, directly or indirectly in fulfilling a customer request. This includes not only the manufacturers and suppliers but also transporters, warehouses, retailers and even customers themselves.	01
	1.2	The three major categories of decisions in a supply chain are supply chain strategy or design, supply chain planning and finally the decisions related to _____.	01
	1.3	Graphically depict the four cycles of a supply chain processes according to the cycle view of supply chain processes.	01
	1.4	Consider the automotive sector. Where do you position the issues in managing supply and demand for an existing automobile model already in the market? Answer with reference to the implied uncertainty (demand and supply) spectrum.	01
	1.5	The first phase of the process of designing a distribution network is to decide on the broad structure of the supply chain structure and develop a visualization of the distribution network. Based on this what is the critical decision in the second phase?	01
	1.6	Indian $FMCG$ distribution channels like the agriculture channels are among the longest in the country. The product moves from manufacturers through C & F agents or warehouses to distributors for further sale to wholesalers or stockists wherefrom these move to retailers to reach eventual customers. The term C & F stands for _____.	01
	1.7	The allocation of Supply Sources and markets to facilities has a significant impact on performance because it affects total production, inventory and transportation costs incurred by the Supply Chain to satisfy _____.	01
	1.8	The set of factors that influence the network design decisions that include items like taxes, tariff's exchange rates and shipping costs is known as the _____ factors as a collective entity of factors.	01
	1.9	The modeling structure used to formulate and solve the decisions related to network design with special emphasis on the aspects of decisions related to faculty locations and consequent capacity allocation decisions is known by the name _____ in the optimization literature.	01

	1.10	A _____ the quantity that a stage of supply chain either produces a purchases at a time. This definition helps in determining the cycle inventory in a supply chain.	01
	1.11	The _____ cost in the context of estimating cycle inventory related costs in practice, that includes all incremental costs associated with placing or receiving an extra order that are incurred regardless of the sizes of the order.	01
	1.12	_____ replenishment across products, retailers or suppliers in a single order allows for reduction in lot sizes for individual products because fixed ordering and transportation costs are now spread across multiple products, retailers or suppliers.	01
	1.13	There are two reasons due to which the quantity discounts can increase the supply chain surplus. One of the reasons is extraction of surplus through price discrimination. Mention the other reason.	01
	1.14	The _____ is an important supply chain driver because products are rarely produced and consumed in the same location.	01
	1.15	Intermodal transportation is the use of more than one mode of transport to move a shipment to its destination. A variety of intermodal combinations are possible with the most common being _____.	01
	1.16	Cheaper modes of transport typically have longer lead times and larger minimum shipment quantities both of which result in higher levels of _____ in the supply chain.	01
	1.17	While selecting a mode of transportation managers must account for unit costs and cycle, safety and in transit inventory costs that result from using each mode. Modes with high transportation costs can be justified if they result in significantly _____ inventory costs.	01
	1.18	A vital step in India's further development is expanding the _____ network and modernizing harbors and airports.	01
	1.19	_____ is essential to making good supply chain decisions because it provides the broad view needed to make optimal decisions.	01
	1.20	The focus that allows a supply chain to better serve more environmentally conscious customers while often improving supply chain performance is known as _____ initiatives.	01

PART-B

2	a	Supply chain decision phases may be categorized as design, planning or operational, depending on the time frame during which decisions made apply. Design decisions constrain or enable good planning which in turn constraints or enables effective operation. Discuss with examples the decision phases and its impact on design and operations of supply chain.	08
	b	Consider the following scenarios from customer needs point of view and discuss the effect of these on the implied demand uncertainty. Give proper reasoning: i) Range of quantity required increases ii) Lead time decreases iii) Variety of product required increases iv) Number of channels through which product may be acquired increases.	08
		OR	
3	a	Sourcing plays a critical role in supply chain context. List and elaborate on the components of sourcing decisions. Give examples of their importance from the firm perspectives.	08

	b	What are some industries in which products have proliferated and life cycles have shortened? How have the supply chains in these industries adapted? Give your view points on the strategies to manage the supply chains.	08
4	a	Diagrammatically represent the following design options for a distribution network: i) Manufacturer storage with direct shipping ii) Distributor storage with last-mile delivery Discuss the cost factor and service factor implications of these two design options.	08
	b	One of the critical phases of network design in the supply chain context requires that managers identifies potential locations in each region where the company has decided to locate a plant: i) Identify the inputs required for making this critical decision ii) Discuss the process involved in applying gravity location models to solve the location decision problems. iii) Give the mathematical structure of the problem.	08
		OR	
5		Reliable is a cell phone manufacturer serving the Asian and North American markets. Current annual demand of its product in Asia is 2 million, whereas the demand in North America is 4 million. Over the next 2 years, demand in Asia is expected to go up with either 50 percent with a probability of 0.7 and by 20 percent with probability of 0.3. Over the same period, demand in North America is expected to go up by 10 percent with a probability of 0.5 or go down by 10 percent with a probability of 0.5. Reliable currently has a production facility in Asia with a capacity of 2.4 million units per year and a facility in north America with a capacity of 4.2 million per year. The variable production cost per phone in Asia is $15 and the variable cost per phone in North America is $17. It costs $3 to ship a phone between the two markets. Each phone sells for $40 in both markets. Reliable is debating whether to add 2 million units or 1.5 million units of capacity to the Asia plant. The larger plant increase will cost $18 million, whereas the smaller addition will cost $15 million. Assume that reliable uses a discount factor of 10 percent. What do you recommend?	16
6	a	"Flexibility should be valued by taking into account uncertainty in demand and economic factors. In general the value of flexibility increase with an increase in uncertainty". Discuss the key concepts enshrined in this and give examples to illustrate this point.	08
	b	What are some major nonfinancial uncertainties that a company should consider when making decisions on where to source product? Discuss with the help of an example the impacts of decisions.	08
		OR	
7	a	The orange company prices I-pods at $550 per unit. Good buy sells the *I*-pods at $775. Annual demand at this retail price turns out to be 4,50,000 units. Good buy incurs ordering, receiving and transporting costs of $10,000 for each lot of *I*-pods ordered. The holding cost used by the retailer is 20 percent.	

		i) What is the optimal lot size that good buy should order? ii) The orange company has discounted *I* pods by $40 for the short term (the next two weeks). Good buy has decided not to change the retail price but may change the lot size ordered from orange. How should good buy adjust its lot size given this discount? How much does the lot size increase because of the discount?	10
	b	Define product fill rate, order fill rate and cycle service level. What are their significance in safety inventory determination?	06
8	a	Elaborate the concept of milk runs. Discuss in detail the design option for a transportation network, known as direct shipping with milk runs.	08
	b	In the context of supply chain, discuss the modes of transportation and their performance characteristics.	08
		OR	
9	a	A publisher sells books to Barnes and Nobles at $12 each. The marginal production cost for the publisher is $1 per book. Darnes and Nobles prices the book to its customers at $24 and expects demand over the next two months to be normally distributed with a mean of 20,000 and a standard deviation of 5000. Barnes and Noble places a single order with the publisher for delivery at the beginning of the two months period. Currently Barnes and nobles discounts any unsold books at the end of two months to $3 and any books that did not sell at full price sell at this price. i) How many books should Barnes and Nobles order? What is its expected profit? How many books does it expect to sell at a discount? ii) What is the profit that the publisher makes given Barnes and nobles action?	10
	b	Discuss the factors that affect decisions to outsource a supply chain function.	06
10	a	Discuss the role of Information Technology in a supply chain. What are the characteristics of information for effective supply chain decisions?	08
	b	Give a few examples illustrating how availability of real time information can be used to improve supply chain performance. Discuss risk management in information technology.	08
		OR	
11	a	Consider the following key metrics for sustainability: i) Energy consumption ii) Water consumption iii) Greenhouse gas emissions iv) Waste generation. Discuss the importance of these metrics in building the initiatives towards sustainability. What are the units of measurements for these metrics commonly deployed?	10
	b	What is a closed loop supply chain? What are the challenges in designing a closed loop supply chain?	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VII Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical table is permitted.

PART-A

1	1.1	_____ is the uncertainty that exists due to the portion of demand that the supply chain is required to meet.	01
	1.2	_____ reflects the uncertainty of customer demand for a product.	01
	1.3	_____ refers to the probability of having a product in stock when a customer order arrives.	01
	1.4	The ease with which a customer can return unsatisfactory merchandise and the ability of the supply chain network to handle such returns is called _____.	01
	1.5	_____ is the fraction of replenishment cycles that end with all the customer demand being met.	01
	1.6	_____ is the fraction of product demand that is satisfied from product in inventory.	01
	1.7	Returns clause is a _____ contract.	01
	1.8	A _____ is a route in which a truck either delivers product from a single supplier to multiple retailers or goes from multiple suppliers to a single retailer.	01
	1.9	Manufacturer–driven substitution reduces inventory requirements for the same level of _____.	01
	1.10	Supply chain decision phase has supply chain design, supply chain planning and _____.	01
	1.11	In _____ system, the inventory is held in large quantities.	01
	1.12	_____ is a process in replenishment cycle.	01
	1.13	Satisfying customer needs is not a global network _____ decision phase.	01
	1.14	If ROP id reorder point, σ_L is standard deviation during lead time and DL is expected demand during lead time, then Cycle service level is given by _____.	01
	1.15	The major disadvantage of aggregating all inventory in one location is _____.	01
	1.16	_____ is a step in savings matrix method.	01
	1.17	The biggest advantage of drop-shipping is the ability to _____ inventories at the manufacturer.	01

	1.18	The ability to aggregate inventories and _____ product customization is a significant advantage of in-transit merge.	01
	1.19	Revenue- sharing contracts counter _____ by decreasing the cost per unit charged to the retailer.	01
	1.20	_____ replenishment policies require more safety inventory than continuous review policies for same lead time and availability.	01

PART-B

2	a	Examine the three key supply chain decision phases and explain the significance of each one.	08
	b	Analyze the various stages of a cyclic view of supply chain? Explain with examples.	08
		OR	
3	a	Explain the value chains linking supply chain with business strategy with illustrations.	08
	b	Compare and contrast a supply chain's responsiveness and efficiency with examples.	08
4	a	Discuss the five key factors to be considered when designing the distribution network.	08
	b	Given 8 market locations and demand at each location, use centre of gravity method and suggest a possible warehouse location.	08
		OR	
5	a	With neat sketches, highlight the importance of any three Design Options for a Distribution Network.	08
	b	List and explain the factors influencing supply chain network design decisions? Illustrate with an example.	08
6	a	Find EOQ, Total cost and Number of orders for a product for which price breaks are as follows:	
		The monthly demand for a product is 400 units. Cost of storage is 20% of unit cost and the cost of ordering is Rs. 25.	08

$Market\ Area$	$Demand\ (Units)$	X_i	Y_i
$A1$	8	2.5	10
$A2$	20	3	5
$A3$	12	6.5	8
$A4$	10	11	10
$A5$	30	11	8
$A6$	20	10	4
$A7$	40	13	3.5
$A8$	30	12	2

$Order\ Quantity$	$Price/Unit\ (Rs.)$
$0 \leq q1 \leq 100$	20
$100 \leq q2 \leq 200$	18
$20 \leq q3$	16

	b	Evaluate the different measures of product availability with suitable examples.	08
		OR	
7	a	The manager at sportmart, a sporting goods store, has to decide on the number of skis to purchase for the winter season. Based on past demand data and weather forecasts for the year, management has forecast demand to be normally distributed, with a mean of $\mu = 350$ and a standard deviation of $\sigma = 100$. Each pair of skis cost $C = \$100$ and retails for $p = \$250$, Any unsold skis at the end of the season are disposed of for \$85. Assume that it costs \$5 to hold a pair of skis in inventory for the season. How many skis should the manager order to maximize expected profits?	08
	b	Evaluate the product fill rate for the following data: Safety stock = 1,000, order quantity, $Q = 10{,}000$, Standard deviation, $S_L = 707$, Product fill rate (f_r) =?	08
8	a	"Sourcing processes is critical to supply chain profitability". Explain with illustrations.	08
	b	"Third party logistics increases the supply chain surplus"- substantiate your answers.	08
		OR	
9	a	Define tailored transportation? Explain its importance in distribution process.	06
	b	Identify the importance of various factors that affect the transportation decisions.	10
10	a	Illustrate with examples how managerial levers could be used to achieve supply chain coordination.	08
	b	Describe the supply chain *IT* framework with suitable examples.	08
		OR	
11	a	How warehouse management could be made more efficient by the application of *RFID*?	08
	b	Highlight the impact of bull whip effect on supply chain profitability. How to overcome it?	08